



Lab5 Students' Learning Level Prediction

ENGG1910 Demystifying AI





Introduction

Background information and
setup of Colab

Supporting projects

- Teaching Development and Learning Enhancement Grant: An AI-enhanced Adaptive and Individualized eLearning System for Mathematics Foundation Courses at the Faculty of Engineering

Principal Supervisors



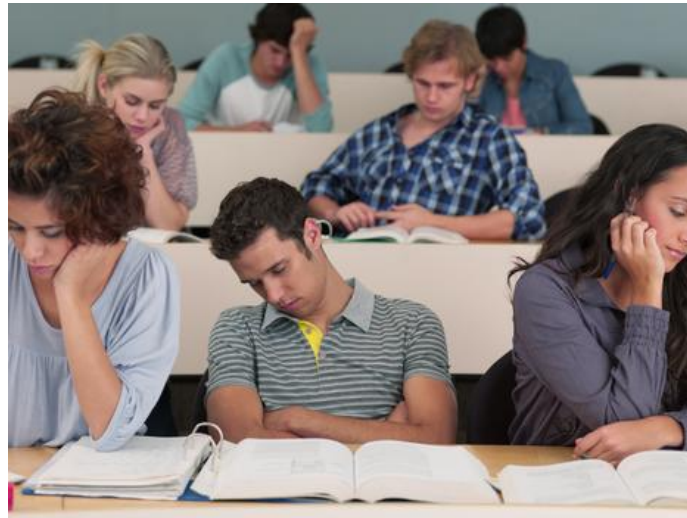
Prof. Tan LEE



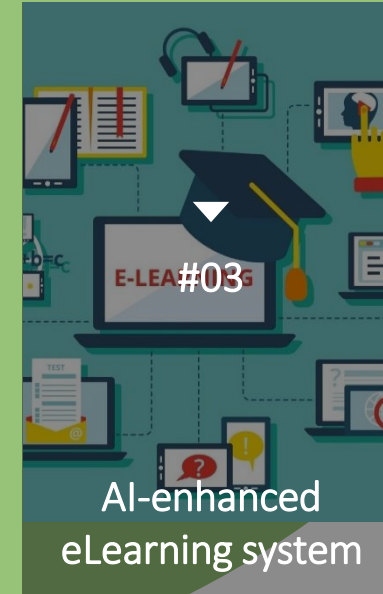
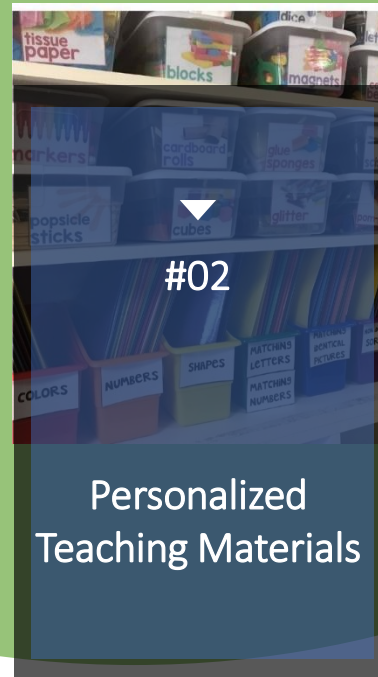
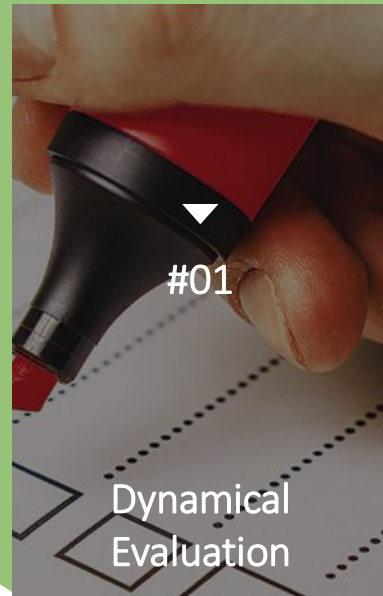
Dr. Dongkun HAN

Main motivations

- Students are different (**variety of learners**)
- Learning materials are different (**variety of knowledge and skills**)
- It is hard to provide suitable knowledge and training to suitable students at a certain learning level (**proper individualized education**)

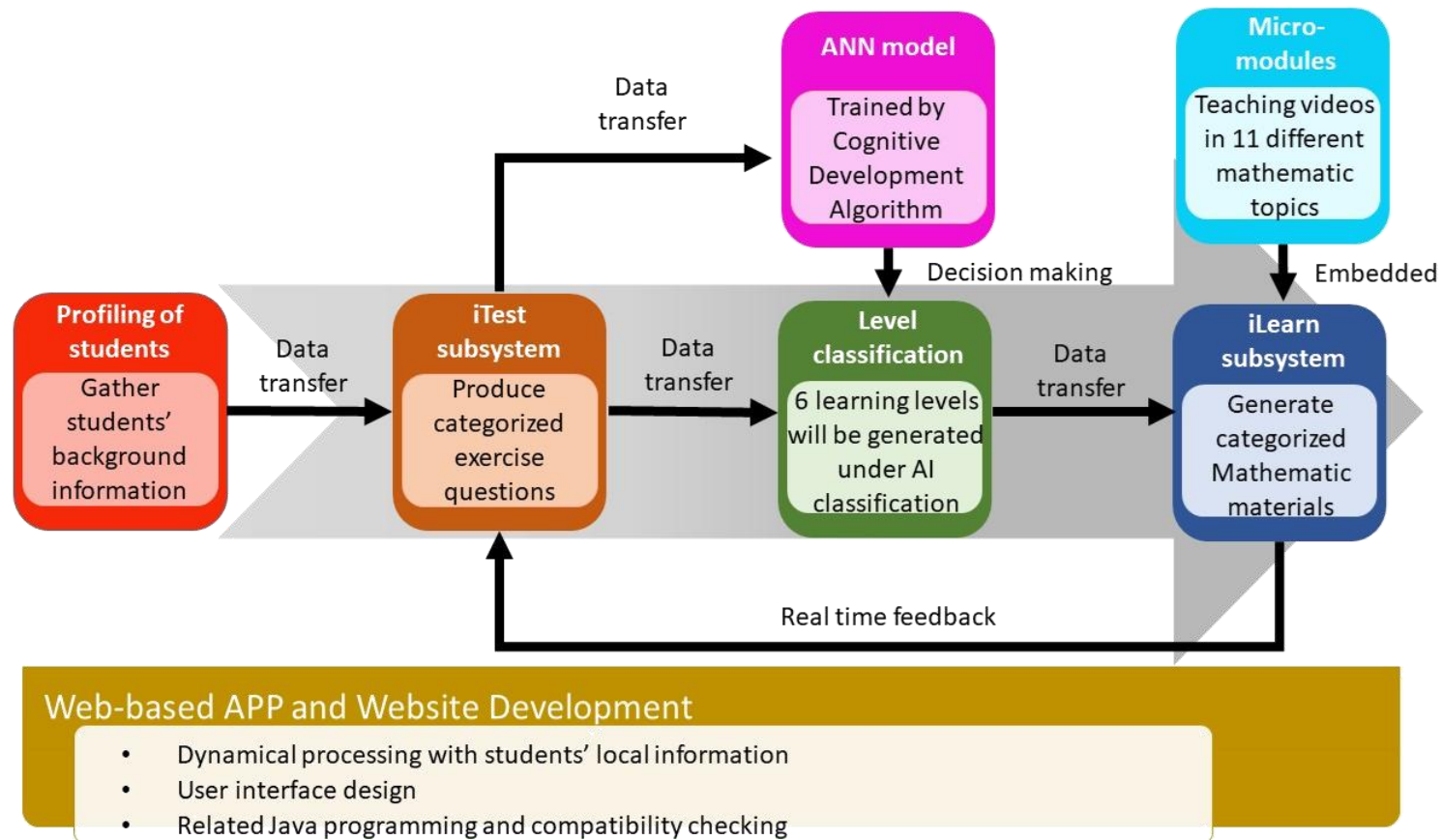


Targets of the Project

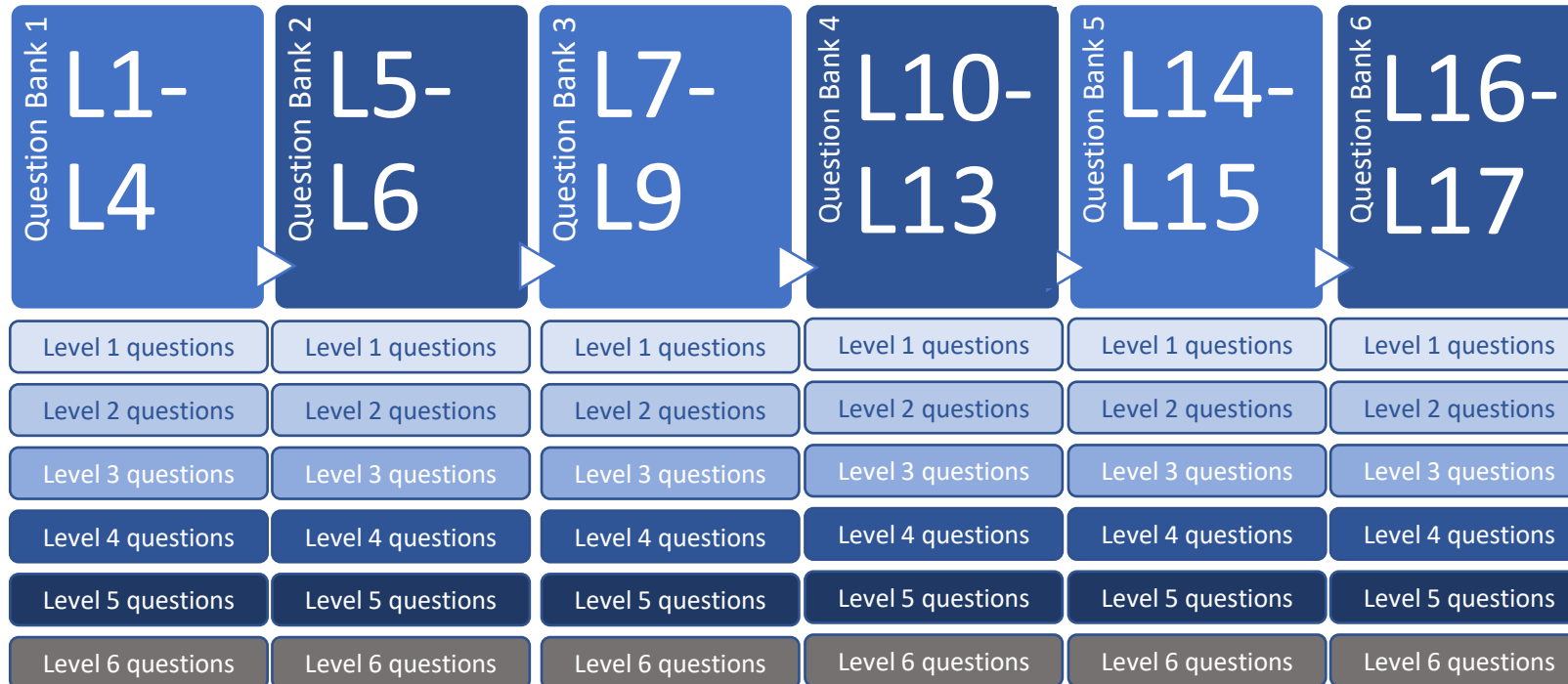


- **Dynamically evaluate** students on different contents in learning mathematics
- Supply personalized and appropriate **teaching materials**
- Develop an **AI-enhanced eLearning system** in teaching mathematic foundation courses

Structure of eLearning system



iTest question bank of ENGG2020



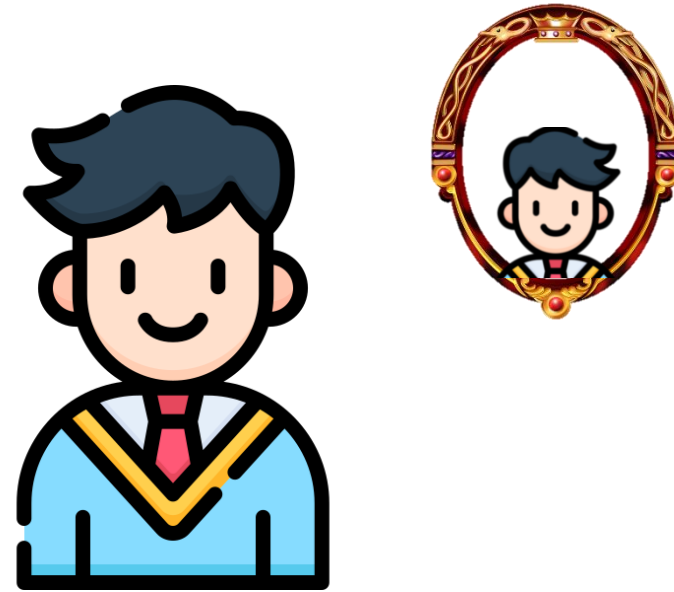
So tired



Prediction is important



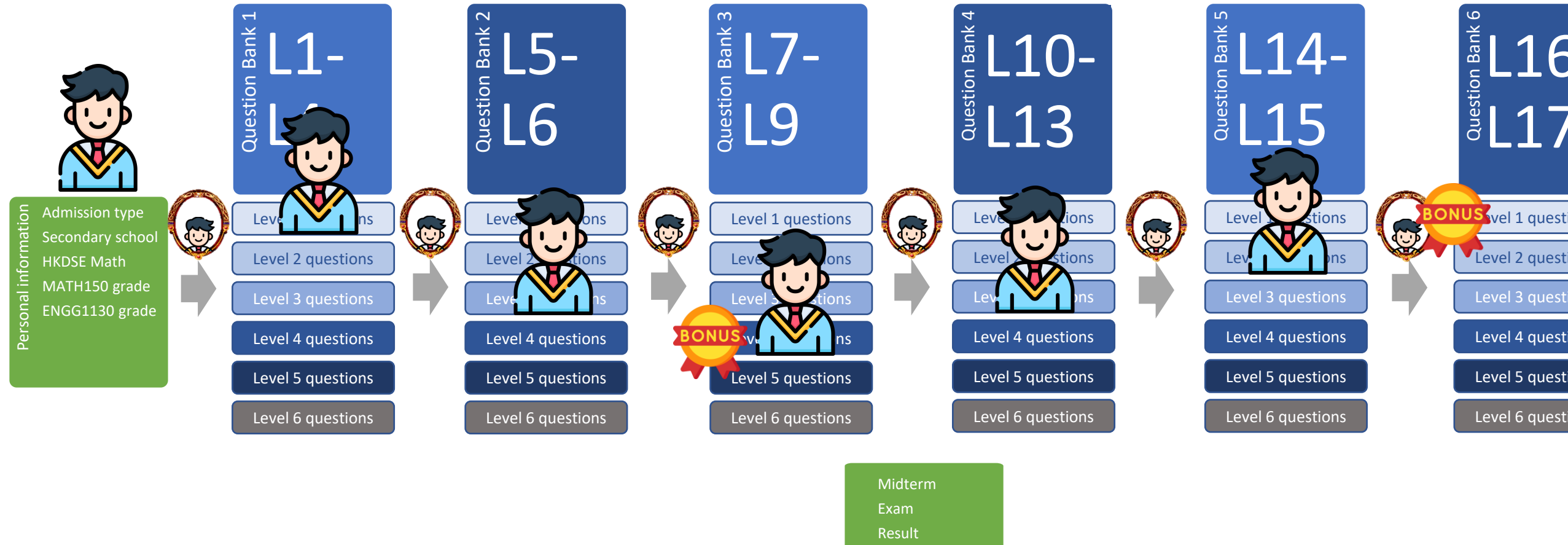
"Mirror, mirror on the wall, Who is the fairest of us all?"



"Mirror, mirror on the wall, Who will get a Fail?"

"Mirror, mirror on the wall, what is my grade of this course overall?"

Dynamical evaluation

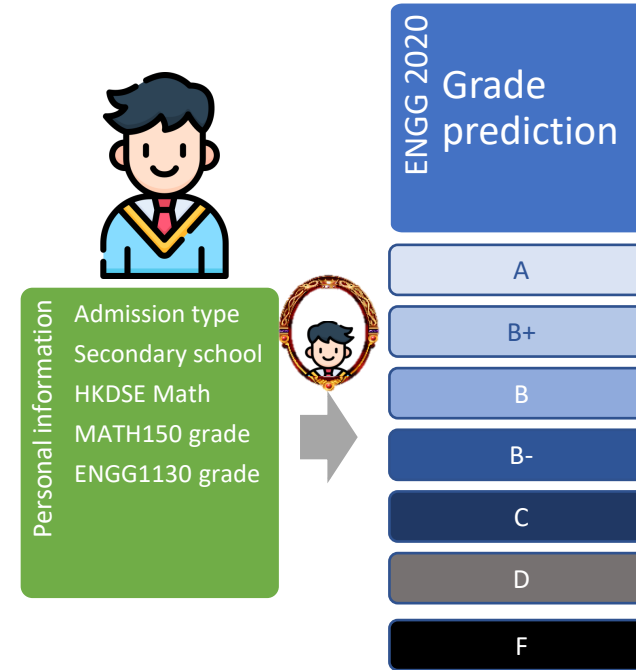


Possible benefits

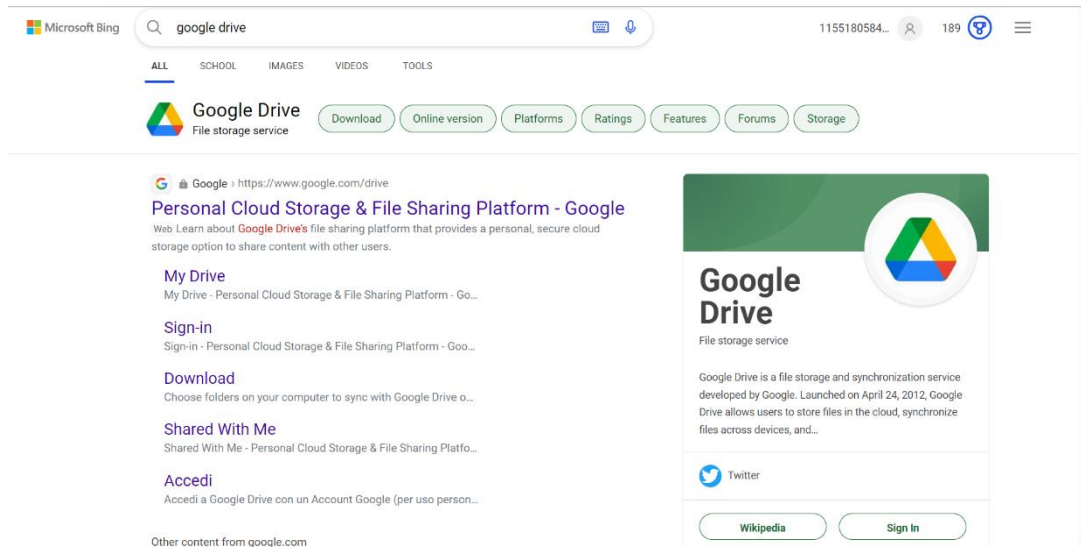
- **Motivations** for self-learning (boosted learning outcome)
- Automatic warning and intervention system (less F)
- Big data analysis for other generative AI tools (digital resources)

Objectives of this lab

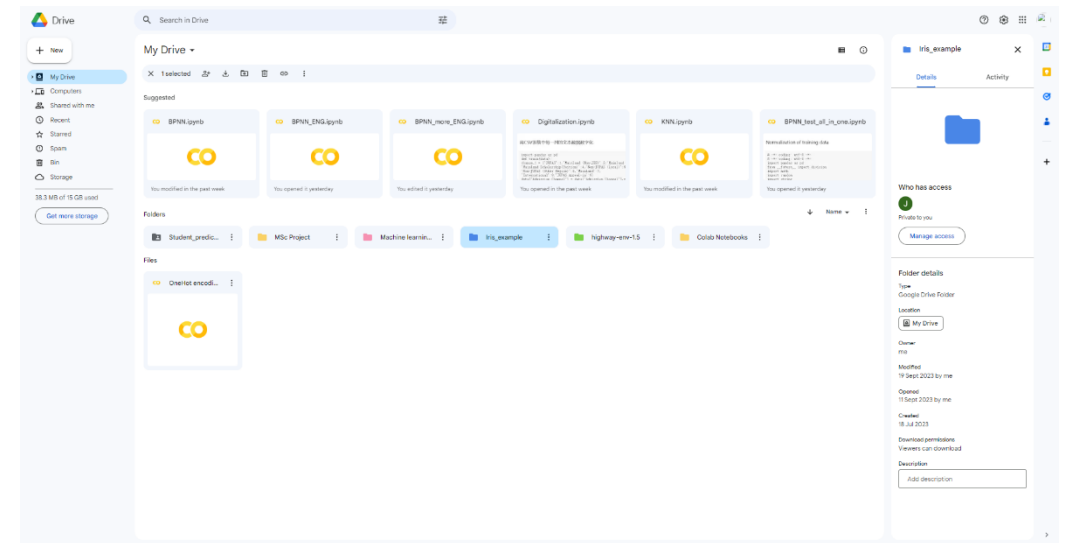
- Setup a programming **platform** for learner level prediction
- Introduce **4 methods** for learner level prediction
- Implement 4 methods on Colab with given data and codes
- Select an optimal method and parameters for your own case



Register a google drive



Register a Google drive: <https://drive.google.com/>



Google drive interface

Download files

- Download the files from Blackboard:

Course Contents/Laboratories/Lab 5 Student Learning Level Prediction

BPNN	11/1/2024 11:15 AM	File folder
Comparison	11/1/2024 11:15 AM	File folder
KNN	11/3/2024 6:13 PM	File folder
Main_code	11/1/2024 11:15 AM	File folder
NB	11/1/2024 11:15 AM	File folder
SVC	11/1/2024 11:15 AM	File folder

Data explanation

- Real information of 2019-2021 undergraduate students at Faculty of Engineering.
- The **features** include Admission channel, Secondary school, Admission exam, Math background, Grade of MATH1510, ENGG1130, ESTR1006 and ENGG2020.

	A	B	C	D	E	F	
1	No	Admission Channel	Secondary School	Exam	Math Background	MATH1510 Grade (2020-21 T1)	EN
2	373	International	#N/A	Kazakhstan High School	Kazakhstan High School - 5/5 in all Math subjects	B+	
3	374	JUPAS	LOK SIN TONG KU CHIU MAN SEC SCH	HKDSE	HKDSE Math: 5	B	
4	375	JUPAS	TWGH WONG FUT NAM COLLEGE	HKDSE	HKDSE Math: 5	B-	
5	376	Mainland	#N/A	Mainland JEE	Mainland JEE Math: 136.5	B+	
6	377	Mainland	#N/A	Mainland JEE	Mainland JEE Math: 147	A-	
7	378	International	#N/A	SAT + SAT Subject Test	SAT Math Lv. 2: 800/800	A-	
8	379	International	#N/A	SAT + SAT Subject Test	SAT Math Lv. 2: 800/800	B-	
9	380	JUPAS	KOWLOON SAM YUK SECONDARY SCHOOL	HKDSE	HKDSE Math: 4, M1: 4	A-	
10	381	JUPAS	Non-school applicant	HKDSE	HKDSE Math: 5	P	
11	382	JUPAS	HKWMA CHU SHEK LUN SECONDARY SCHOOL	HKDSE	HKDSE Math: 5, M2: 3	C-	
12	383	JUPAS	LAI KING CATHOLIC SECONDARY SCHOOL	HKDSE	HKDSE Math: 4, M2: 3	C-	
13	384	JUPAS	TWGH CHEN ZAO MEN COLLEGE	HKDSE	HKDSE Math: 4	F	
14	385	JUPAS	Non-school applicant	HKDSE	HKDSE Math: 5*, M1: 5*	A-	
15	386	JUPAS	KING LING COLLEGE	HKDSE	HKDSE Math: 5, M2: 4	B	
16	387	JUPAS	TSUEN WAN PUB HO CHUEN YIU MEMORIAL COLL	HKDSE	HKDSE Math: 5*, M2: 4	C+	
17	388	JUPAS	KING'S COLLEGE	HKDSE	HKDSE Math: 5*, M2: 5	B	
18	389	JUPAS	Non-school applicant	HKDSE	HKDSE Math: 4, M2: Abs	U	
19	390	JUPAS	POOI TO MIDDLE SCHOOL	HKDSE	HKDSE Math: 5, M2: 3	P	
20	391	JUPAS	SKH BISHOP BAKER SECONDARY SCHOOL	HKDSE	HKDSE Math: 5, M1: 3	B-	
21	392	JUPAS	CARMEL SECONDARY SCHOOL	HKDSE	HKDSE Math: 5	B	
22	393	JUPAS	Non-school applicant	HKDSE	HKDSE Math: 5, M2: 4	A-	
23	394	JUPAS	STFA LEUNG KAU KUI COLLEGE	HKDSE	HKDSE Math: 5**, M2: 4	B+	
24	395	JUPAS	Non-school applicant	HKDSE	HKDSE Math: 4	P	
25	396	JUPAS	SKH LAM WOO MEMORIAL SECONDARY SCHOOL	HKDSE	HKDSE Math: 5*, M2: 4	B-	
26	397	JUPAS	CCC CHUEN YUEN COLLEGE	HKDSE	HKDSE Math: 5*, M2: 4	B	
27	398	JUPAS	STFA LEE SHAU KEE COLLEGE	HKDSE	HKDSE Math: 5	F	
28	399	JUPAS	METHODIST CHURCH HK WESLEY COLLEGE	HKDSE	HKDSE Math: 4, M2: 4	B-	
29	400	JUPAS	CCC HEIP WOH COLLEGE	HKDSE	HKDSE Math: 4	P	

Data preparation

- Only keep the data of students admitted by HKDSE and Gaokao.
- Only keep the data of students without selecting P/F option.

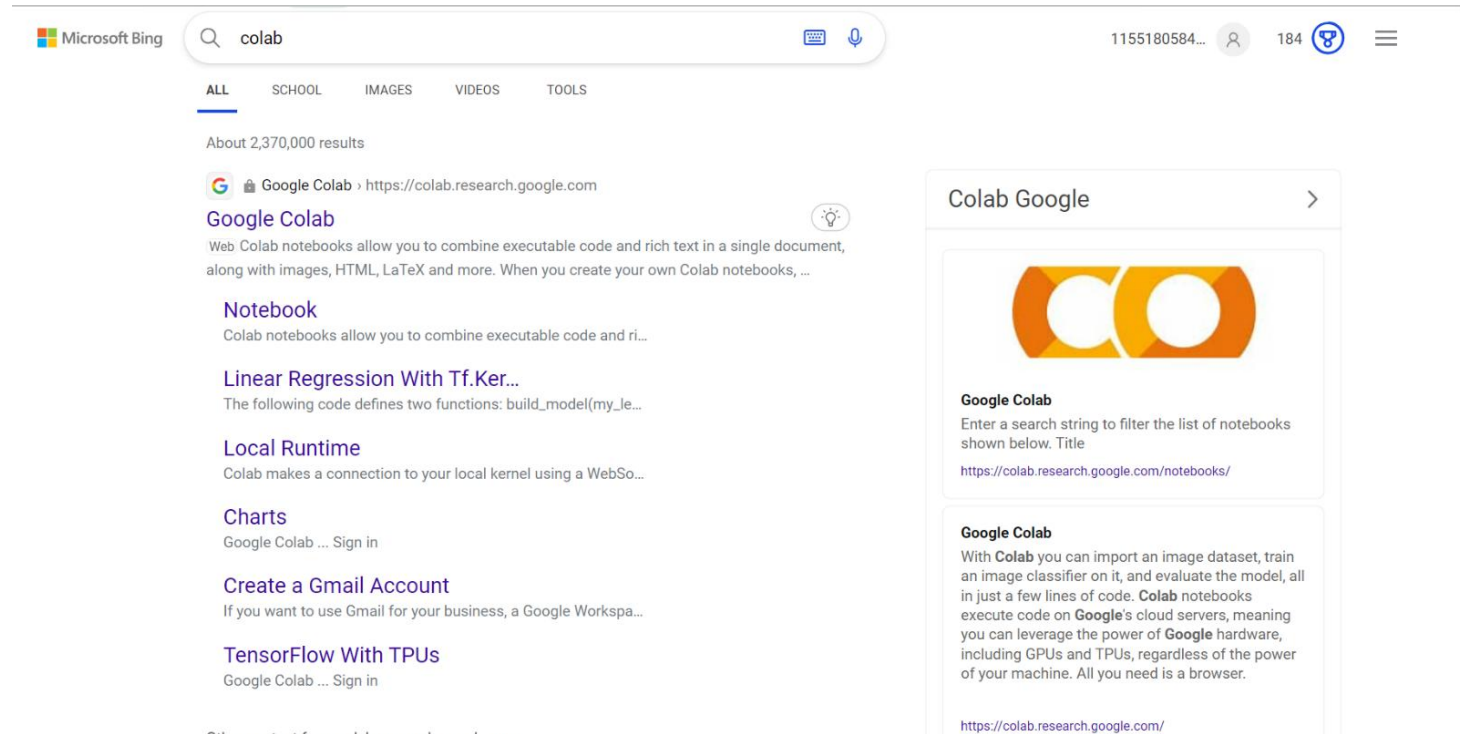
#	A	B	C	D	E	F	G	H
1	Original No.	Admission_Channel	Secondary_School	Exam	Math_Background	MATH1510	ENG1130	Level
2	17	JUPAS	TSUEN WAN PUB HO CHUEN YIU MEMORIAL COLL	HKDSE	HKDSE 5	B+	B+	B+
3	20	JUPAS	CHRISTIAN ALLIANCE S C CHAN MEM COLL	HKDSE	HKDSE 5+	A-	B	B
4	25	JUPAS	MUNSANG COLLEGE (HONG KONG ISLAND)	HKDSE	HKDSE 4	B	A-	A
5	29	JUPAS	PO LEUNG KUK LAWS FOUNDATION COLLEGE	HKDSE	HKDSE 4	B+	B-	B-
6	36	JUPAS	STFA LEUNG KAU KUI COLLEGE	HKDSE	HKDSE 5	B-	B+	B+
7	37	JUPAS	METHODIST CHURCH HK WESLEY COLLEGE	HKDSE	HKDSE 5	D	D+	D
8	50	JUPAS	Others	HKDSE	HKDSE 5	B-	C+	C
9	55	JUPAS	KWOK TAK SENG CATHOLIC SECONDARY SCHOOL	HKDSE	HKDSE 5	B	B	B
10	62	JUPAS	YWCA HIOE TJO YOENG COLLEGE	HKDSE	HKDSE 4	B	B+	B+
11	82	JUPAS	CCC CHUEN YUEN COLLEGE	HKDSE	HKDSE 5	B	F	F
12	88	JUPAS	HKMA K S LO COLLEGE	HKDSE	HKDSE 4	C	C+	C
13	92	JUPAS	LIONS COLLEGE	HKDSE	HKDSE 4	B	C-	C
14	94	JUPAS	YCH LIM POR YEN SECONDARY SCHOOL	HKDSE	HKDSE 5**	B+	A-	A
15	98	JUPAS	HK SKH BISHOP HALL SECONDARY SCHOOL	HKDSE	HKDSE 5**	B+	B+	B+
16	101	JUPAS	CHRISTIAN ALLIANCE S W CHAN MEM COLL	HKDSE	HKDSE 5	B-	B-	B-
17	109	JUPAS	VALTORTA COLLEGE	HKDSE	HKDSE 5**	A-	A	A
18	119	JUPAS	AOG HEBRON SECONDARY SCHOOL	HKDSE	HKDSE 5	B-	A-	A
19	129	JUPAS	PUI CHING MIDDLE SCHOOL	HKDSE	HKDSE 5**	A-	A	A
20	130	JUPAS	TWGH LI KA SHING COLLEGE	HKDSE	HKDSE 5	C+	C-	C
21	142	JUPAS	TANG KING PO SCHOOL	HKDSE	HKDSE 4	B+	B-	B-
22	144	JUPAS	PLK MRS MA KAM MING CHEUNG FOOK SIEN COL	HKDSE	HKDSE 4	B+	B-	B-
23	147	JUPAS	STFA YUNG YAU COLLEGE	HKDSE	HKDSE 5	B-	C	C
24	168	JUPAS	FUKIEN SECONDARY SCHOOL	HKDSE	HKDSE 5+	A-	B+	B+
25	172	JUPAS	SKH LAM WOO MEMORIAL SECONDARY SCHOOL	HKDSE	HKDSE 4	B	C+	C
26	177	JUPAS	CHAN SUI KI (LA SALLE) COLLEGE	HKDSE	HKDSE 5	A-	B-	B-
27	182	JUPAS	TSUEN WAN GOVERNMENT SECONDARY SCHOOL	HKDSE	HKDSE 5+	B-	D	D
28	183	JUPAS	HONG KONG CHINESE WOMENS CLUB COLLEGE	HKDSE	HKDSE 5	B-	F	F
29	184	JUPAS	TACK CHING GIRLS SECONDARY SCHOOL	HKDSE	HKDSE 4	B	B-	B-
30	187	JUPAS	CHINESE FOUNDATION SECONDARY SCHOOL	HKDSE	HKDSE 4	A-	B	B
31	189	JUPAS	MUNSANG COLLEGE (HONG KONG ISLAND)	HKDSE	HKDSE 4	B	A-	A
32	202	JUPAS	SHAU KEI WAN GOVERNMENT SECONDARY SCHOOL	HKDSE	HKDSE 5+	B	C-	C
33	208	JUPAS	YWCA HIOE TJO YOENG COLLEGE	HKDSE	HKDSE 4	B-	F	F
34	213	JUPAS	SING YIN SECONDARY SCHOOL	HKDSE	HKDSE 5	A-	A-	A
35	219	JUPAS	SALEM-IMMANUEL LUTHERAN COLLEGE	HKDSE	HKDSE 5+	B+	B	B
36	224	JUPAS	ELEGANTIA COLLEGE (SPONSORED BY ED CONV)	HKDSE	HKDSE 5+	B-	F	F
37	225	JUPAS	PLK LEE SHING PIK COLLEGE	HKDSE	HKDSE 4	D	C-	C
38	229	JUPAS	Others	HKDSE	HKDSE 5+	A-	A	A
39	232	JUPAS	STFA YUNG YAU COLLEGE	HKDSE	HKDSE 5	D	D+	D
40	240	JUPAS	SKH BISHOP MOK SAU TSENG SEC SCH	HKDSE	HKDSE 5+	A-	A-	A
41	241	JUPAS	PUI CHING MIDDLE SCHOOL	HKDSE	HKDSE 5+	A	A-	A

#	A	B	C	D	E	F	G	H
406	1079	JUPAS	Others	HKDSE	HKDSE 5	C	A-	A
407	1080	JUPAS	TAK OI SECONDARY SCHOOL	HKDSE	HKDSE 5	C	B+	B+
408	1081	JUPAS	SHAU KEI WAN GOVERNMENT SECONDARY SCHOOL	HKDSE	HKDSE 5	D+	B	B
409	1082	JUPAS	TSEUNG KWAN O GOVERNMENT SEC SCHOOL	HKDSE	HKDSE 5	A	B+	B+
410	1083	JUPAS	LA SALLE COLLEGE	HKDSE	HKDSE 4	D	B-	B-
411	1086	JUPAS	KIANGSU-CHEKIANG COLLEGE (KWAI CHUNG)	HKDSE	HKDSE 5	C	B-	B-
412	1088	JUPAS	HEUNG TO MIDDLE SCHOOL	HKDSE	HKDSE 4	C	B-	B-
413	1090	JUPAS	PENTECOSTAL LAM HON KWONG SCHOOL	HKDSE	HKDSE 5+	B-	A-	A
414	10	Mainland	Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)>140	A	A	A
415	120	Mainland	Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)120-129	A	A	A
416	131	Mainland	Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A	A	A
417	132	Mainland	Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A	A	A
418	201	Mainland	Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)120-129	A	A	A
419	207	Mainland	Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A-	B-	B-
420	212	Mainland	Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)120-129	A	A-	A
421	214	Mainland	Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A	B	B
422	216	Mainland	Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)110-119	A	B-	B-
423	226	Mainland	Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)120-129	A	B+	B+
424	227	Mainland	Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)120-129	A-	B+	B+
425	234	Mainland	Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)120-129	A	A-	A
426	260	Mainland	Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A	B	B
427	261	Mainland	Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A	B	B
428	265	Mainland	Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A	B-	B-
429	269	Mainland	Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A	A-	A
430	278	Mainland	Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A	B+	B+
431	281	Mainland	Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)120-129	A-	B+	B+
432	287	Mainland	Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)>140	A	A-	A
433	303	Mainland	Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A-	B+	B+
434	304	Mainland	Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A-	B+	B+
435	305	Mainland	Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A	A-	A
436	306	Mainland	Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)>140	A-	A-	A
437	307	Mainland	Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)120-129	A	B-	B-
438	325	Mainland	Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)>140	A-	B-	B-
439	327	Mainland	Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A	A	A
440	329	Mainland	Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A	B-	B-
441	330	Mainland	Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	B+	A	A
442	331	Mainland	Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)>140	A	B+	B+
443	351	Mainland	Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A-	C	C
444	355	Mainland	Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A-	A	A
445	357	Mainland	Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)130-139	A	A	A
446	365	Mainland	Others	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)>140	A-	B+	B+

Manipulated data

Visit and log-in Colab

- Search “colab” in your browser (<https://drive.google.com/>).
- Click the first search result.
- Sign in with your Google account if there is one.



Log-in with your google account

Programming skills

■ Engineering students



■ Non-engineering students





4 methods

Mechanism and implementations on
learner level prediction

Method overview

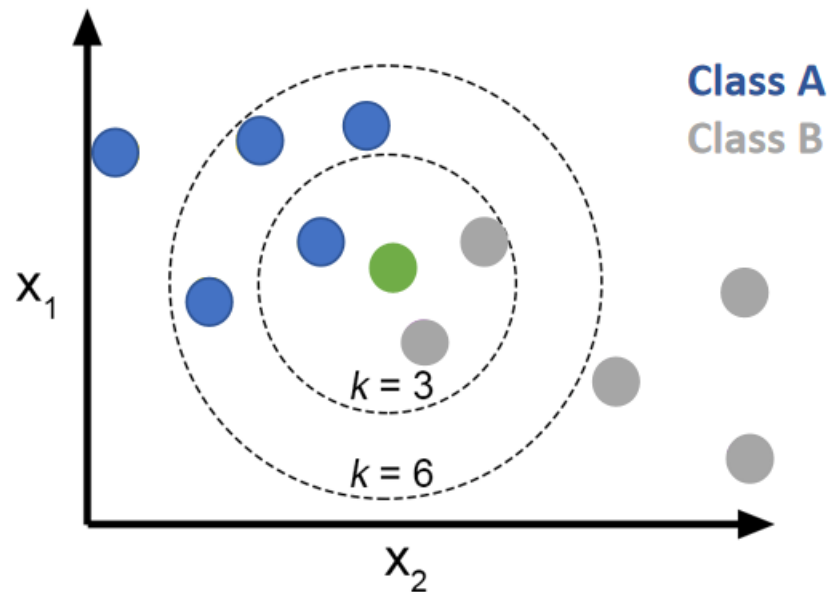
- To find the most suitable algorithm, we investigate four popular machine learning algorithms and make a **comparison** among them.



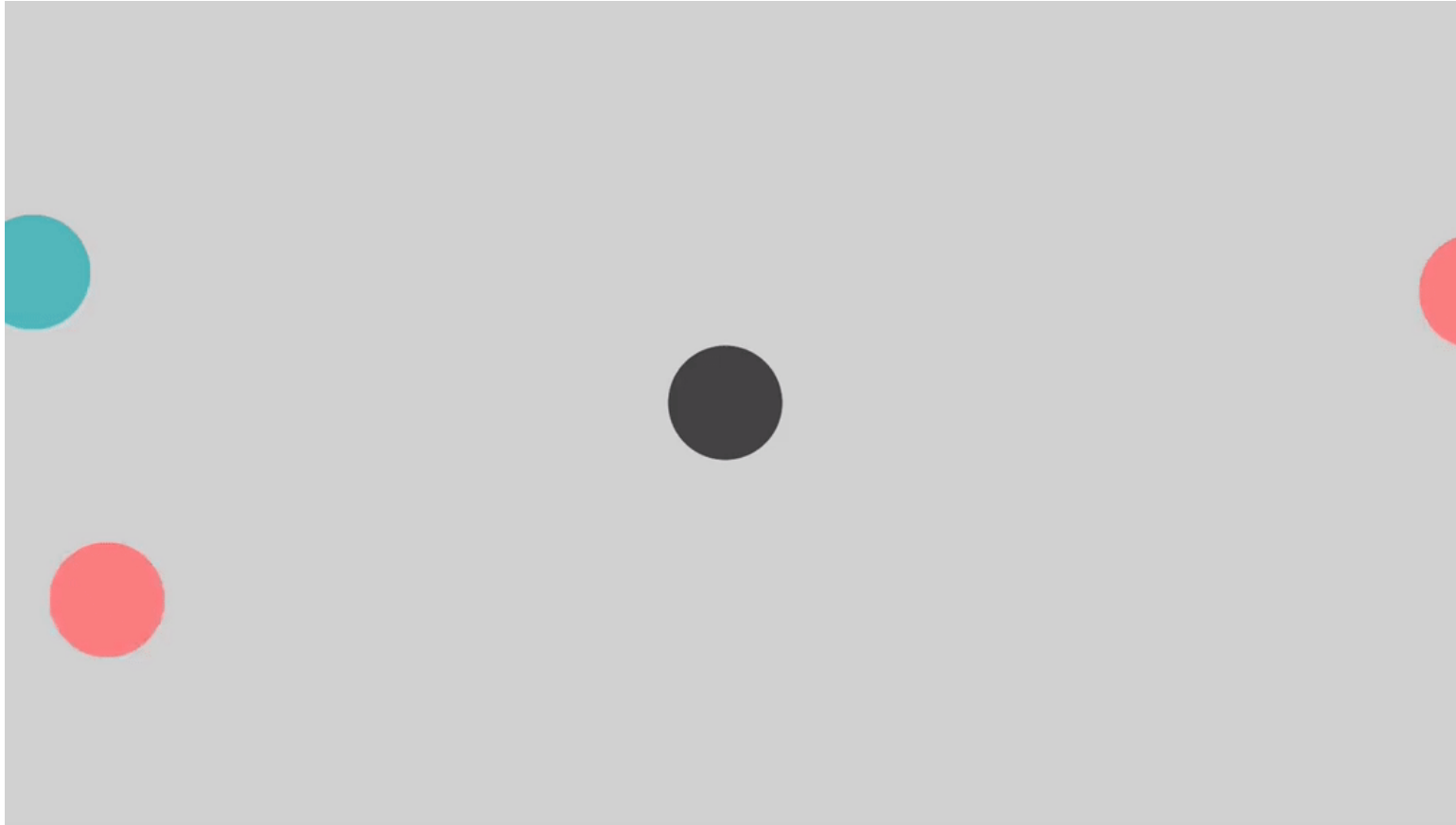
- K Nearest Neighbor (KNN)
- Back Propagate Neural Network (BPNN)
- Support Vector Classification (SVC)
- Naïve Bayes (NB)

K-nearest neighbors method

- One of the simplest **machine learning algorithms**
- Used for **classification**
- Based on the **similarity (distance)**



K-nearest neighbors method



(kNN in a classification with $k=3$.) An object is classified by a plurality vote of its 3 neighbors, with the object being assigned to the class most common among its 3 nearest neighbors.

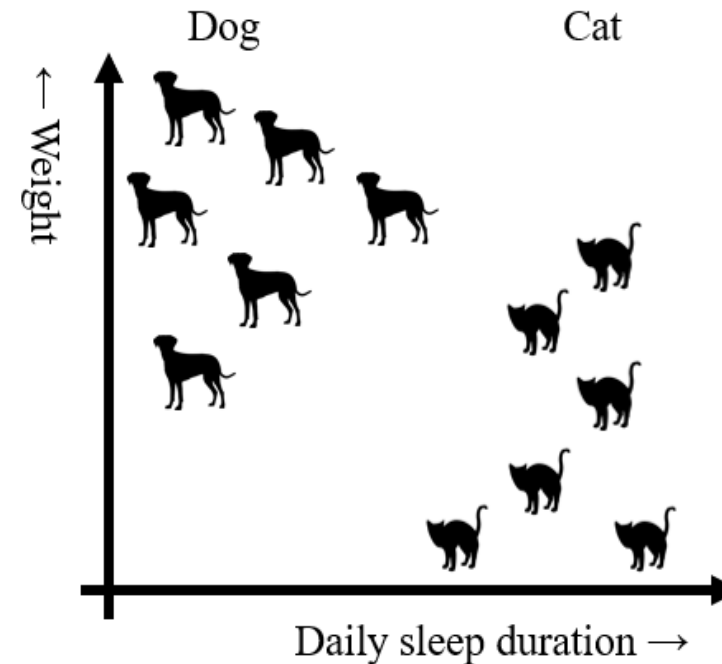
An example

■ Step 1: Choose **factors**

Weight, daily sleep duration.....

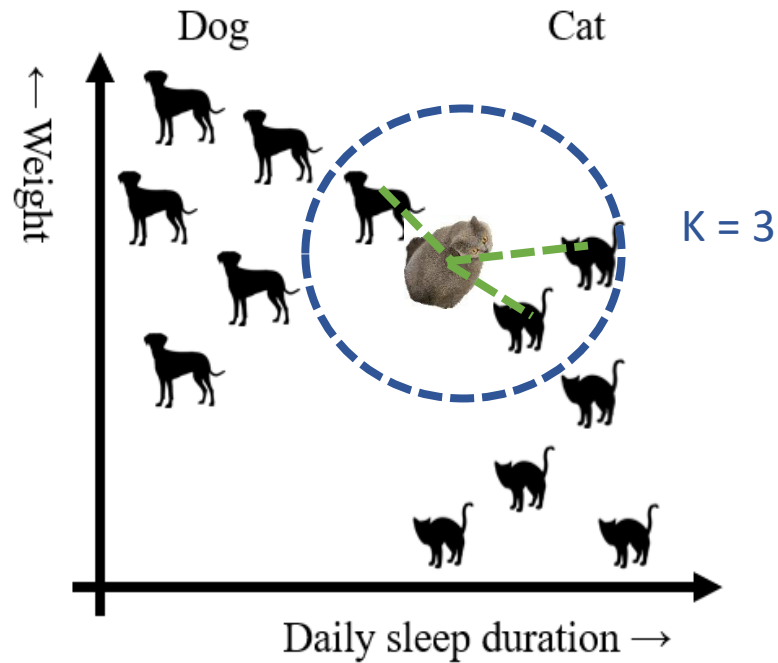
■ Step 2: Prepare **training data**

Number	Type	Weight (kg)	Daily Sleep Duration (hour)
1	dog	17	8
2	cat	6	16
3	dog	10	10
.....			

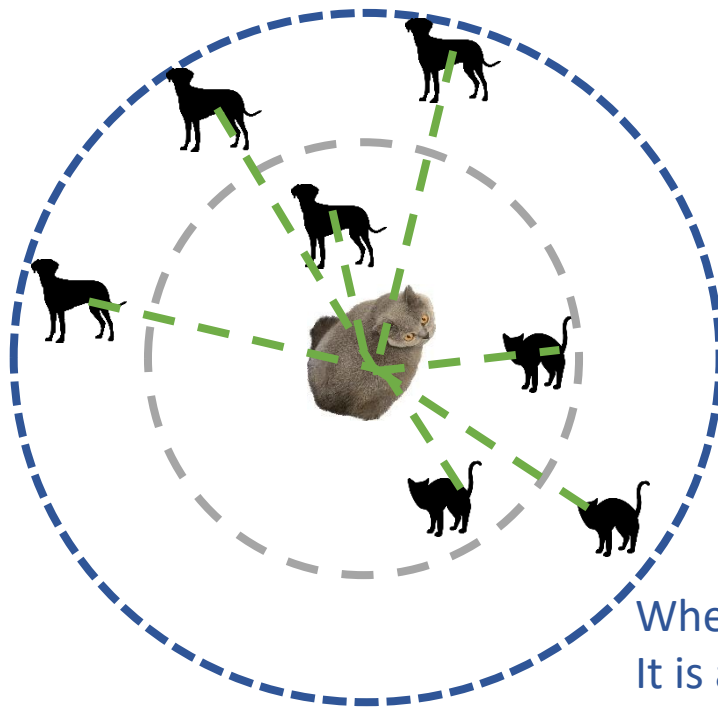


An example

- Step 3: Input **testing data**
- Step 4: Choose the value of **K** and make predictions



The value of K



When $K = 7$

It is assigned to the class of dogs

When $K = 3$

It is assigned to the class of cats

- A **user-defined** constant
- Represents the **number of the nearest neighbors**
- Affects the **accuracy** of prediction

KNN implementation

Now, let's try to use KNN to predict students' grade in ENGG2020.

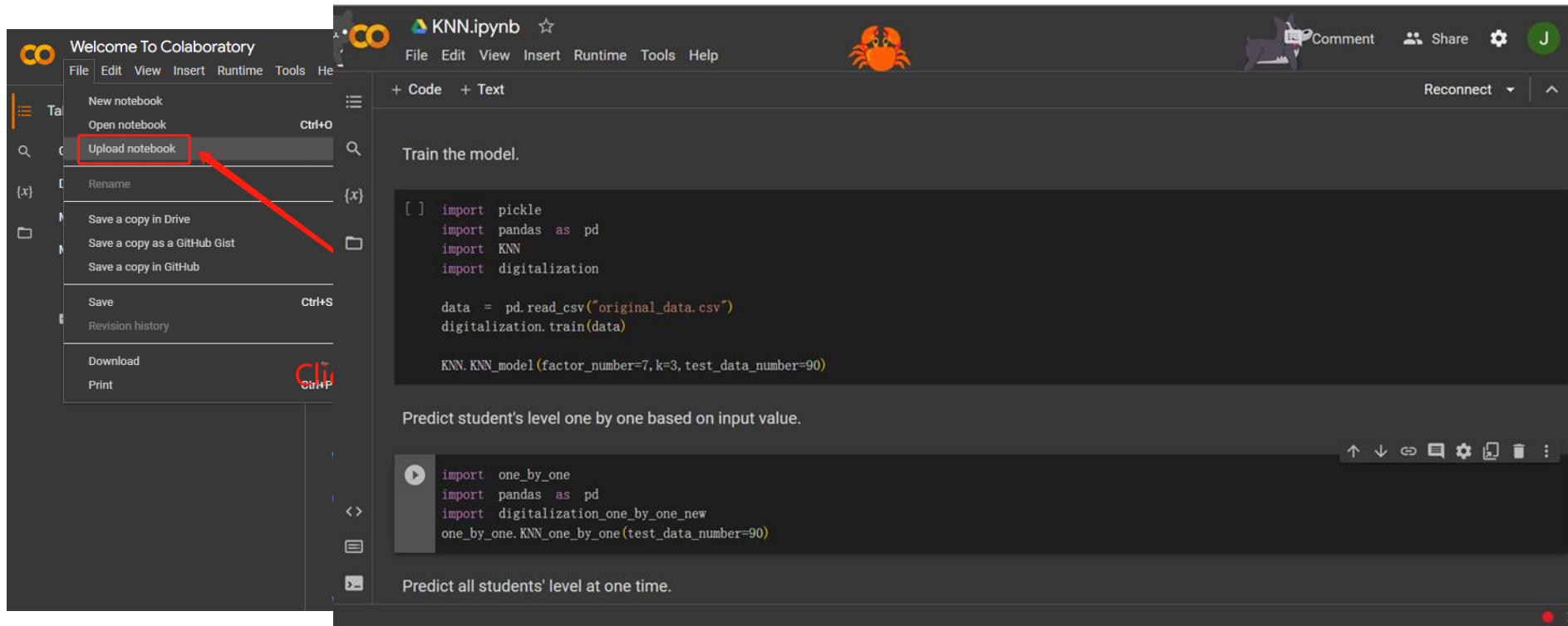
- Step 1: Choose **factors**
- Step 2: Prepare **training data**
- Step 3: Input **test data**
- Step 4: Choose different values of **K** and make predictions

Now it is your turn!



Preparations

- Upload the “KNN.ipynb” from the file “Main_code”.



类型	大小
PYNB 文件	15 KB
PYNB 文件	137 KB
PYNB 文件	84 KB
PYNB 文件	82 KB

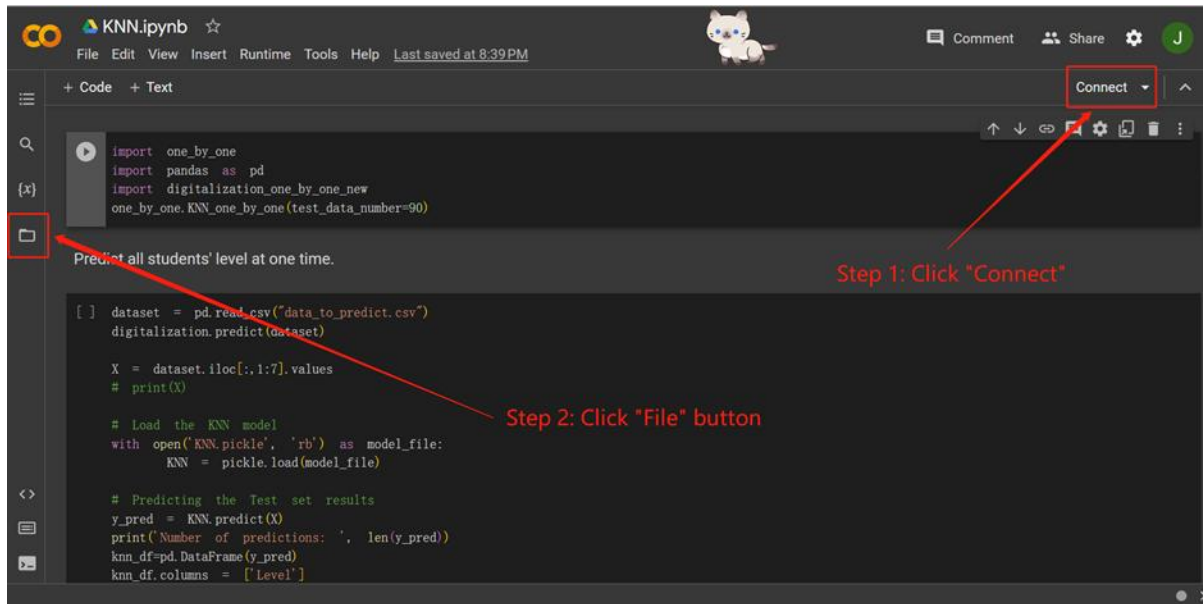
Upload the program

Upload complete

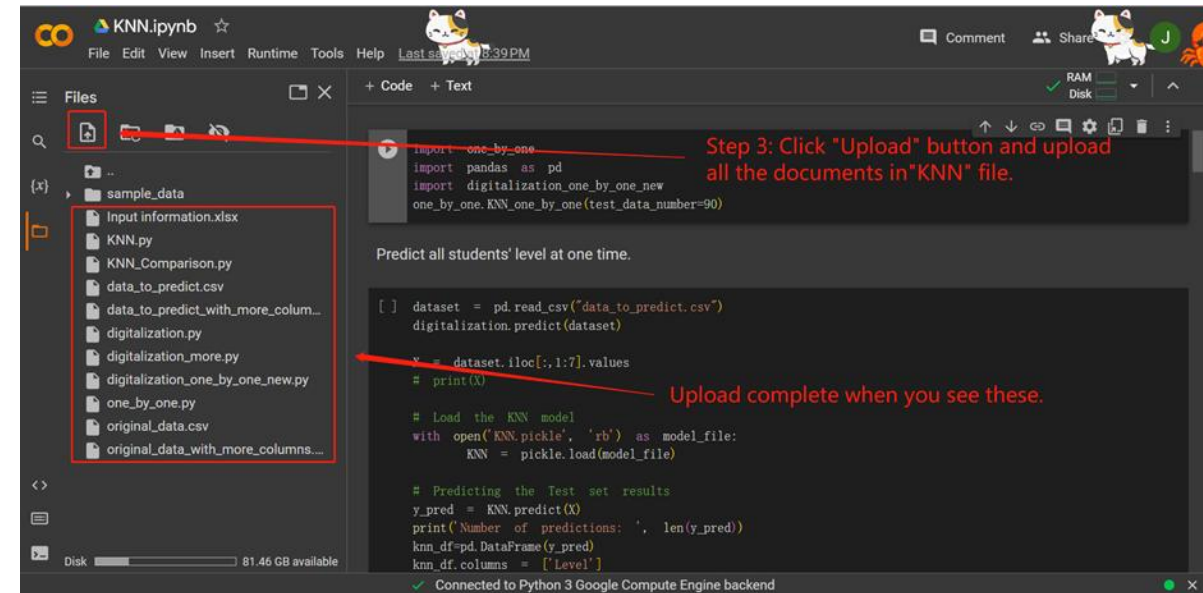
Select “KNN.ipynb”

Preparations

- **Connect** the program to online processor.
- **Upload** all the documents from “KNN” file to the notebook.



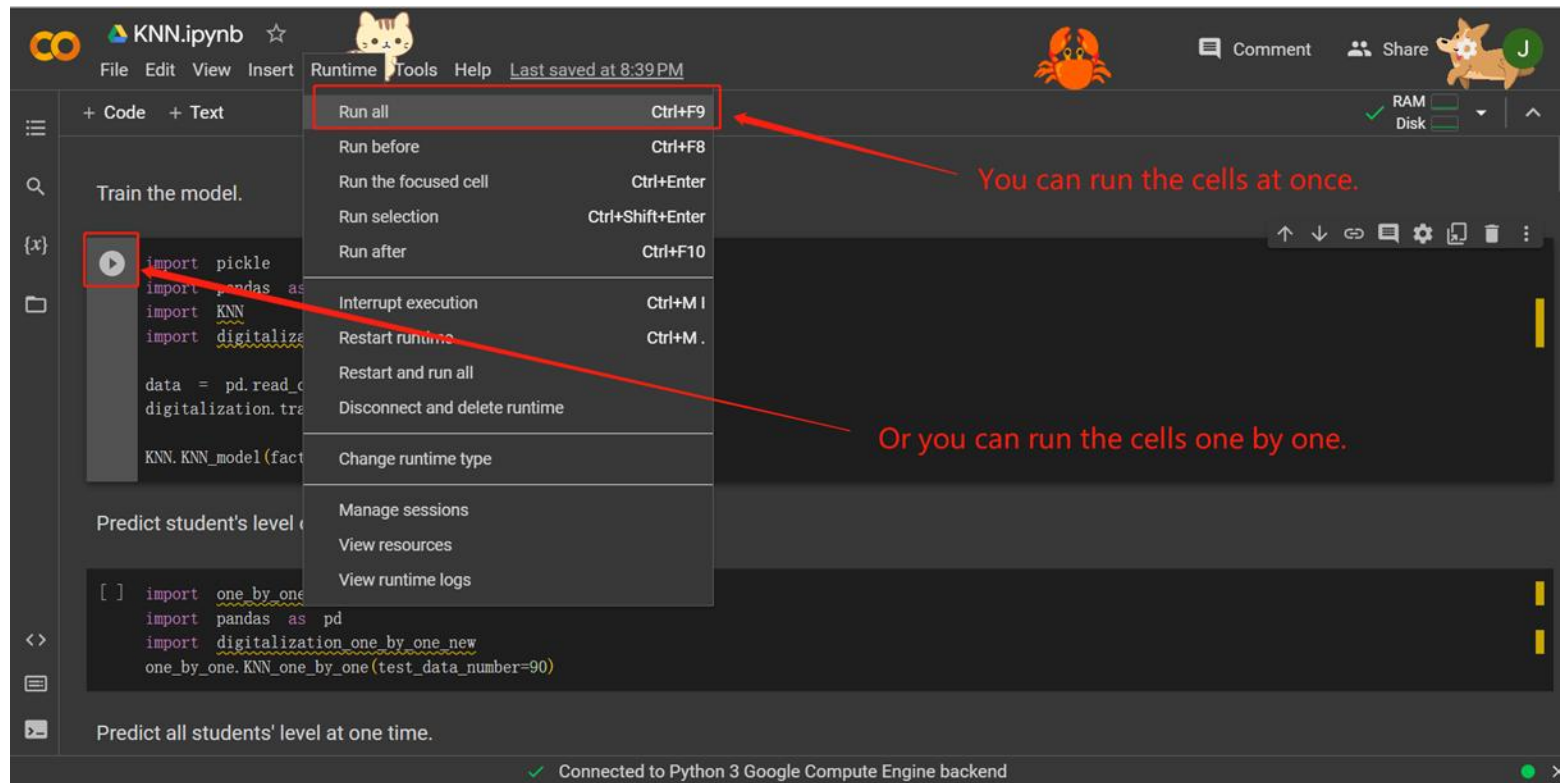
Connect the notebook



Upload our files

Run the program

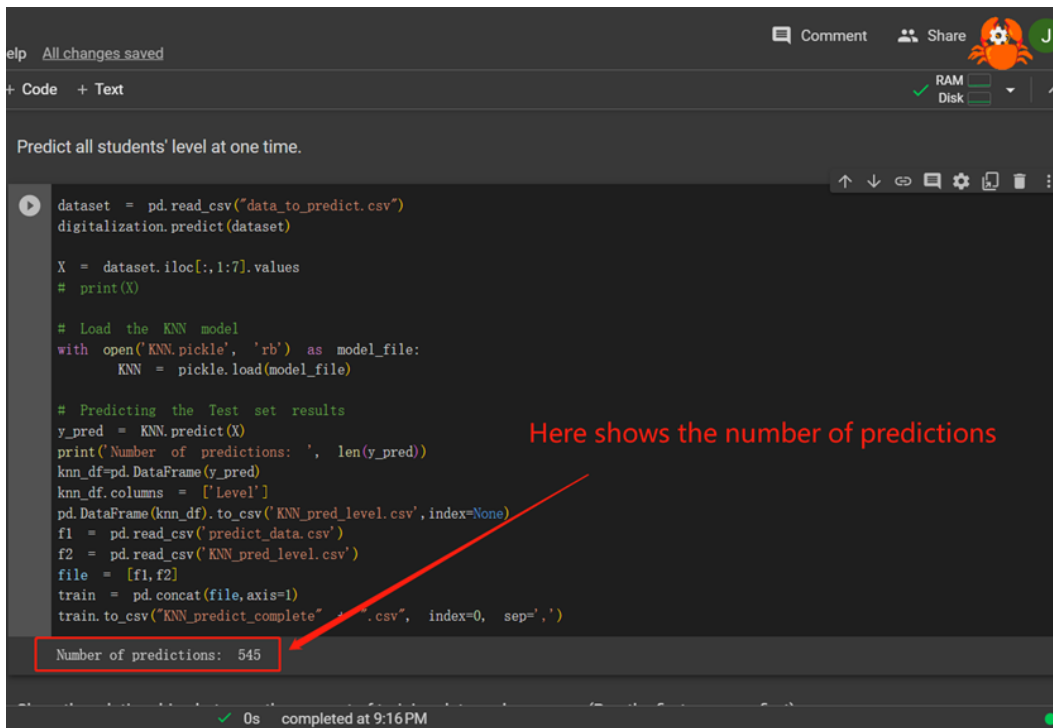
- Click “Run” button and **run the cells one by one** or you can run them all in one click. (One by one is recommended)



Run the KNN program

Check the results

- Click “Run” button and **run the cells one by one** or you can run them all in one click. (One by one is recommended)



```
dataset = pd.read_csv("data_to_predict.csv")
digitalization.predict(dataset)

X = dataset.iloc[:, 1:7].values
# print(X)

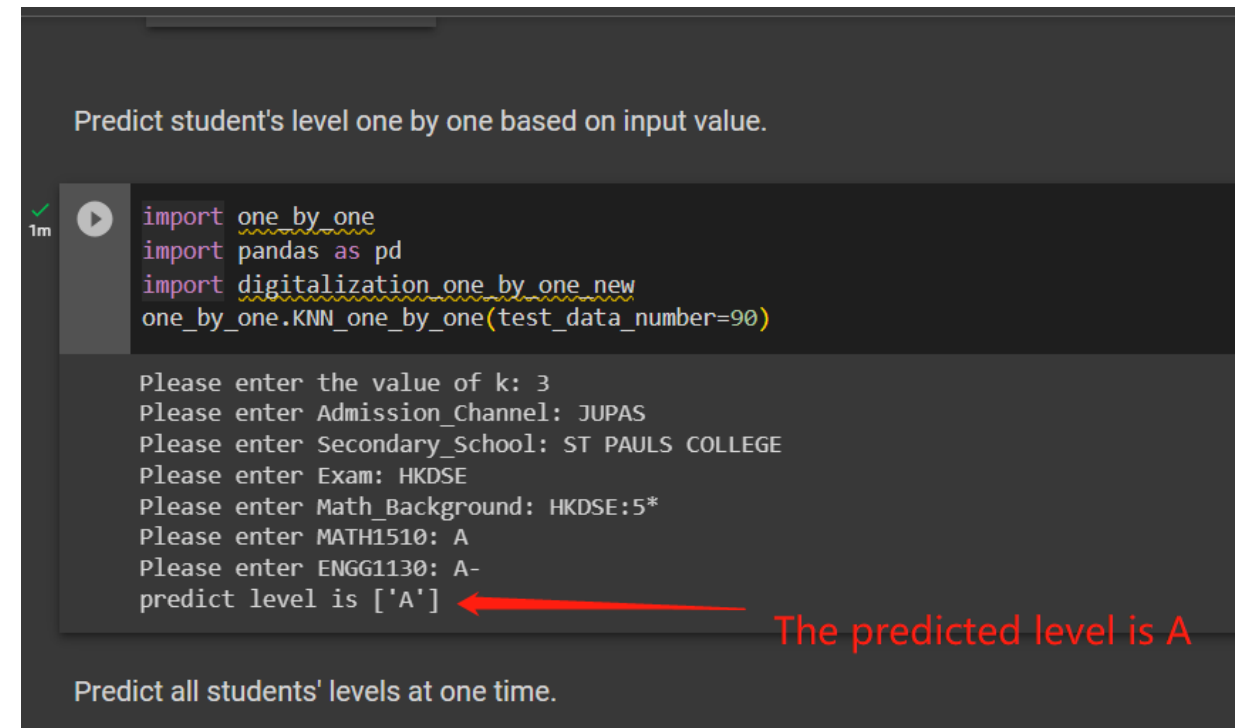
# Load the KNN model
with open('KNN.pickle', 'rb') as model_file:
    KNN = pickle.load(model_file)

# Predicting the Test set results
y_pred = KNN.predict(X)
print('Number of predictions: ', len(y_pred))
knn_df=pd.DataFrame(y_pred)
knn_df.columns = ['Level']
pd.DataFrame(knn_df).to_csv('KNN_pred_level.csv', index=None)
f1 = pd.read_csv('predict_data.csv')
f2 = pd.read_csv('KNN_pred_level.csv')
file = [f1, f2]
train = pd.concat(file,axis=1)
train.to_csv("KNN_predict_complete" + ".csv", index=0, sep=',')
```

Number of predictions: 545

0s completed at 9:16 PM

Output of first cell



```
import one_by_one
import pandas as pd
import digitalization one_by_one new
one_by_one.KNN_one_by_one(test_data_number=90)
```

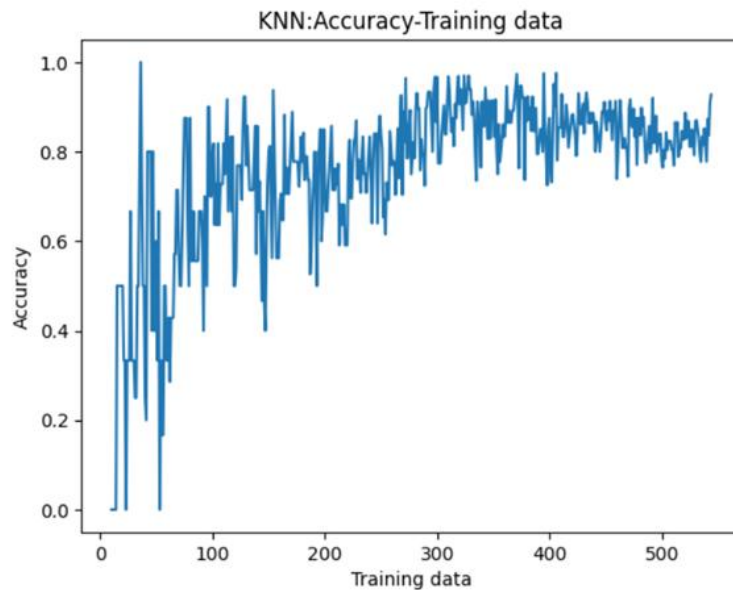
Please enter the value of k: 3
Please enter Admission_Channel: JUPAS
Please enter Secondary_School: ST PAULS COLLEGE
Please enter Exam: HKDSE
Please enter Math_Background: HKDSE:5*
Please enter MATH1510: A
Please enter ENGG1130: A-
predict level is ['A']

Predict all students' levels at one time.

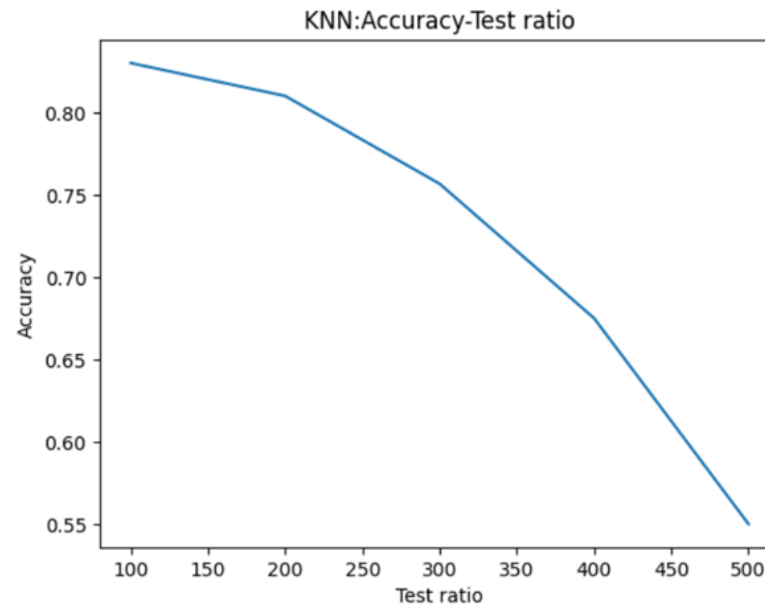
Output of second cell

Check the results

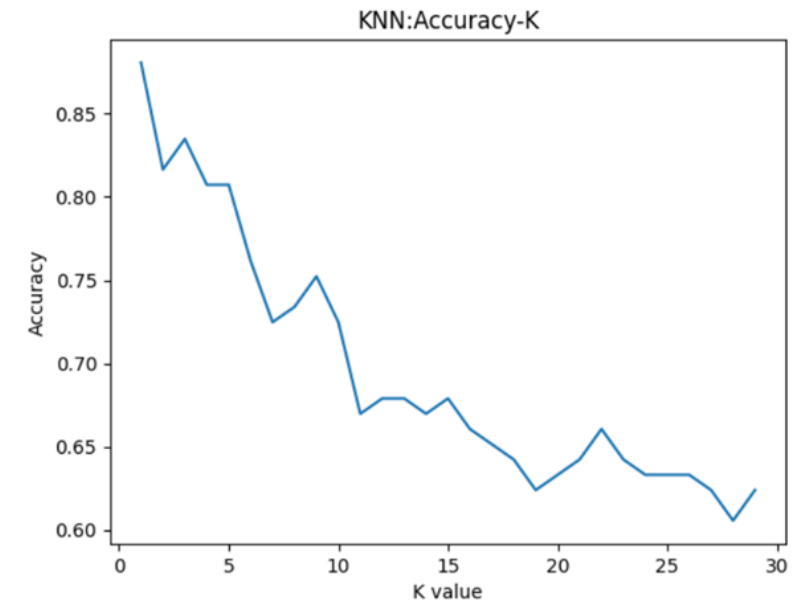
■ Click “Run” button and [run the cells one by one](#) or you can run them all in one click.



Output of third cell



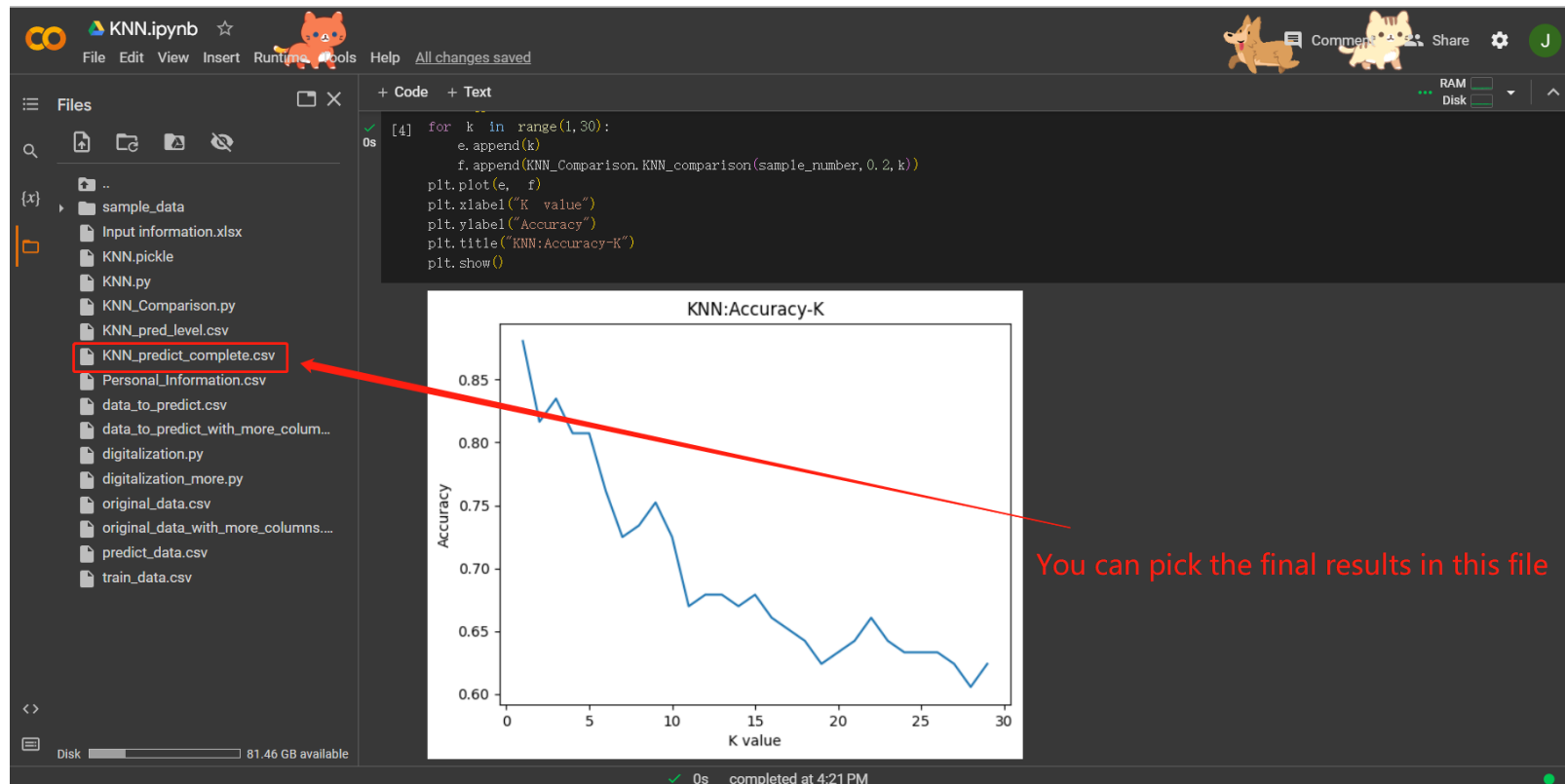
Output of fourth cell



Output of fifth cell

Check the outputs

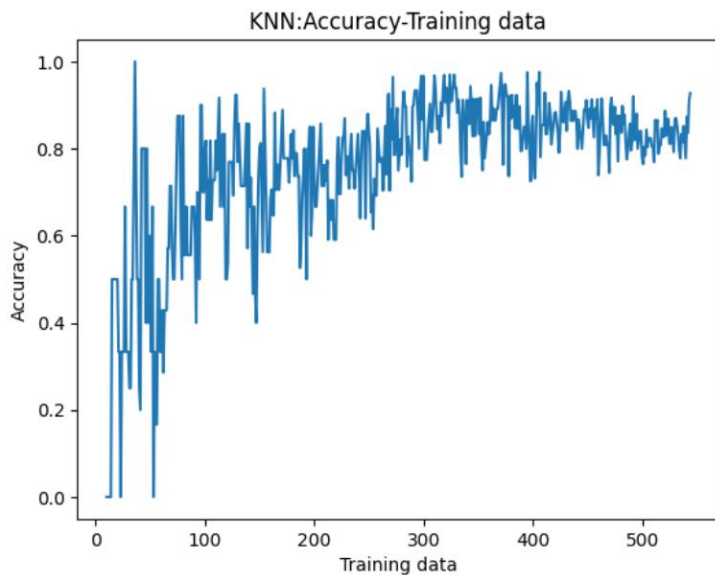
- The final output can be found on the left, named “KNN_predict_complete.csv”, after opening the file we can find that the predict level is on the rightest column.



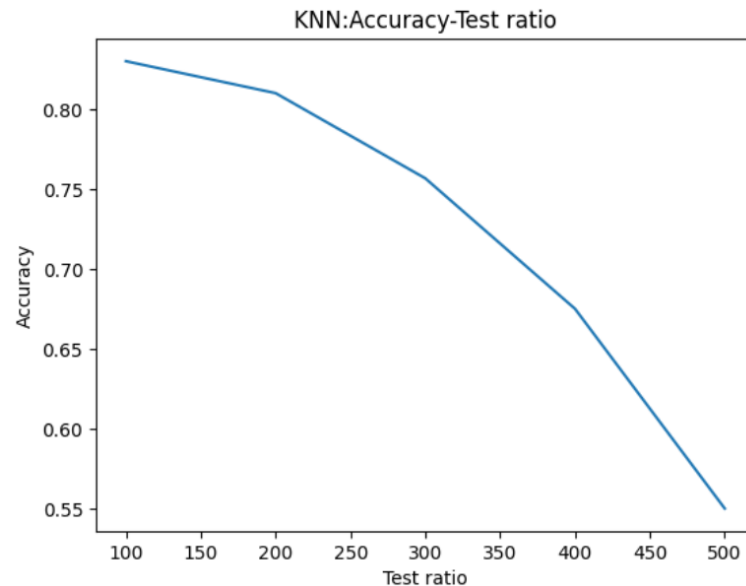
Outcome file

Select the optimal parameters

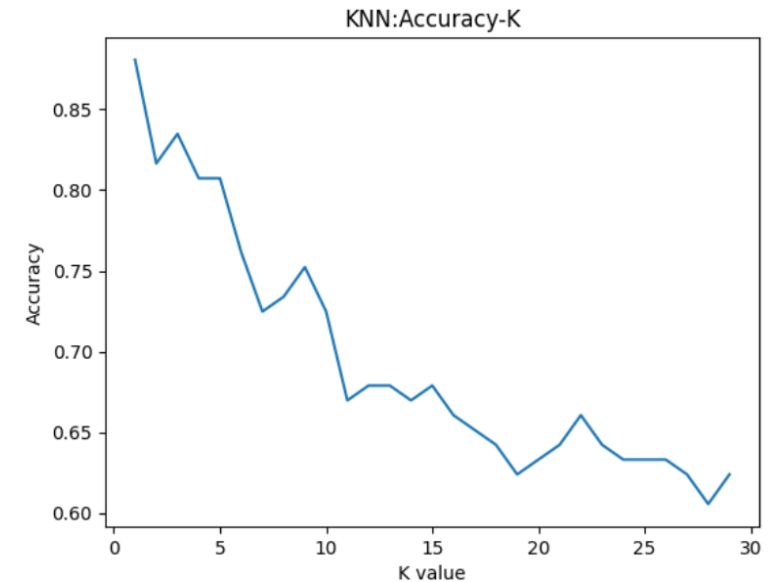
- By comparing the prediction data and the target, we can calculate the **accuracy** of the trained model.
- For each algorithm, we can also find out in which set of **parameters** the model can reach the highest accuracy.



Generally, more training data represents higher accuracy



Generally, less test data represents higher accuracy



K=3 or K=1 provides a desired model

Select the optimal parameters

- After comparison, you can adjust the K value and ratio of test data like in the picture below.

```
Train the model.
```

```
[1] import pickle
import pandas as pd
import KNN
import digitalization

data = pd.read_csv("original_data.csv")
digitalization.train(data)

KNN.KNN_model(factor_number=7, k=3, test_data_number=90)
```

Accuracy of the KNN Clasification is: 0.8444444444444444

```
Predict student's level one by one based on input value.
```

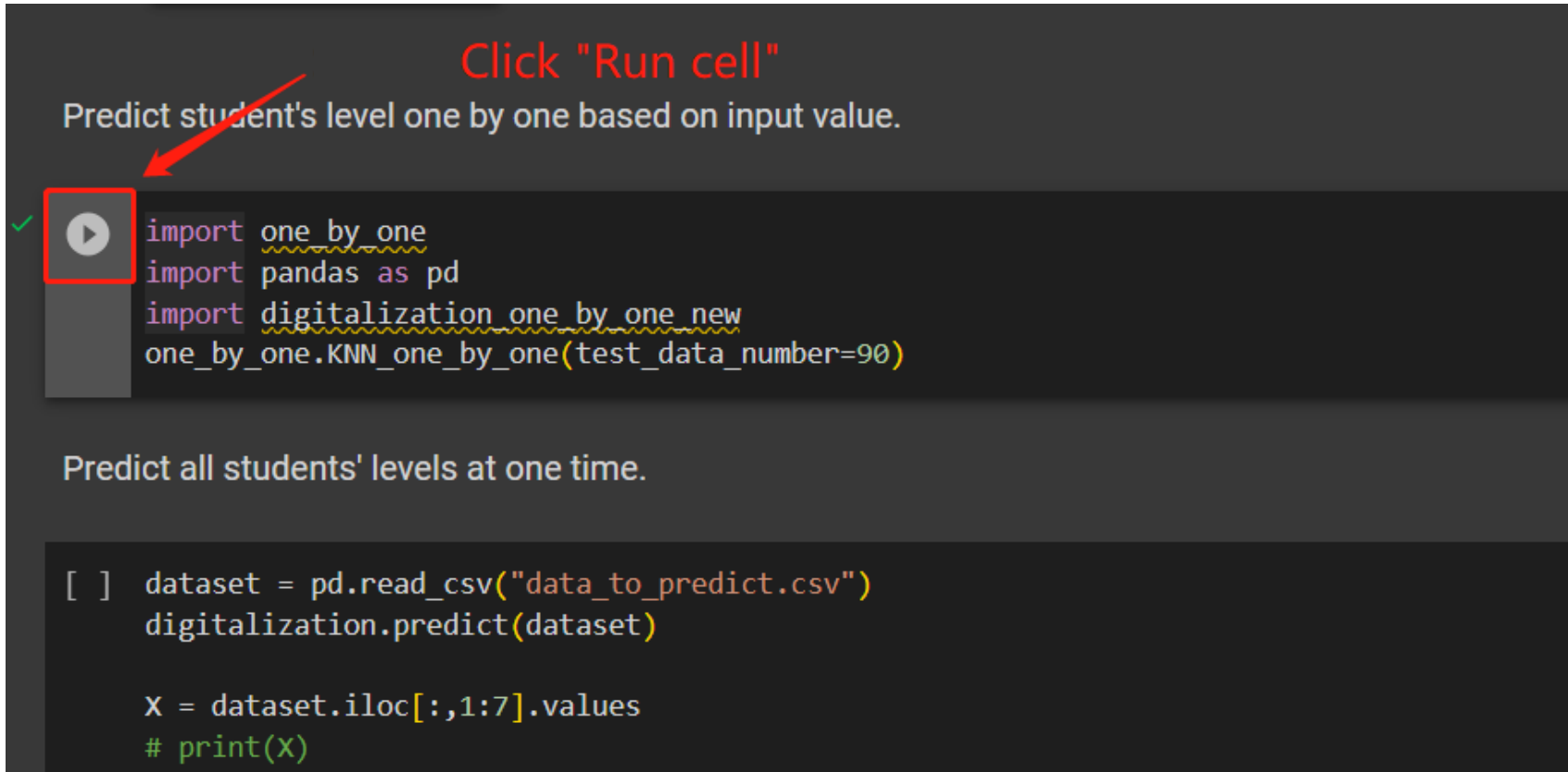
```
import one_by_one
import pandas as pd
import digitalization_one_by_one_new
one_by_one.KNN_one_by_one(test_data_number=90)
```

You can adjust k and ratio of test data here.

You can adjust the parameters here based on the comparison

Predict one by one

- Run the following code



The screenshot shows a Jupyter Notebook interface with a dark background. At the top, there is a text prompt: "Predict student's level one by one based on input value." Below this, there is a code cell. To the left of the code cell is a play button icon (a circle with a right-pointing triangle) which is highlighted with a red square. A red arrow points from the text "Click 'Run cell'" to this play button. The code in the cell is as follows:

```
import one_by_one
import pandas as pd
import digitalization_one_by_one_new
one_by_one.KNN_one_by_one(test_data_number=90)
```

Below the code cell, there is another text prompt: "Predict all students' levels at one time." Underneath this, there is another code cell containing the following code:

```
[ ] dataset = pd.read_csv("data_to_predict.csv")
    digitalization.predict(dataset)

X = dataset.iloc[:,1:7].values
# print(X)
```

Predict one by one

- Select a value of K

Predict student's level one by one based on input value.

```
import one_by_one
import pandas as pd
import digitalization_one_by_one_new
one_by_one.KNN_one_by_one(test_data_number=90)
```

Please enter the value of k:

Predict all students' levels at one time.

Choose a value of k

```
[ ] dataset = pd.read_csv("data_to_predict.csv")
    digitalization.predict(dataset)
```

Predict one by one

- Enter **Admission_Channel**: JUPAS/Mainland Scholarship(Tuition)/Mainland
- And then click “Enter”

Predict student's level one by one based on input value.

```
import one_by_one
import pandas as pd
import digitalization_one_by_one_new
one_by_one.KNN_one_by_one(test_data_number=90)
```

Please enter the value of k: 3

Please enter Admission_Channel:

Predict all students' levels at one time.

Choose an Admission_Channel

```
[ ] dataset = pd.read_csv("data_to_predict.csv")
    digitalization.predict(dataset)

X = dataset.iloc[:,1:7].values
```

Predict one by one

- Enter **Secondary_School**: For example: LA SALLE COLLEGE/ ST PAULS COLLEGE/SHA TIN METHODIST COLLEGE/Others
- And then click “Enter”

Predict student's level one by one based on input value.

```
import one_by_one
import pandas as pd
import digitalization_one_by_one_new
one_by_one.KNN_one_by_one(test_data_number=90)
```

Please enter the value of k: 3
Please enter Admission_Channel: JUPAS
Please enter Secondary_School:

Choose a Secondary_School

Predict all students' levels at one time.

```
[ ] dataset = pd.read_csv("data_to_predict.csv")
digitalization_predict(dataset)
```

Predict one by one

- Enter **Exam type:** HKDSE/Mainland Gaokao (JEE)/IB/GCE-AL...
- And then click “Enter”

Predict student's level one by one based on input value.

```
import one_by_one
import pandas as pd
import digitalization_one_by_one_new
one_by_one.KNN_one_by_one(test_data_number=90)
```

Please enter the value of k: 3
Please enter Admission_Channel: JUPAS
Please enter Secondary School: ST PAULS COLLEGE
Please enter Exam:

Predict all students' levels at one time. Choose an exam

```
[ ] dataset = pd.read_csv("data_to_predict.csv")
    digitalization.predict(dataset)
```

Predict one by one

- **Math_Background:** HKDSE:5**/HKDSE:5/Mainland Gaokao (JEE):>140/ Mainland Gaokao (JEE):130-139...
- And then click “Enter”

Predict student's level one by one based on input value.

```
import one_by_one
import pandas as pd
import digitalization_one_by_one_new
one_by_one.KNN_one_by_one(test_data_number=90)
```

Please enter the value of k: 3
Please enter Admission_Channel: JUPAS
Please enter Secondary_School: ST PAULS COLLEGE
Please enter Exam: HKDSE

Please enter Math_Background:

Choose a Math_Background

Predict all students' levels at one time.

Predict one by one

- **MATH1510:** A/A-/B+/B/B-/C+/C/C-/D+/D
- And then click “Enter”

```
Predict student's level one by one based on input value.
```

```
import one_by_one
import pandas as pd
import digitalization_one_by_one_new
one_by_one.KNN_one_by_one(test_data_number=90)
```

```
Please enter the value of k: 3
Please enter Admission_Channel: JUPAS
Please enter Secondary_School: ST PAULS COLLEGE
Please enter Exam: HKDSE
Please enter Math_Background: HKDSE:5*
Please enter MATH1510: A
```

Choose a grade of MATH1510

```
[ ] dataset = pd.read_csv("data_to_predict.csv")
digitalization_predict(dataset)
```

Predict one by one

- **ENGG1130:** A/A-/B+/B/B-/C+/C/C-/D+/D/F
- And then click “Enter”

Predict student's level one by one based on input value.

```
import one_by_one
import pandas as pd
import digitalization_one_by_one_new
one_by_one.KNN_one_by_one(test_data_number=90)
```

Please enter the value of k: 3
Please enter Admission_Channel: JUPAS
Please enter Secondary_School: ST PAULS COLLEGE
Please enter Exam: HKDSE
Please enter Math_Background: 5*
Please enter MATH1510: A-

Please enter ENGG1130:

Choose a grade of ENGG1130

Predict all students' levels at one time.

Predict one by one

- Then we can get the predicted level of the student.

```
Predict student's level one by one based on input value.
```

```
✓ 1m ▶ import one_by_one
import pandas as pd
import digitalization_one_by_one_new
one_by_one.KNN_one_by_one(test_data_number=90)
```

```
Please enter the value of k: 3
Please enter Admission_Channel: JUPAS
Please enter Secondary_School: ST PAULS COLLEGE
Please enter Exam: HKDSE
Please enter Math_Background: HKDSE:5*
Please enter MATH1510: A
Please enter ENGG1130: A-
predict level is ['A']
```

The predicted level is A

```
Predict all students' levels at one time.
```

Want to try your own data?

■ Follow the steps below:

1. First, open the “**Input information.xlsx**” from the “KNN” file we have already downloaded, fill in your personal information. Notice that in “Exam” and “Math_Background” column, only Gaokao and HKDSE are accepted.

	A	B	C	D	E	F	G
1	Student_ID	Admission_Channel	Secondary_School	Exam	Math_Background	MATH1510	ENGG1130
2	1	JUPAS	CCC HEEP WOH COLLEGE	Mainland Gaokao (JEE)	Mainland Gaokao (JEE):>140	A	A
3	2	Non-JUPAS (Local)	YUEN WAN PUB HO CHUEN YIU MEMORIAL CO	HKDSE	HKDSE:5*	B+	C
4							
5							
6							
7			SKH LEUNG KWAI YEE SECONDARY SCHOOL				
8			SKH BISHOP BAKER SECONDARY SCHOOL				
9			SKH BISHOP MOK SAU TSENG SEC SCH				
10			SKH TANG SHIU KIN SECONDARY SCHOOL				
11			SKH CHAN YOUNG SECONDARY SCHOOL				
12			ST STEPHEN'S GIRLS' COLLEGE				
13			ST BONAVENTURE COLLEGE & HIGH SCHOOL				
14			IMMACULATE HEART OF MARY COLLEGE				
15			OUR LADY OF THE ROSARY COLLEGE				
16			ST ROSE OF LIMA'S COLLEGE				
17			ST FRANCIS XAVIER'S COLLEGE				
18			ST JOSEPH'S COLLEGE				
19							
20							

Input your information

Want to try your own data?

2. **Upload** the “Input information.xlsx” to program like before and run the **fifth cell**.

3. Then pick up the “**Personal_KNN_predict_complete.csv**” file in the left to check the prediction of your grade!

```
import digitalization
df = pd.read_excel('Input information.xlsx')
df1 = df.drop(['School_Information'], axis=1)
pd.DataFrame(df1).to_csv('Personal_Information.csv', index=None)
dataset = pd.read_csv('Personal_Information.csv')
digitalization.predict(dataset)
student_number = int(input('Please input the number of students: '))
X = dataset.iloc[:student_number, 1:7].values
# print(X)

# Load the KNN model
with open('KNN.pickle', 'rb') as model_file:
    KNN = pickle.load(model_file)

# Predicting the Test set results
y_pred = KNN.predict(X)
print('Number of predictions: ', len(y_pred))
print('Predict Level: ', y_pred)
knn_df = pd.DataFrame(y_pred)
knn_df.columns = ['Level']
pd.DataFrame(knn_df).to_csv('KNN_pred_level.csv', index=None)
f1 = pd.read_csv('predict_data.csv')
f2 = pd.read_csv('KNN_pred_level.csv')
file = [f1, f2]
train = pd.concat(file, axis=1)
train.to_csv('Personal_KNN_predict_complete' + '.csv', index=0, sep=',')
```

Please input the number of students: 2
Number of predictions: 2
Predict Level: ['A' 'C']

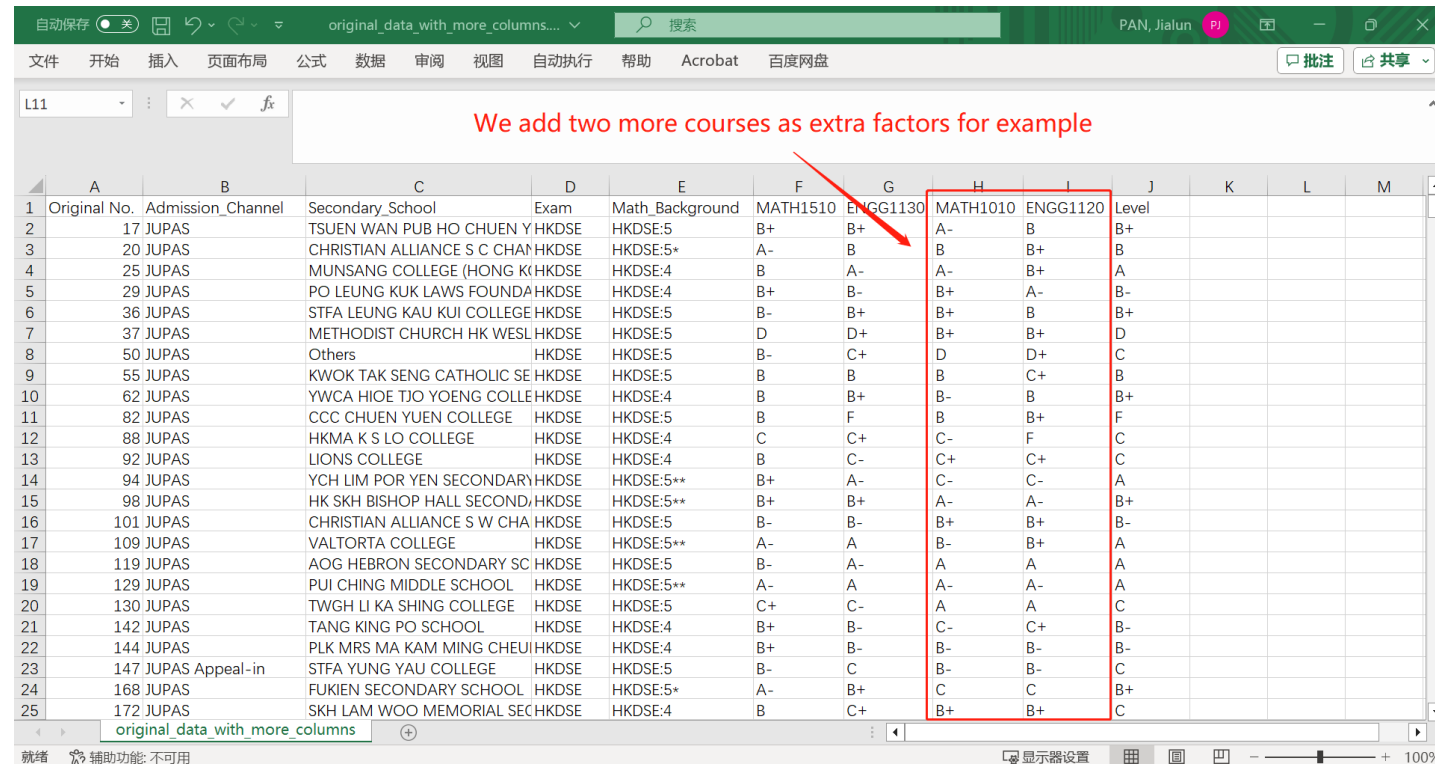
Run the seventh cell.

Two ways to see the predictions

Want to add extra factors?

■ Follow the steps below:

1. We can take a look at “original_data_with_more_columns.csv” in “KNN” file.



We add two more courses as extra factors for example

	A	B	C	D	E	F	G	H	I	J	K	L	M
	Original No.	Admission_Channel	Secondary_School	Exam	Math_Background	MATH1510	ENGG1130	MATH1010	ENGG1120	Level			
1	17	JUPAS	TSUEN WAN PUB HO CHUEN Y HKDSE	HKDSE:5	B+	B+		A-	B	B+			
2	20	JUPAS	CHRISTIAN ALLIANCE S C CHAN HKDSE	HKDSE:5*	A-	B		B	B+	B			
3	25	JUPAS	MUNSANG COLLEGE (HONG K HKDSE	HKDSE:4	B	A-		A-	B+	A			
4	29	JUPAS	PO LEUNG KUK LAWS FOUNDA HKDSE	HKDSE:4	B+	B-		B+	A-	B-			
5	36	JUPAS	STFA LEUNG KAU KUI COLLEGE HKDSE	HKDSE:5	B-	B+		B+	B	B+			
6	37	JUPAS	METHODIST CHURCH HK WESL HKDSE	HKDSE:5	D	D+		B+	B+	D			
7	50	JUPAS	Others	HKDSE	HKDSE:5	B-	C+	D	D+	C			
8	55	JUPAS	KWOK TAK SENG CATHOLIC SE HKDSE	HKDSE:5	B	B		B	C+	B			
9	62	JUPAS	YWCA HIOE TJO YOENG COLLE HKDSE	HKDSE:4	B	B+		B-	B	B+			
10	82	JUPAS	CCC CHUEN YUEN COLLEGE	HKDSE	HKDSE:5	B	F	B	B+	F			
11	88	JUPAS	HKMA K S LO COLLEGE	HKDSE	HKDSE:4	C	C+	C-	F	C			
12	92	JUPAS	LIONS COLLEGE	HKDSE	HKDSE:4	B	C-	C+	C+	C			
13	94	JUPAS	YCH LIM POR YEN SECONDARY HKDSE	HKDSE:5**	B+	A-		C-	C-	A			
14	98	JUPAS	HK SKH BISHOP HALL SECOND HKDSE	HKDSE:5**	B+	B+		A-	A-	B+			
15	101	JUPAS	CHRISTIAN ALLIANCE S W CHA HKDSE	HKDSE:5	B-	B-		B+	B+	B-			
16	109	JUPAS	VALTORTA COLLEGE	HKDSE	HKDSE:5**	A-	A	B-	B+	A			
17	119	JUPAS	AOG HEBRON SECONDARY SC HKDSE	HKDSE:5	B-	A-		A	A	A			
18	129	JUPAS	PUI CHING MIDDLE SCHOOL	HKDSE	HKDSE:5**	A-	A	A-	A-	A			
19	130	JUPAS	TWGH LI KA SHING COLLEGE	HKDSE	HKDSE:5	C+	C-	A	A	C			
20	142	JUPAS	TANG KING PO SCHOOL	HKDSE	HKDSE:4	B+	B-	C-	C+	B-			
21	144	JUPAS	PLK MRS MA KAM MING CHEUI HKDSE	HKDSE:4	B+	B-		B-	B-	B-			
22	147	JUPAS Appeal-in	STFA YUNG YAU COLLEGE	HKDSE	HKDSE:5	B-	C	B-	B-	C			
23	168	JUPAS	FUKIEN SECONDARY SCHOOL	HKDSE	HKDSE:5*	A-	B+	C	C	B+			
24	172	JUPAS	SKH LAM WOO MEMORIAL SEC HKDSE	HKDSE:4	B	C+		B+	B+	C			

You can replace the content as you like

Want to add extra factors?

2. After training the model with the dataset in previous slide.
3. You can make a csv document to **contain the students' information** you want to predict their level.

After running the sixth cell, the level of these students will be added here

	A	B	C	D	E	F	G	H	I	J	K
1	Student_ID	Admission_Channel	Secondary_School	Exam	Math_Background	MATH1510	ENGG1130	MATH1010	ENGG1120		
2	627	JUPAS	CHAN SUI KI (LA SALLE) COLLEGE	HKDSE	HKDSE:5*	A-	A-	B-	B-		
3	882	JUPAS	CHRISTIAN ALLIANCE CHENG WING GEE C	HKDSE	HKDSE:4	D	F	B-	B+		
4	801	JUPAS	MA ON SHAN TSUNG TSIN SECONDARY S	HKDSE	HKDSE:5*	B-	B	C	C+		
5	1087	Mainland Fee-paying	Others	Mainland	(Mainland Gaokao (JEE):<110	B-	B	B-	C+		
6	224	JUPAS	ELEGANTIA COLLEGE (SPONSORED BY ED	HKDSE	HKDSE:5*	B-	F	B-	B-		
7	240	JUPAS	SKH BISHOP MOK SAU TSENG SEC SCH	HKDSE	HKDSE:5*	A-	A-	B-	B+		
8	911	JUPAS	Others	HKDSE	HKDSE:5**	A	A	B-	B+		
9	917	JUPAS	DIOCESAN BOYS SCHOOL	HKDSE	HKDSE:4	D	B-	B	B+		
10	624	Mainland	Others	Mainland	(Mainland Gaokao (JEE):120-1A-	A-	A	B	A		
11	724	JUPAS	SING YIN SECONDARY SCHOOL	HKDSE	HKDSE:4	B	B	B-	B+		
12	546	JUPAS	KWUN TONG MARYKNOLL COLLEGE	HKDSE	HKDSE:5*	A	A	B	B+		
13	410	Mainland	Others	Mainland	(Mainland Gaokao (JEE):130-1B+	B+	B	A-	B		
14	492	JUPAS	Others	HKDSE	HKDSE:5	B-	A-	B	B-		
15	986	JUPAS	Others	HKDSE	HKDSE:4	C+	B+	A-	B+		
16	1046	JUPAS	HELEN LIANG MEMORIAL SEC SCH (SHATIN	HKDSE	HKDSE:5	A-	B+	B	B		
17	272	JUPAS	HEUNG TO MIDDLE SCHOOL	HKDSE	HKDSE:4	B-	C-	A-	A-		
18	803	JUPAS	Others	HKDSE	HKDSE:4	B-	B+	B	B-		
19	383	JUPAS	LAI KING CATHOLIC SECONDARY SCHOOL	HKDSE	HKDSE:4	C-	B	B-	B		
20	881	JUPAS	YAN OI TONG TIN KA PING SECONDARY S	HKDSE	HKDSE:4	B-	C+	B+	B		
21	932	JUPAS	SKH TANG SHIU KIN SECONDARY SCHOOL	HKDSE	HKDSE:4	B	B+	B-	A-		
22	120	Mainland	Others	Mainland	(Mainland Gaokao (JEE):120-1A	A	A	A-	A		
23	732	JUPAS	HKSYC&JA WONG TAI SHAN MEM COLL	HKDSE	HKDSE:4	C	B	B+	B+		
24	474	JUPAS	SING YIN SECONDARY SCHOOL	HKDSE	HKDSE:4	B	A	B-	C-		
25	776	JUPAS	STFA LEUNG KAU KUI COLLEGE	HKDSE	HKDSE:5*	A	A	B-	B-		

The level of students will be added automatically

Want to add extra factors?

4. Then give the file to our model.

Files

- sample_data
- Input information.xlsx
- KNN.pickle
- KNN.py
- KNN_Comparison.py
- KNN_pred_level.csv
- KNN_predict_complete.csv
- More_KNN_predict_complete.csv
- Personal_Information.csv
- Personal_KNN_predict_complete.csv
- data_to_predict.csv
- data_to_predict_with_more_columns.csv
- digitalization.py
- digitalization_more.py
- original_data.csv
- original_data_with_more_columns.csv
- predict_data.csv
- train_data.csv

```
[6] import pickle
import pandas as pd
import KNN
import digitalization_more

data = pd.read_csv("original_data_with_more_columns.csv")
digitalization_more.train(data)

KNN.KNN_model(8)

dataset = pd.read_csv("data_to_predict_with_more_columns.csv")
digitalization_more.predict(dataset)

X = dataset.iloc[:, 1:8].values
# print(X)

# Load the KNN model
with open('KNN.pickle', 'rb') as model_file:
    knn = pickle.load(model_file)

# Predicting the Test set results
y_pred = KNN.predict(X)
print('Number of predictions: ', len(y_pred))
knn_df=pd.DataFrame(y_pred)
knn_df.columns = ['Level']
pd.DataFrame(knn_df).to_csv('KNN_pred_level.csv', index=None)
f1 = pd.read_csv('predict_data.csv')
f2 = pd.read_csv('KNN_pred_level.csv')
file = [f1,f2]
train = pd.concat(file,axis=1)
train.to_csv('More_KNN_predict_complete' + ".csv", index=0, sep=',')
```

Accuracy of the KNN Clasification is: 0.6330275229357798
Number of predictions: 545

Step 1: Upload two csv files to the program.

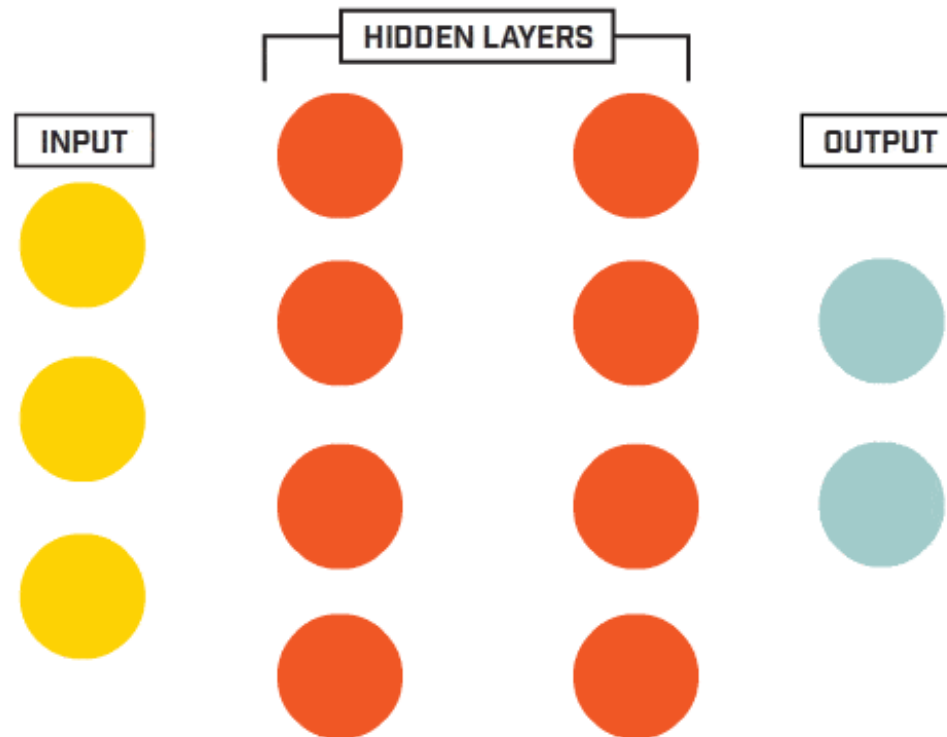
Step 2: You can check the accuracy of the predictions.

Step 3: You can pick the final results in this csv file after running the cell.

Run the eighth cell and get your results

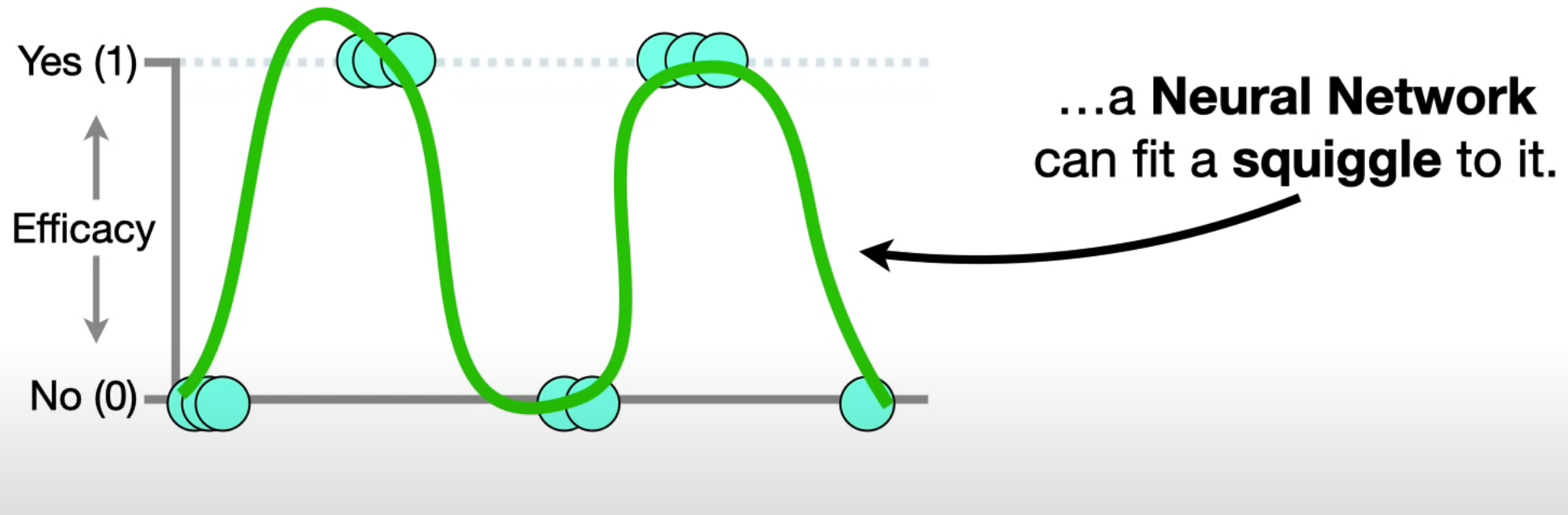
Back Propagate Neural Network

- To find out how BPNN works, let's first drop "Back propagate" away and focus on "Neural network".
- Neural network consists of many nodes, all of the nodes are distributed in input layer, hidden layer and output layer, and all of them are fully connected.

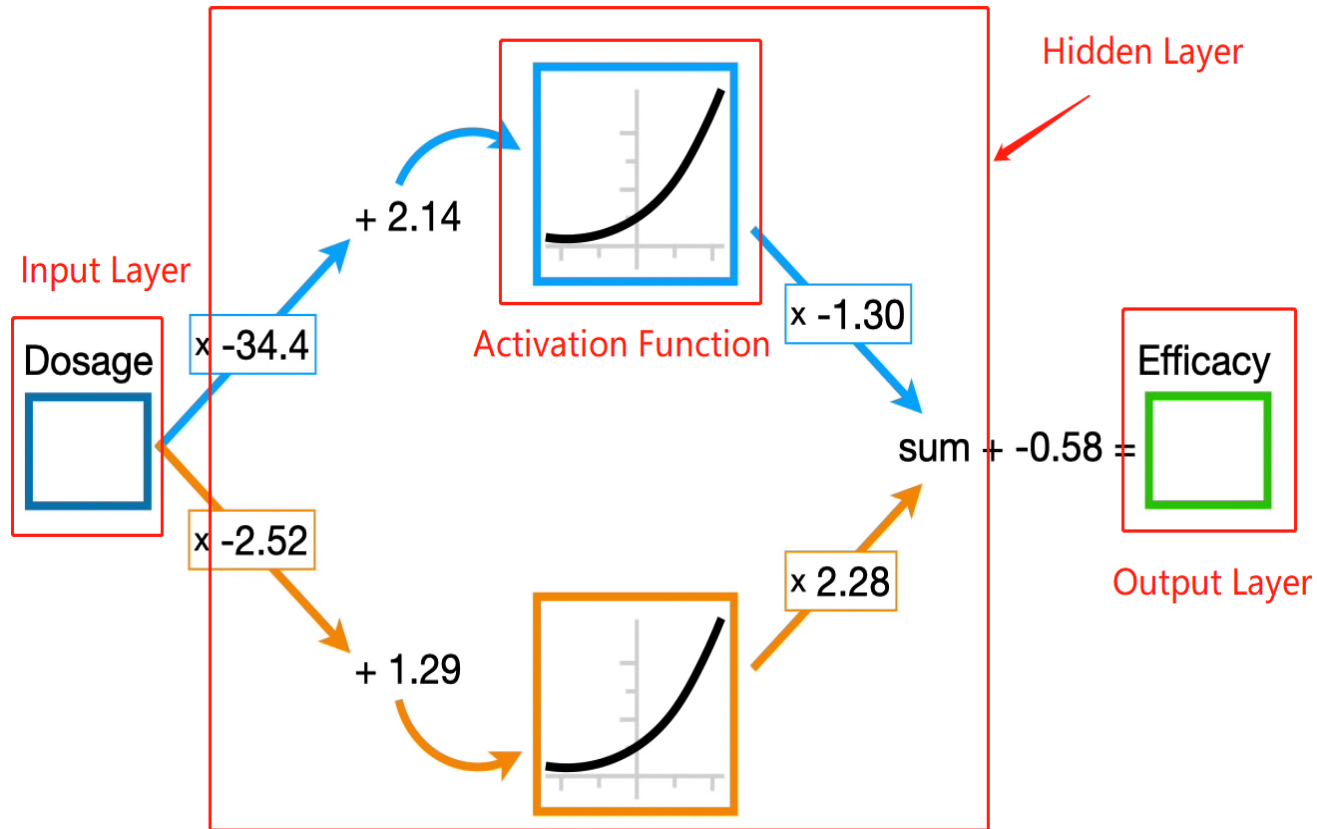


Back Propagate Neural Network

- Compared to [linear regression](#). A BPNN can fit a very zigzag line when the given dataset are very discrete, or the prediction task is very difficult to complete.



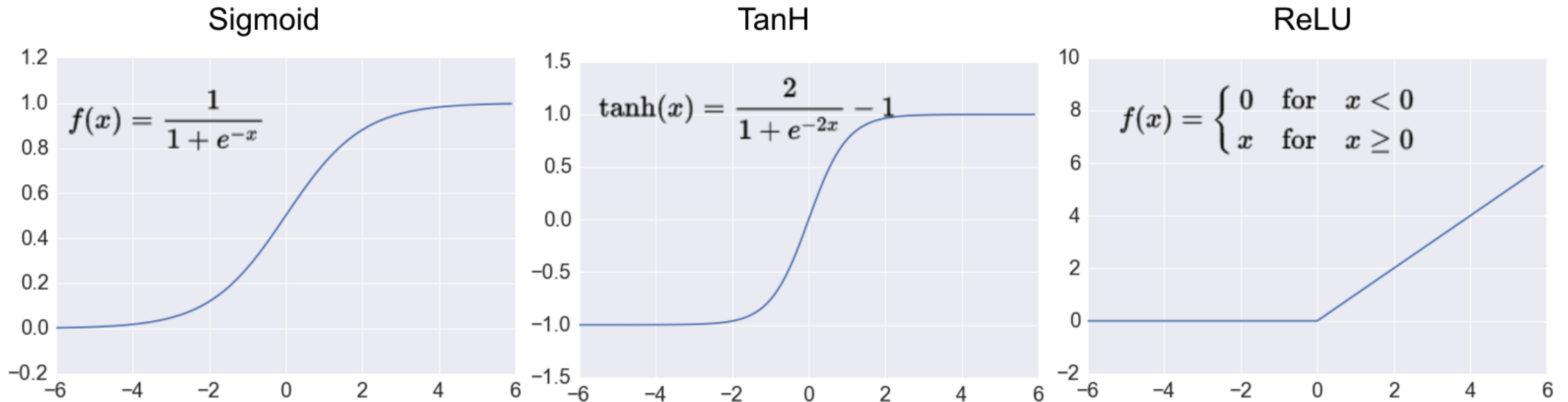
Activation function



- Which means the function inside the network can be **nonlinear**. Let's take a look at a simple neural network.
- For simplicity, the neural network below includes one node in one input layer, two nodes in one hidden layer and one node in one output layer.

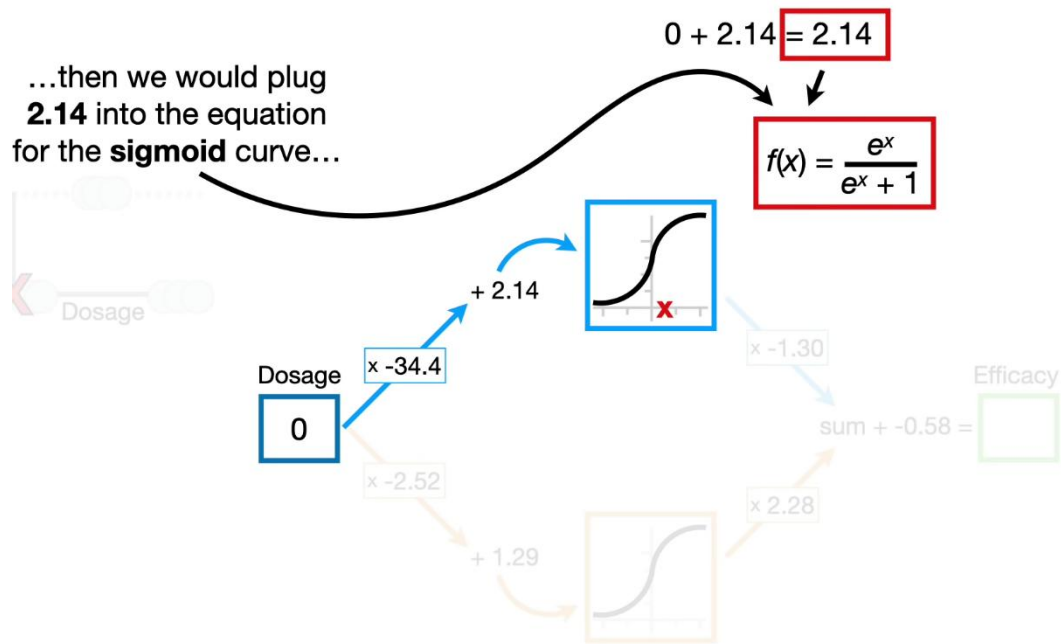
Activation function

- The shape of activation function can be changed, popular activation function includes (sigmoid, relu, soft plus and tanh).

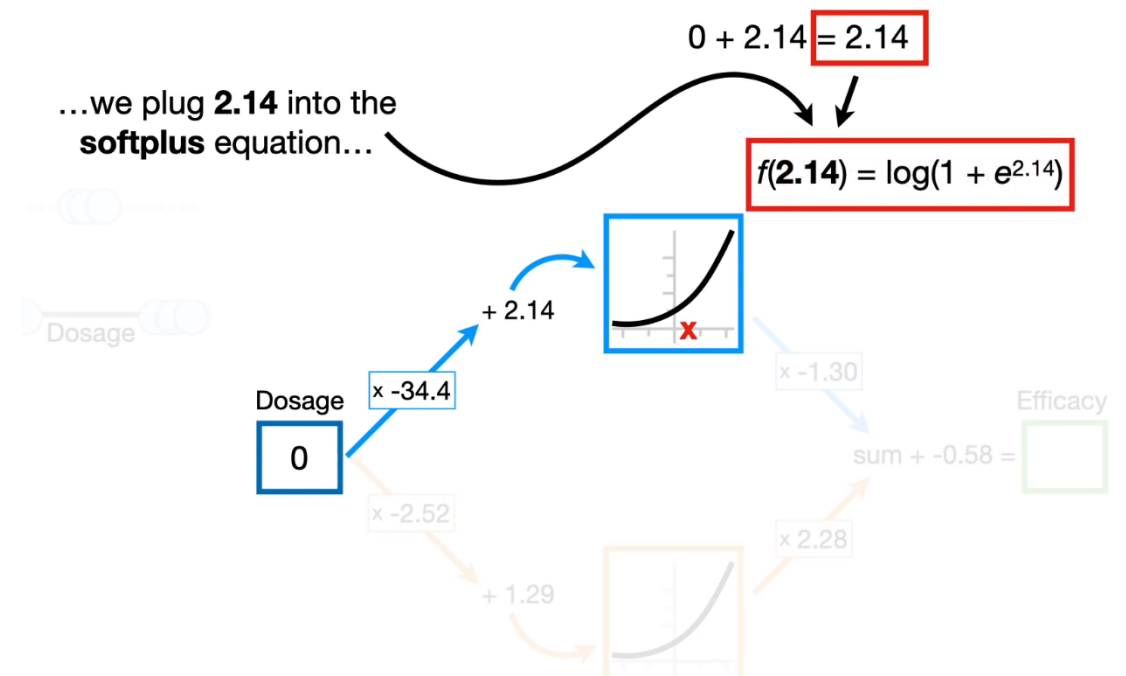


Activation function

- How the function helps network to fit datasets.



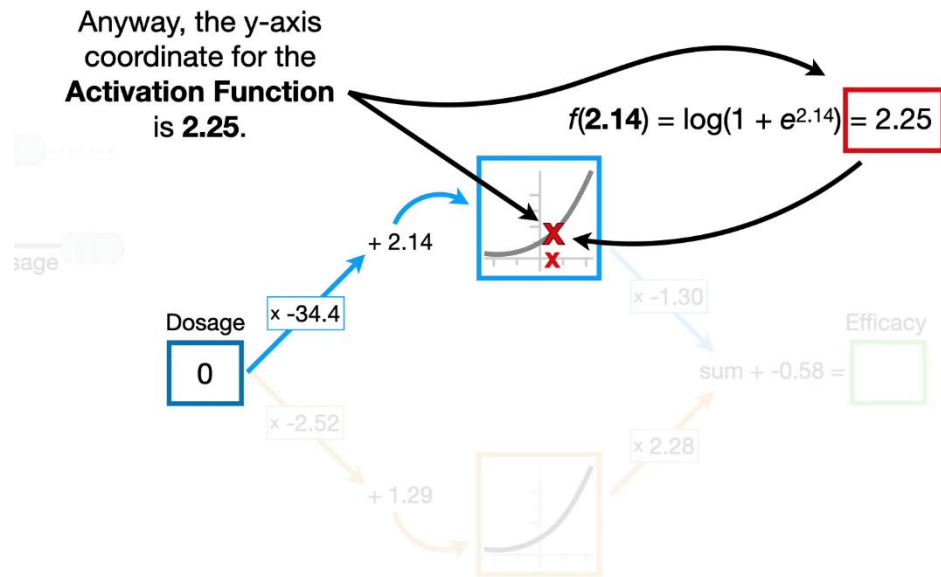
Sigmoid as activation function



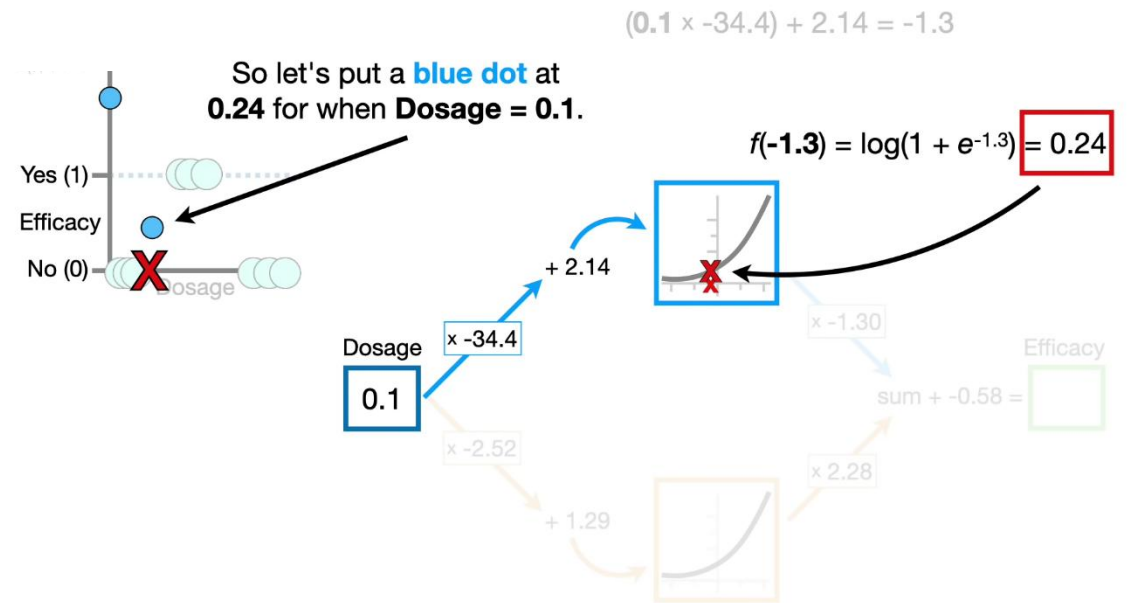
Softplus as activation function

Activation function

- How the function helps network to fit datasets.



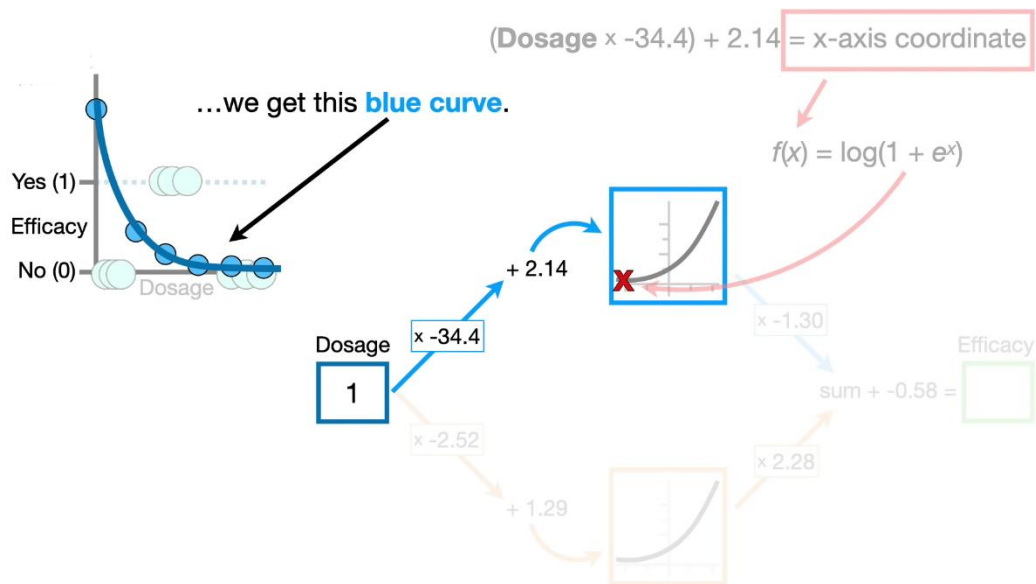
Calculate the first sample



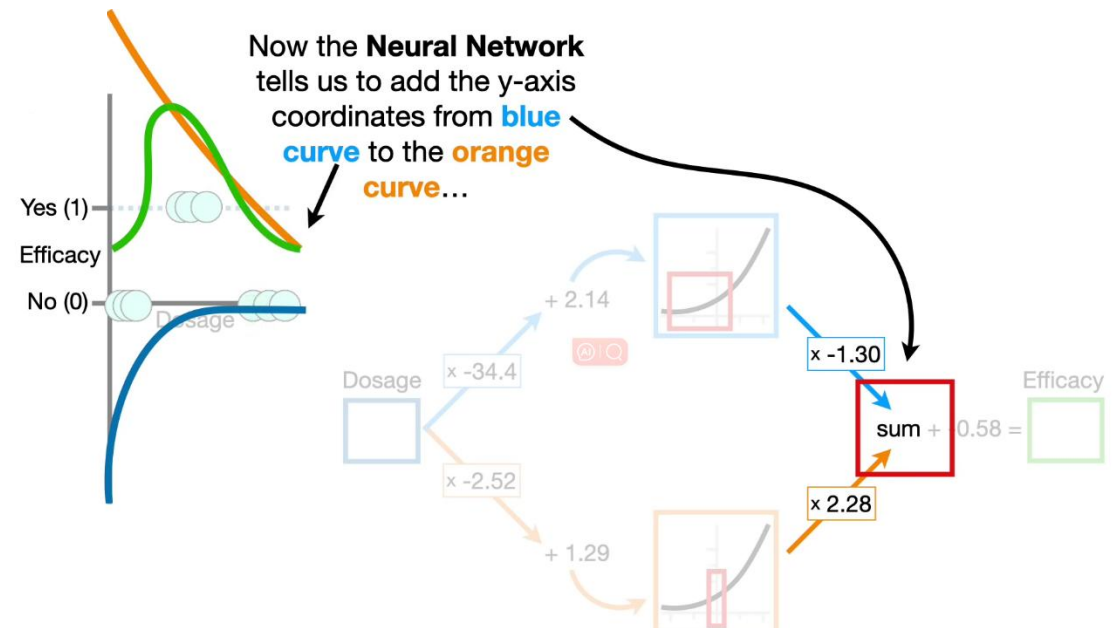
Calculate the second sample

Activation function

- How the function helps network to fit datasets.



Calculate the inputs one by one

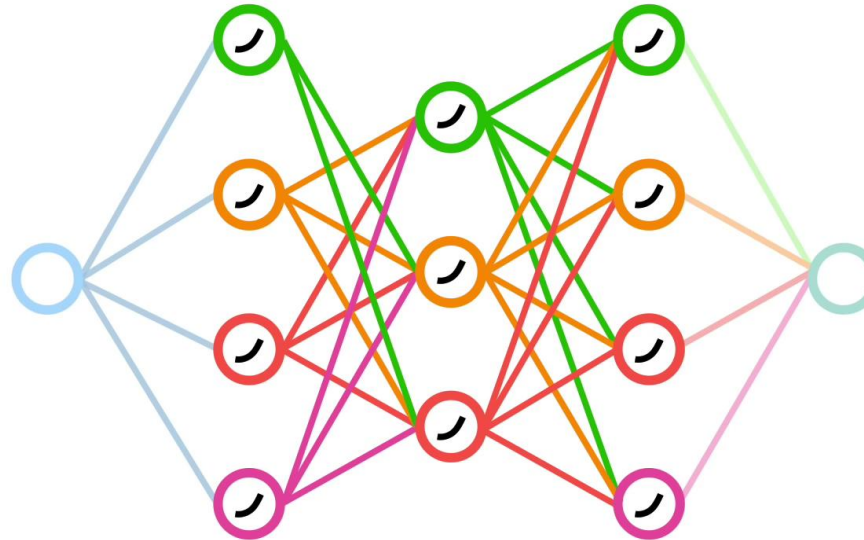
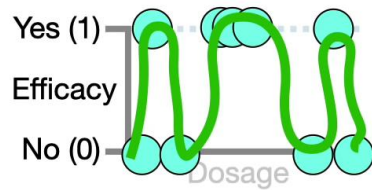


Calculate the second nodes of hidden layer in the same way and add them up

Back Propagate Neural Network

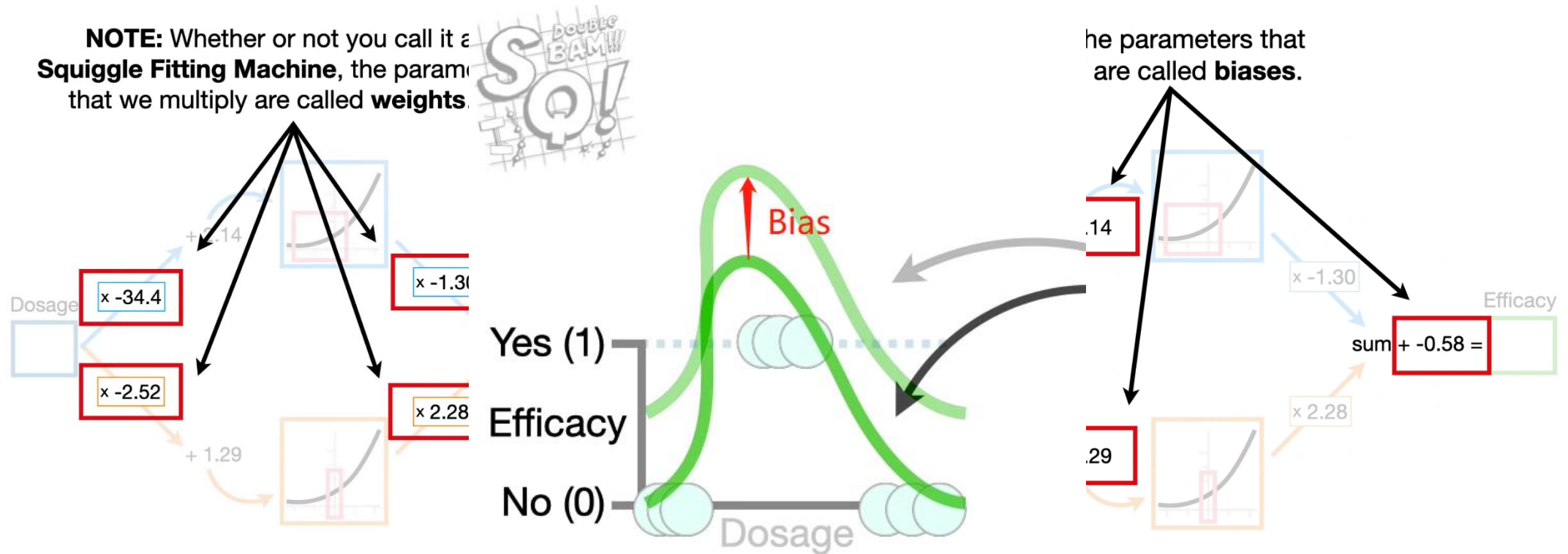
- With more layers and nodes, Neural networks can do much more!

In theory, **Neural Networks** can fit a **green squiggle** to just about any dataset, no matter how complicated!



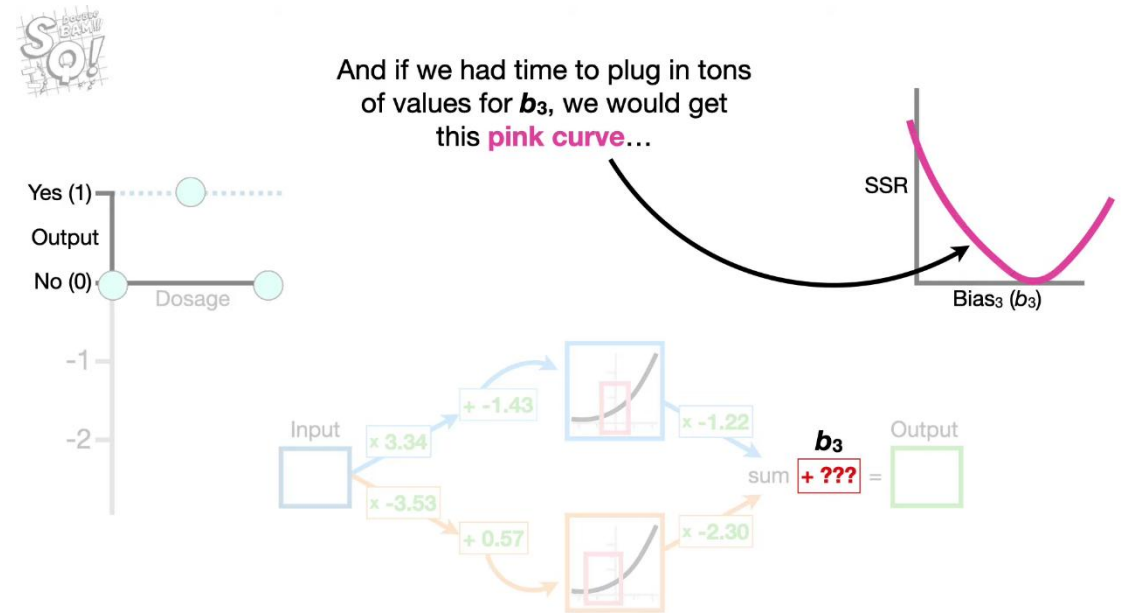
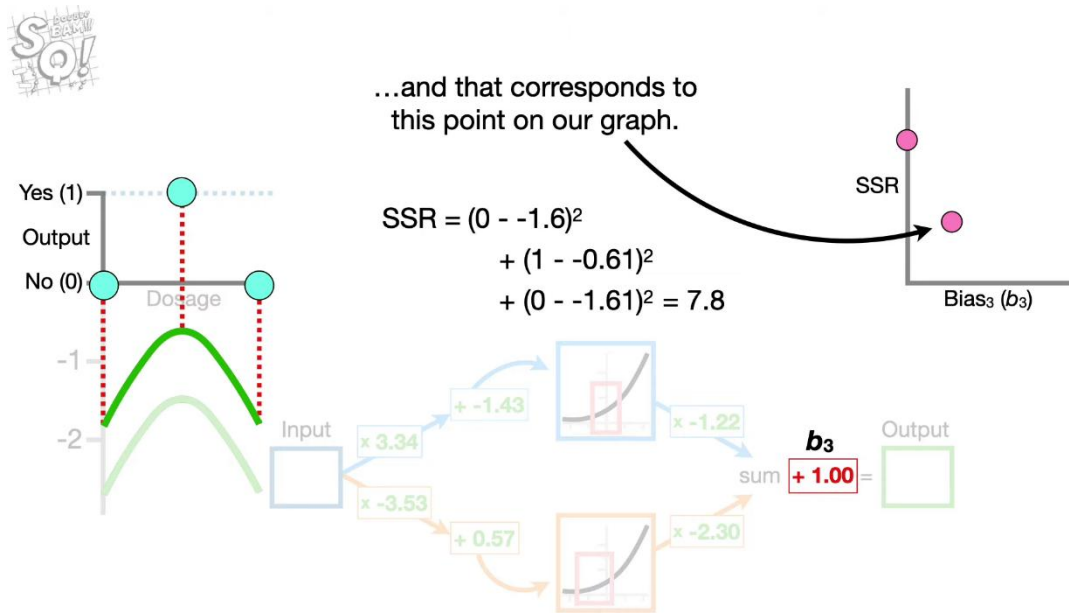
Back Propagate Neural Network

- You may want to ask what's the number that appears between each layers? Here is the answer.
- Weights and bias are updated through each iterations in "Back propagate". Weights are a bit like "P" in PID control.
- Weights and bias reshape the activation function.



Back Propagate Neural Network

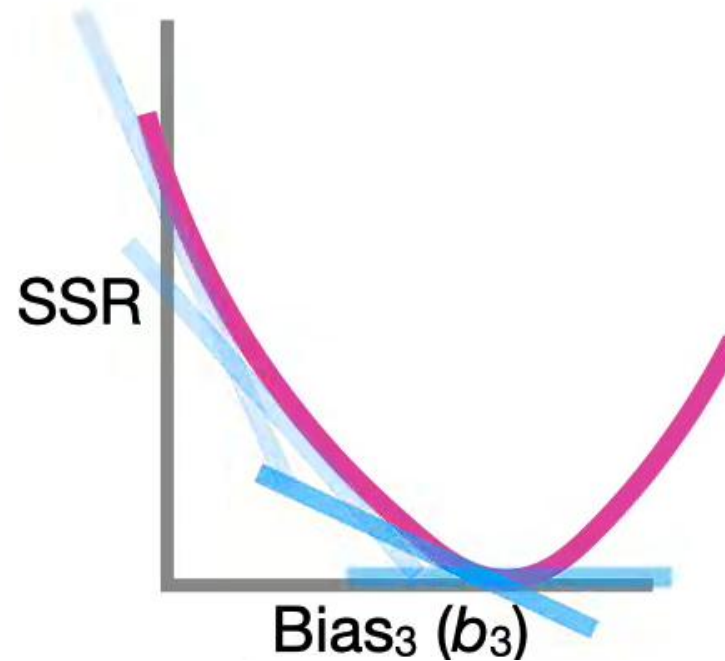
- Now let's talk about the "Back propagate"
- Back propagate optimize the weights or bias backwards, which means we start with bias3 (b_3) and find out the relationship how b_3 effect the SSR (Sum of the Squared Residuals)



Calculate the SSR one by one and we get the relationship

Back Propagate Neural Network

- Gradient descent can help us find the lowest point faster, which means we need to find the derivative of SSR with respect to b_3 .



Find the gradient or slope of the SSR respective with b_3

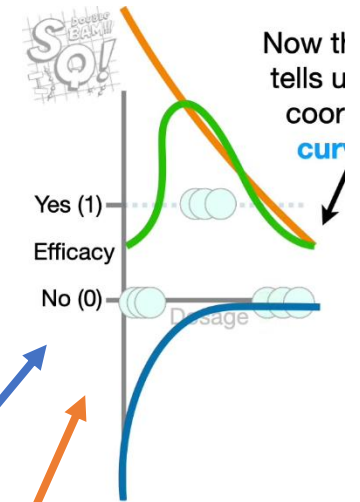
Back Propagate Neural Network

■ Now let's introduce the chain rule.

$$\frac{d SSR}{d b_3} = \frac{d SSR}{d \text{Predicted}} \times \frac{d \text{Predicted}}{d b_3}$$

$$\frac{d SSR}{d \text{Predicted}} = \frac{d}{d \text{Predicted}} \sum_{i=1}^{n=3} (\text{Observed}_i - \text{Predicted}_i)^2 = \sum_{i=1}^{n=3} 2 \times (\text{Observed}_i - \text{Predicted}_i) \times -1$$

$$\frac{d \text{Predicted}}{d b_3} = \frac{d}{d b_3} \text{green squiggle} = \frac{d}{d b_3} (\text{blue} + \text{orange} + b_3) = 1$$



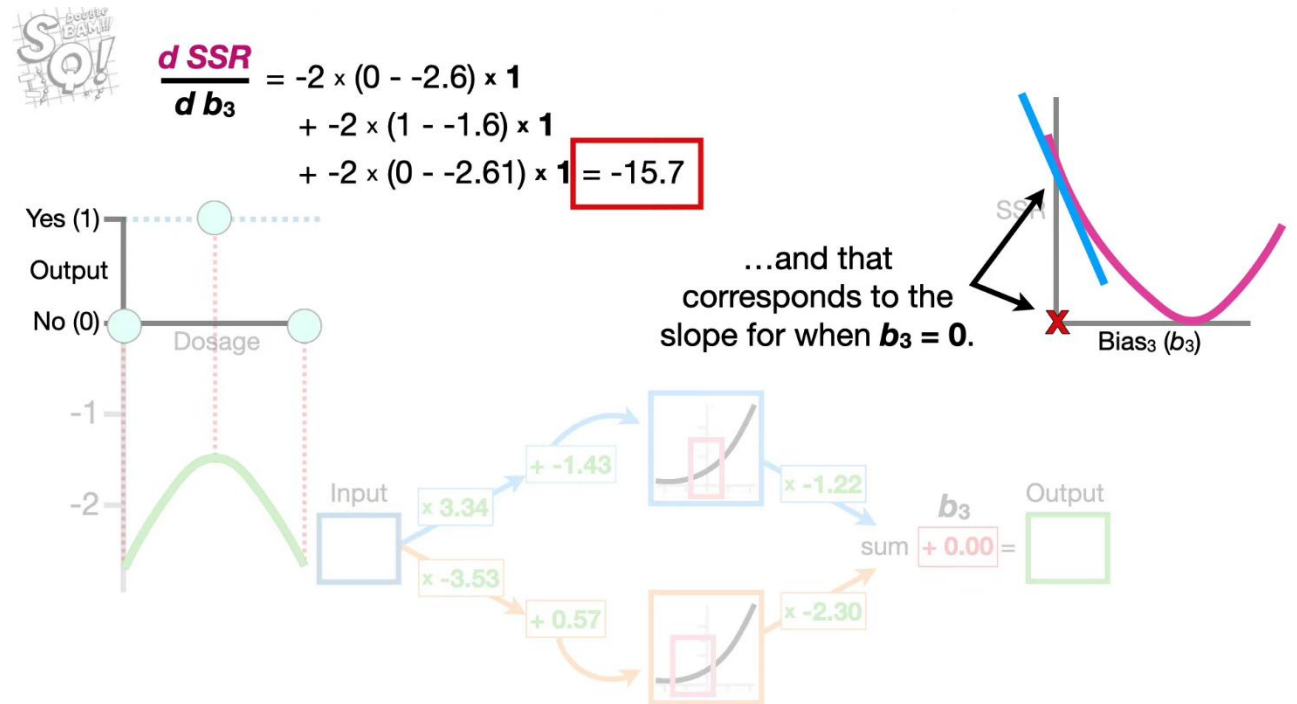
Independent with b_3

Back Propagate Neural Network

- Now we get the equation and can calculate the derivative of initial $b_3 = 0$.
- Step size = Slope * Learning rate
- New b_3 = Previous b_3 – Step size

$$\frac{d SSR}{d b_3} = \frac{d SSR}{d Predicted} \times \frac{d Predicted}{d b_3}$$

- Step size = Slope * Learning rate = $-15.7 * 0.1 = -1.57$
 - New b_3 = Previous b_3 – Step size = $0 - (-1.57) = 1.57$
- $$= \sum_{i=1}^{n=3} -2 \times (\text{Observed}_i - \text{Predicted}_i) \times 1$$



Back Propagate Neural Network

■ Now we get the equation and can calculate the derivative of initial $b_3 = 0$.

■ Step size = Slope * Learning rate

■ New b_3 = Previous b_3 – Step size

From previous page:

■ Step size = Slope * Learning rate = $-15.7 * 0.1 = -1.57$

■ New b_3 = Previous b_3 – Step size = $0 - (-1.57) = 1.57$

Now another iteration:

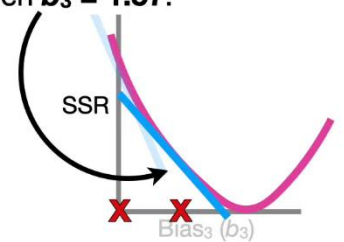
■ Step size = Slope * Learning rate = $-6.26 * 0.1 = -0.626$

■ New b_3 = Previous b_3 – Step size = $1.57 - (-0.626) = 2.19$

Continue circulation until the slope is smaller than the threshold, in another word: close enough to 0, then we get an updated b_3 .

$$\frac{d SSR}{d b_3} = -2 \times (0 - -1.03) \times 1$$
$$+ -2 \times (1 - -0.03) \times 1$$
$$+ -2 \times (0 - -1.04) \times 1 = -6.26$$

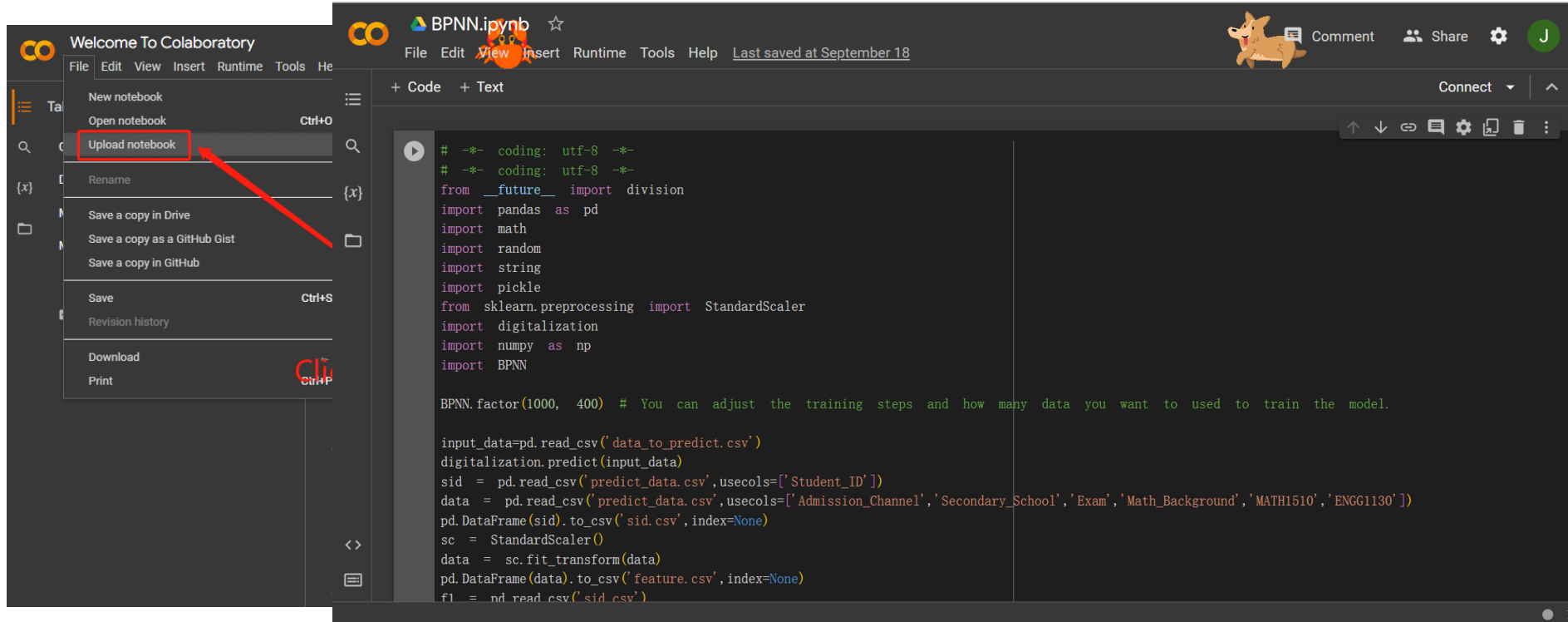
...which corresponds to the slope when $b_3 = 1.57$.



Now do it by yourself!

Preparations

- Upload the “BPNN.ipynb” from the file “Main_code”.



Upload the program

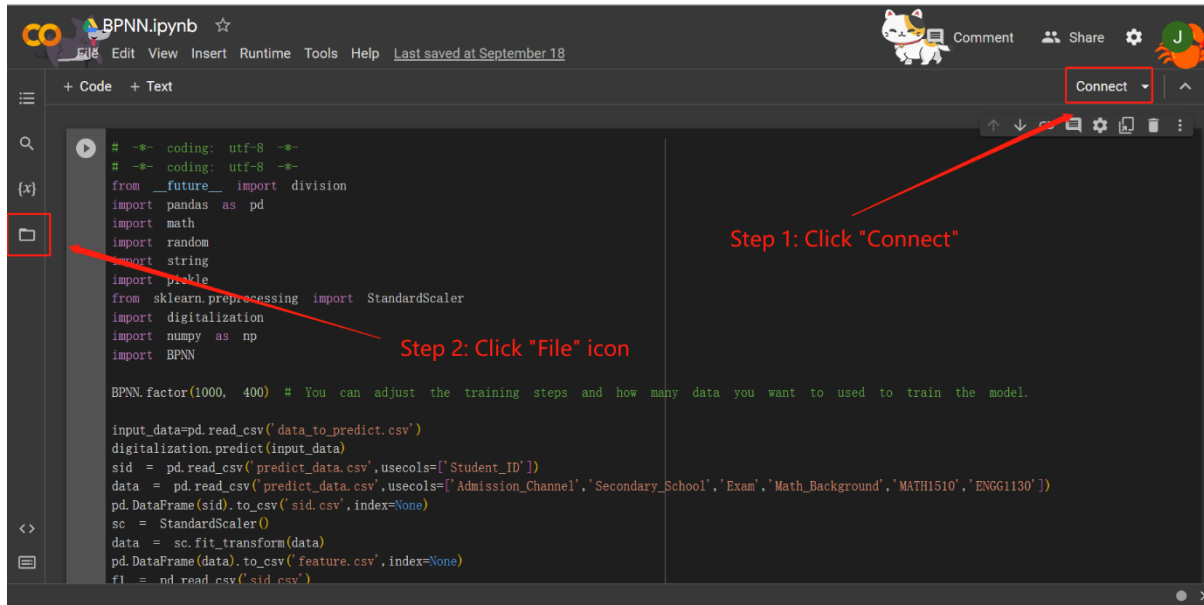
Upload complete

Select “BPNN.ipynb”

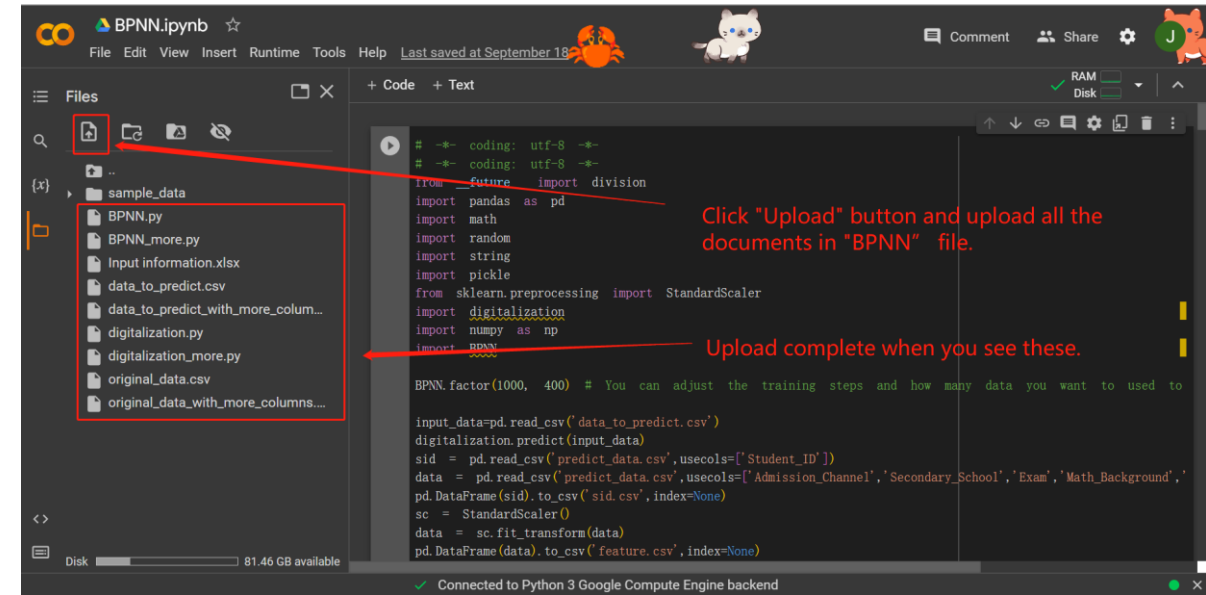
型	大小
YNNB 文件	15 KB
YNNB 文件	137 KB
YNNB 文件	84 KB
YNNB 文件	82 KB

Preparations

- Connect the program to online processor.
- Upload all the documents from “BPNN” file to the notebook.



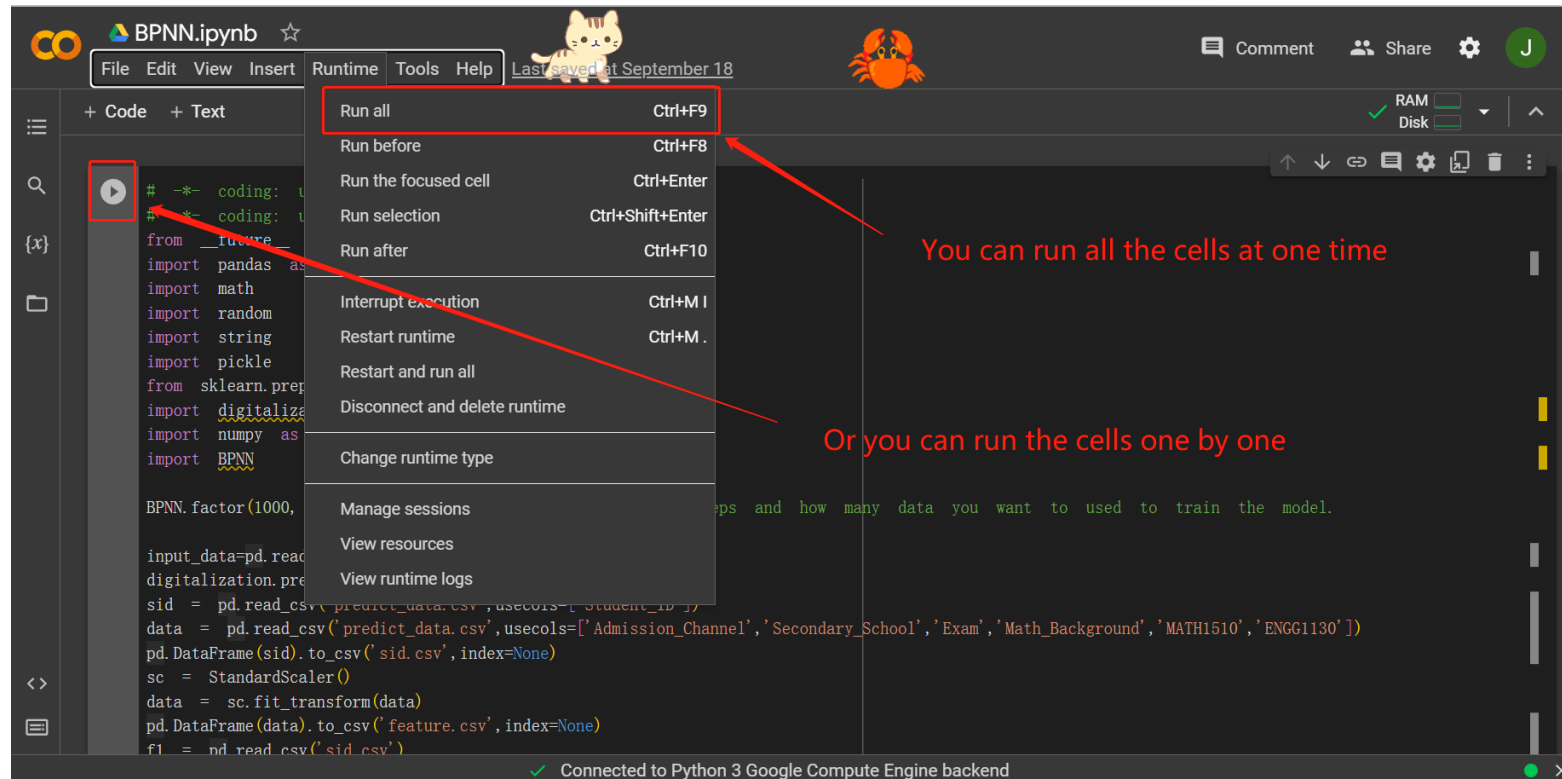
Connect the notebook



Upload our files

Run the program

- Click “Run” button and run the cells **one by one** or you can run them all in one click. (One by one is recommended)



Run the BPNN program

Check the result

- Click “Run” button and run the cells **one by one** or you can run them all in one click. (One by one is recommended)

```
raw_data = raw.values
raw_feature = raw_data[:,1:7]
for i in range(len(raw_feature)):
    ele = []
    ele.append(list(raw_feature[i]))
    data.append(ele)
test = data[0:]
nn.test(test)

if __name__ == '__main__':
    factor()
```

error 8.100187388
error 3.991341323
error 2.990424677
error 2.818203462
error 2.419634204
error 1.953786973
error 1.592854441
error 1.335460151
error 1.162391376
error 1.043945217
accuracy: 0.986111111
545

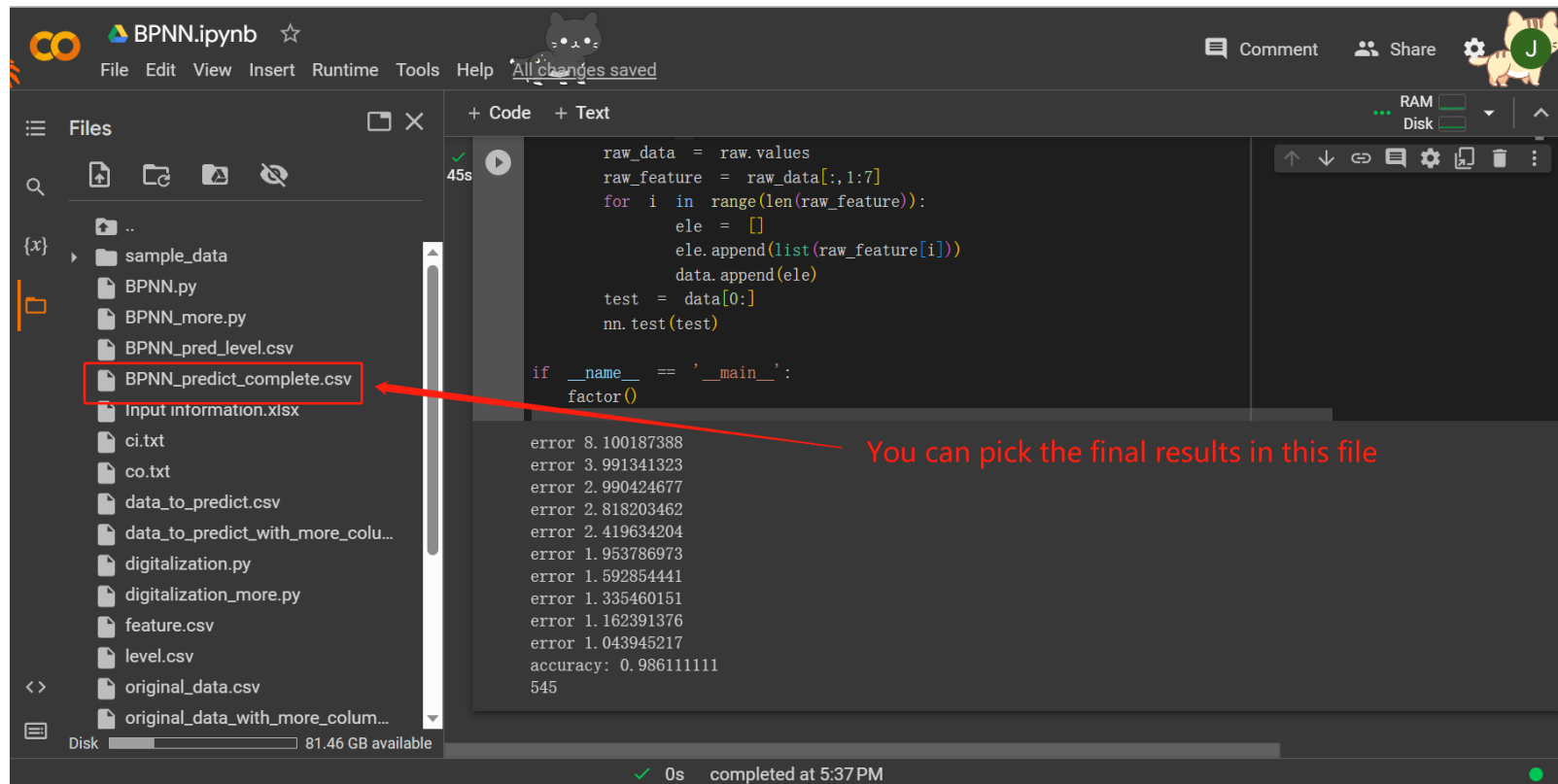
Here will show the loss per 100 iterations

Here will show the accuracy of the algorithm and the number of predictions

Output of first cell

Check the output file

- The final output can be found on the left, named “BPNN_predict_complete.csv”, after opening the file we can find that the predict level is on the rightest column.



The screenshot shows a Jupyter Notebook interface with a dark theme. On the left, the 'Files' sidebar is open, showing a directory structure. The file 'BPNN_predict_complete.csv' is highlighted with a red box. A red arrow points from this box to the output of the code cell. The code cell contains the following Python code:

```
raw_data = raw.values
raw_feature = raw_data[:,1:7]
for i in range(len(raw_feature)):
    ele = []
    ele.append(list(raw_feature[i]))
    data.append(ele)
test = data[0:]
nn.test(test)

if __name__ == '__main__':
    factor()
```

The output of the code cell is displayed below the code, showing a list of error messages and an accuracy value:

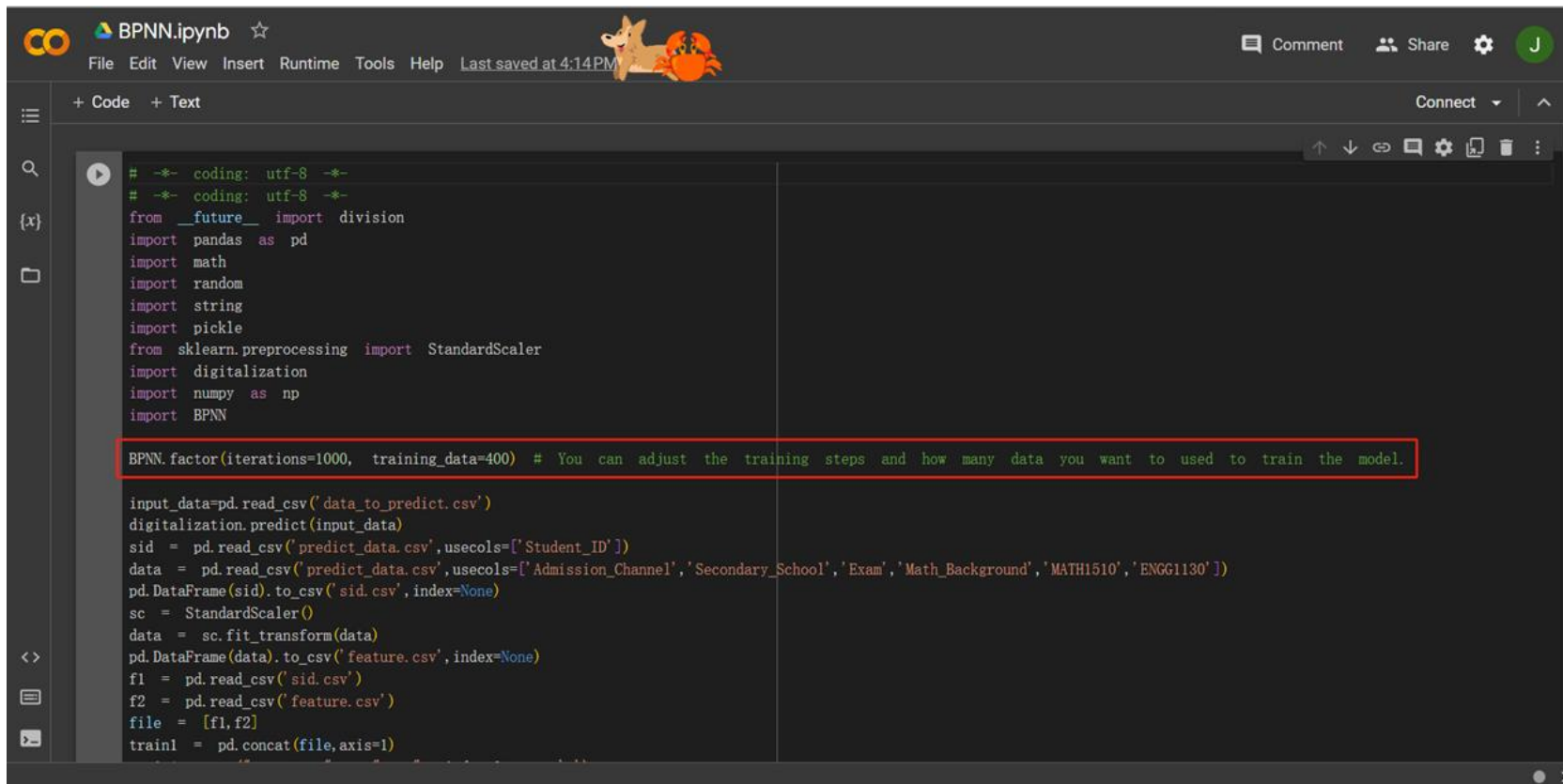
```
error 8. 100187388
error 3. 991341323
error 2. 990424677
error 2. 818203462
error 2. 419634204
error 1. 953786973
error 1. 592854441
error 1. 335460151
error 1. 162391376
error 1. 043945217
accuracy: 0. 986111111
545
```

A red arrow points from the text 'You can pick the final results in this file' to the output of the code cell.

Outcome file

Try to adjust some parameters

- After [comparison](#), you can [adjust the training iterations and ratio of test data](#) like in the picture below. (Since training data is generally the more the better so there's no need to change)



```
BPNN.ipynb ☆
File Edit View Insert Runtime Tools Help Last saved at 4:14PM
+ Code + Text
Connect ^

# -*- coding: utf-8 -*-
# -*- coding: utf-8 -*-
from __future__ import division
import pandas as pd
import math
import random
import string
import pickle
from sklearn.preprocessing import StandardScaler
import digitalization
import numpy as np
import BPNN

BPNN.factor(iterations=1000, training_data=400) # You can adjust the training steps and how many data you want to used to train the model.

input_data=pd.read_csv('data_to_predict.csv')
digitalization.predict(input_data)
sid = pd.read_csv('predict_data.csv',usecols=['Student_ID'])
data = pd.read_csv('predict_data.csv',usecols=['Admission_Channel','Secondary_School','Exam','Math_Background','MATH1510','ENGG1130'])
pd.DataFrame(sid).to_csv('sid.csv',index=None)
sc = StandardScaler()
data = sc.fit_transform(data)
pd.DataFrame(data).to_csv('feature.csv',index=None)
f1 = pd.read_csv('sid.csv')
f2 = pd.read_csv('feature.csv')
file = [f1,f2]
train1 = pd.concat(file,axis=1)
```

Adjust iterations and ratio of training data

Want to try your own data?

■ Sure! Only need to follow my steps:

1. First, open the “Input information.xlsx” from the “BPNN” file we have already downloaded, fill in your personal information. Notice that in “Exam” and “Math_Background” column, only Gaokao and HKDSE are accepted.

	A	B	C	D	E	F	G	H
1	Student_ID	Admission_Channel	Secondary_School	Exam	Math_Background	MATH1510	ENGG1130	ENGG2020
2	1	JUPAS	CCC HEEP WOH COLLEGE	Mainland Gaokao (JEE)	Mainland Gaokao (JEE):>140	A	A	
3	2	Non-JUPAS (Local)	SUEN WAN PUB HO CHUEN YIU MEMORIAL COL	HKDSE	HKDSE:5*	B+	C	
4								
5								
6			SHUN LEE CATHOLIC SECONDARY SCHOOL					
7			STFA LEE SHAU KEE COLLEGE					
8			STFA LEUNG KAU KUI COLLEGE					
9			STFA YUNG YAU COLLEGE					
10			STFA SEAWARD WOO COLLEGE					
11			STFA TAM PAK YU COLLEGE					
12			STFA CHENG YU TUNG SECONDARY SCHOOL					
13			ELEGANTIA COLLEGE (SPONSORED BY ED CONV)					
14			HEUNG TO MIDDLE SCHOOL					
15			HHCKLA BUDDHIST CHING KOK SEC SCH					
16			HONG KONG CHINESE WOMEN'S CLUB COLLEGE					
17			CHRISTIAN ALLIANCE CHENG WING GEE COLL					
18								
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27								
28								
29								

Input your information

Want to try your own data?

2. Upload the “Input information.xlsx” to program like before and run the second cell.
3. Then pick up the “Personal_BPNN_predict_complete.csv” file in the left to check the prediction of your grade!

The screenshot shows a Jupyter Notebook interface with a dark theme. On the left is a file explorer pane showing a directory structure with files like 'sample_data', 'BPNN.py', 'BPNN_more.py', 'BPNN_pred_level.csv', 'BPNN_predict_complete.csv', 'Input information.xlsx', 'More_BPNN_predict_complete...', 'Personal_BPNN_predict_comple...', 'ci.txt', 'co.txt', 'data_to_predict.csv', 'data_to_predict_with_more_colu...', 'digitalization.py', 'digitalization_more.py', 'feature.csv', and 'level.csv'. The file 'Personal_BPNN_predict_comple...' is highlighted with a red box. The main area shows a code cell with the following Python code:

```
# features 0-3
# labels 4
def factor():
    data = []
    # read dataset
    raw = pd.read_csv('temporary.csv')
    raw_data = raw.values
    raw_feature = raw_data[:, 1:7]
    for i in range(len(raw_feature)):
        ele = []
        ele.append(list(raw_feature[i]))
        data.append(ele)
    test = data[0:]
    nn.test(test)

if __name__ == '__main__':
    factor()
```

Red arrows point from the highlighted file in the file explorer to the code cell and from the code cell to a terminal output box. The terminal output box contains the following text:

```
Please input the number of students: 2
Number of predictions: 2
['A', 'C']
```

Below the terminal output box, there is a text prompt: "If you want to add some extra columns of student features".

Red text annotations are present: "You can check the result in this file." pointing to the highlighted file, and "Or you can just see the predictions here" pointing to the terminal output box.

Run the second cell.

Two ways to see the predictions

Want to add some extra factors?

■ Sure! Only need to follow my steps:

1. We can take a look at “[original_data_with_more_columns.csv](#)” in “BPNN” file.

A1 Original No.

We add two more courses as factors for example

	A	B	C	D	E	F	G	H	I	J
1	Original No.	Admission_Channel	Secondary_School	Exam	Math_Background	MATH1510	ENGG1130	MATH1010	ENGG1120	ENGG2020
2	17	JUPAS	TSUEN WAN PUB HO CHUEN YIU MEMORIAL COLL	HKDSE	HKDSE:5	B+	B+	A-	B	B+
3	20	JUPAS	CHRISTIAN ALLIANCE S C CHAN MEM COLL	HKDSE	HKDSE:5*	A-	B	B	B+	B
4	25	JUPAS	MUNSANG COLLEGE (HONG KONG ISLAND)	HKDSE	HKDSE:4	B	A-	A-	B+	A
5	29	JUPAS	PO LEUNG KUK LAWS FOUNDATION COLLEGE	HKDSE	HKDSE:4	B+	B-	B+	A-	B-
6	36	JUPAS	STFA LEUNG KAU KUI COLLEGE	HKDSE	HKDSE:5	B-	B+	B+	B	B+
7	37	JUPAS	METHODIST CHURCH HK WESLEY COLLEGE	HKDSE	HKDSE:5	D	D+	B+	B+	D
8	50	JUPAS	Others	HKDSE	HKDSE:5	B-	C+	D	D+	C
9	55	JUPAS	KWOK TAK SENG CATHOLIC SECONDARY SCHOOL	HKDSE	HKDSE:5	B	B	B	C+	B
10	62	JUPAS	YWCA HIOE TJO YOENG COLLEGE	HKDSE	HKDSE:4	B	B+	B-	B	B+
11	82	JUPAS	CCC CHUEN YUEN COLLEGE	HKDSE	HKDSE:5	B	F	B	B+	F
12	88	JUPAS	HKMA K S LO COLLEGE	HKDSE	HKDSE:4	C	C+	C-	F	C
13	92	JUPAS	LIONS COLLEGE	HKDSE	HKDSE:4	B	C-	C+	C+	C
14	94	JUPAS	YCH LIM POR YEN SECONDARY SCHOOL	HKDSE	HKDSE:5**	B+	A-	C-	C-	A
15	98	JUPAS	HK SKH BISHOP HALL SECONDARY SCHOOL	HKDSE	HKDSE:5**	B+	B+	A-	A-	B+
16	101	JUPAS	CHRISTIAN ALLIANCE S W CHAN MEM COLL	HKDSE	HKDSE:5	B-	B-	B+	B+	B-
17	109	JUPAS	VALTORTA COLLEGE	HKDSE	HKDSE:5**	A-	A	B-	B+	A
18	119	JUPAS	AOG HEBRON SECONDARY SCHOOL	HKDSE	HKDSE:5	B-	A-	A	A	A
19	129	JUPAS	PUI CHING MIDDLE SCHOOL	HKDSE	HKDSE:5**	A-	A	A-	A-	A
20	130	JUPAS	TWGH LI KA SHING COLLEGE	HKDSE	HKDSE:5	C+	C-	A	A	C
21	142	JUPAS	TANG KING PO SCHOOL	HKDSE	HKDSE:4	B+	B-	C-	C+	B-
22	144	JUPAS	PLK MRS MA KAM MING CHEUNG FOOK SIEN COL	HKDSE	HKDSE:4	B+	B-	B-	B-	B-
23	147	JUPAS Appeal-in	STFA YUNG YAU COLLEGE	HKDSE	HKDSE:5	B-	C	B-	B-	C
24	168	JUPAS	FUKIEN SECONDARY SCHOOL	HKDSE	HKDSE:5*	A-	B+	C	C	B+
25	172	JUPAS	SKH LAM WOO MEMORIAL SECONDARY SCHOOL	HKDSE	HKDSE:4	B	C+	B+	B+	C

You can replace the content as you like

Want to add some extra factors?

2. After **training** the model with the dataset in previous slide.
3. You can make a csv document to **contain the students' information** you want to predict their level.

A1 Student_ID

After running the cell, the level of these students will be added

	A	B	C	D	E	F	G	H	I	J
1	Student_ID	Admission_Channel	Secondary_School	Exam	Math_Background	MATH1510	ENGG1130	MATH1010	ENGG1120	ENGG2020
2	627	JUPAS	CHAN SUI KI (LA SALLE) COLLEGE	HKDSE	HKDSE:5*	A-	A-	B-	B-	
3	882	JUPAS	CHRISTIAN ALLIANCE CHENG WING GEE COLL	HKDSE	HKDSE:4	D	F	B	B+	
4	801	JUPAS	MA ON SHAN TSUNG TSIN SECONDARY SCHOOL	HKDSE	HKDSE:5*	B-	B	C	C+	
5	1087	Mainland Fee-paying	Others	Mainland	(Mainland Gaokao (JEE):<110	B-	B	B-	C+	
6	224	JUPAS	ELEGANTIA COLLEGE (SPONSORED BY ED CONV)	HKDSE	HKDSE:5*	B-	F	B-	B-	
7	240	JUPAS	SKH BISHOP MOK SAU TSENG SEC SCH	HKDSE	HKDSE:5*	A-	A-	B-	B+	
8	911	JUPAS	Others	HKDSE	HKDSE:5**	A	A	B-	B+	
9	917	JUPAS	DIOCESAN BOYS SCHOOL	HKDSE	HKDSE:4	D	B-	B	B+	
10	624	Mainland	Others	Mainland	(Mainland Gaokao (JEE):120-129	A-	A	B	A	
11	724	JUPAS	SING YIN SECONDARY SCHOOL	HKDSE	HKDSE:4	B	B	B-	B+	
12	546	JUPAS	KWUN TONG MARYKNOLL COLLEGE	HKDSE	HKDSE:5*	A	A	B	B+	
13	410	Mainland	Others	Mainland	(Mainland Gaokao (JEE):130-139	B+	B	A-	B	
14	492	JUPAS	Others	HKDSE	HKDSE:5	B-	A-	B	B-	
15	986	JUPAS	Others	HKDSE	HKDSE:4	C+	B+	A-	B+	
16	1046	JUPAS	HELEN LIANG MEMORIAL SEC SCH (SHATIN)	HKDSE	HKDSE:5	A-	B+	B	B	
17	272	JUPAS	HEUNG TO MIDDLE SCHOOL	HKDSE	HKDSE:4	B-	C-	A-	A-	
18	803	JUPAS	Others	HKDSE	HKDSE:4	B-	B+	B	B-	
19	383	JUPAS	LAI KING CATHOLIC SECONDARY SCHOOL	HKDSE	HKDSE:4	C-	B	B-	B	
20	881	JUPAS	YAN OI TONG TIN KA PING SECONDARY SCHOOL	HKDSE	HKDSE:4	B-	C+	B+	B	
21	932	JUPAS	SKH TANG SHIU KIN SECONDARY SCHOOL	HKDSE	HKDSE:4	B	B+	B-	A-	
22	120	Mainland	Others	Mainland	(Mainland Gaokao (JEE):120-129	A	A	A-	A	
23	732	JUPAS	HKSYC&JA WONG TAI SHAN MEM COLL	HKDSE	HKDSE:4	C	B	B+	B+	
24	474	JUPAS	SING YIN SECONDARY SCHOOL	HKDSE	HKDSE:4	B	A	B-	C-	
25	776	JUPAS	STFA LEUNG KAU KUI COLLEGE	HKDSE	HKDSE:5*	A	A	B-	B-	

The level of students will be added automatically

Want to add some extra factors?

4. Then give the file to our model.

The screenshot shows a Jupyter Notebook titled "BPNN.ipynb". The left sidebar displays a file explorer with various files. Three files are highlighted with red boxes: "data_to_predict_with_more_colu...", "original_data_with_more_colum...", and "original_data.csv". A yellow arrow points from the text "Step 3: You can pick the final results here after running the cell." to the output area of the code cell. A red arrow points from the text "Step 1: Upload two csv file to the program." to the file explorer. A blue arrow points from the text "Step 2: You can check the accuracy of the predictions." to the output area of the code cell. The code cell contains the following Python code:

```
[3] raw = pd.read_csv('temporary.csv')
raw_data = raw.values
raw_feature = raw_data[:,1:9]
for i in range(len(raw_feature)):
    ele = []
    ele.append(list(raw_feature[i]))
    data.append(ele)
test = data[0:]
nn.test(test)

if __name__ == '__main__':
    factor()
```

The output of the code cell shows a list of error values and an accuracy value:

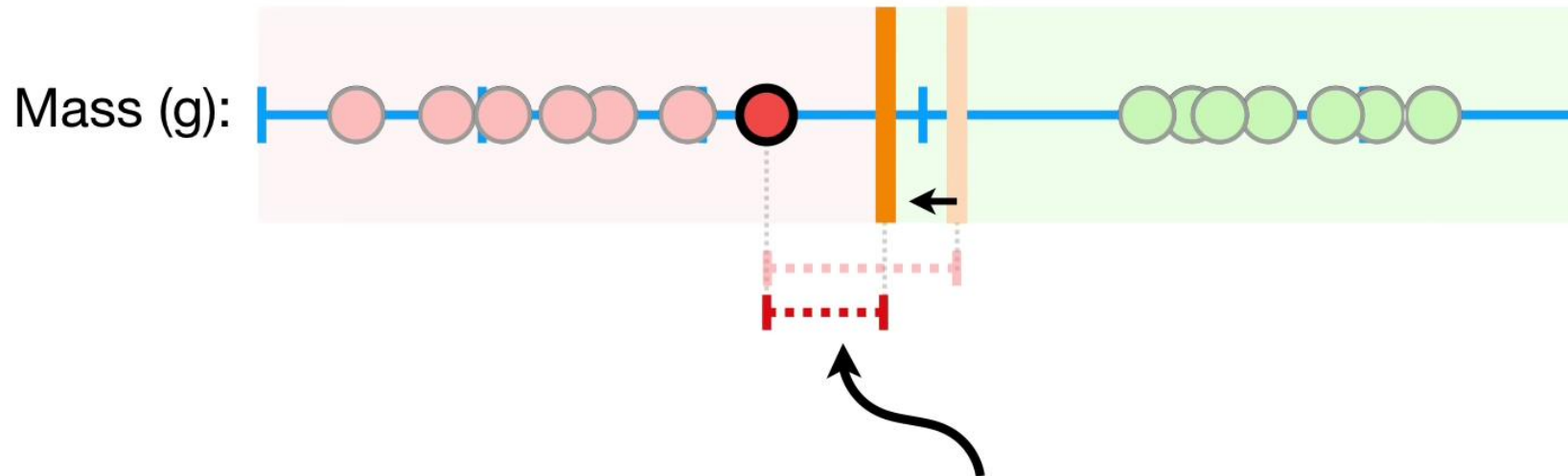
```
error 21.373795558
error 4.507757808
error 3.201286841
error 2.831055563
error 2.449160335
error 2.172280050
error 1.913800399
error 1.656997156
error 1.425042299
error 1.242467155
accuracy: 0.986111111
545
```

The bottom status bar indicates "3m 3s completed at 5:57 PM".

Run the third cell and get your results

Support Vector Machine

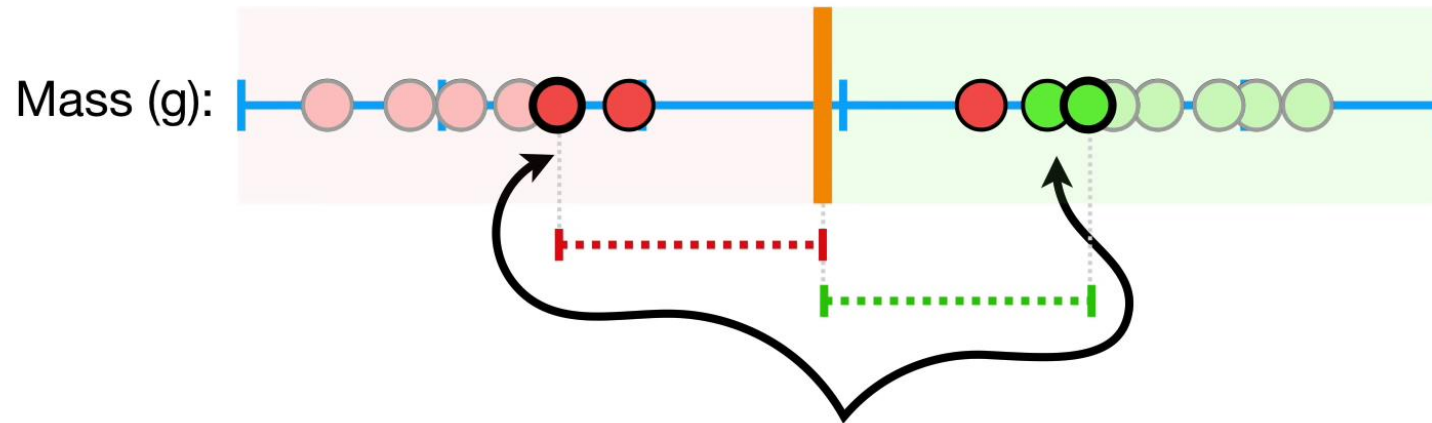
- First, we learn about the margin, when we do classification, we need to find a threshold first, isn't it?
- The further observations to threshold, the more accurate the classification is.



...then the distance between the threshold and the observation that is **not obese** would be smaller...

Support Vector Machine

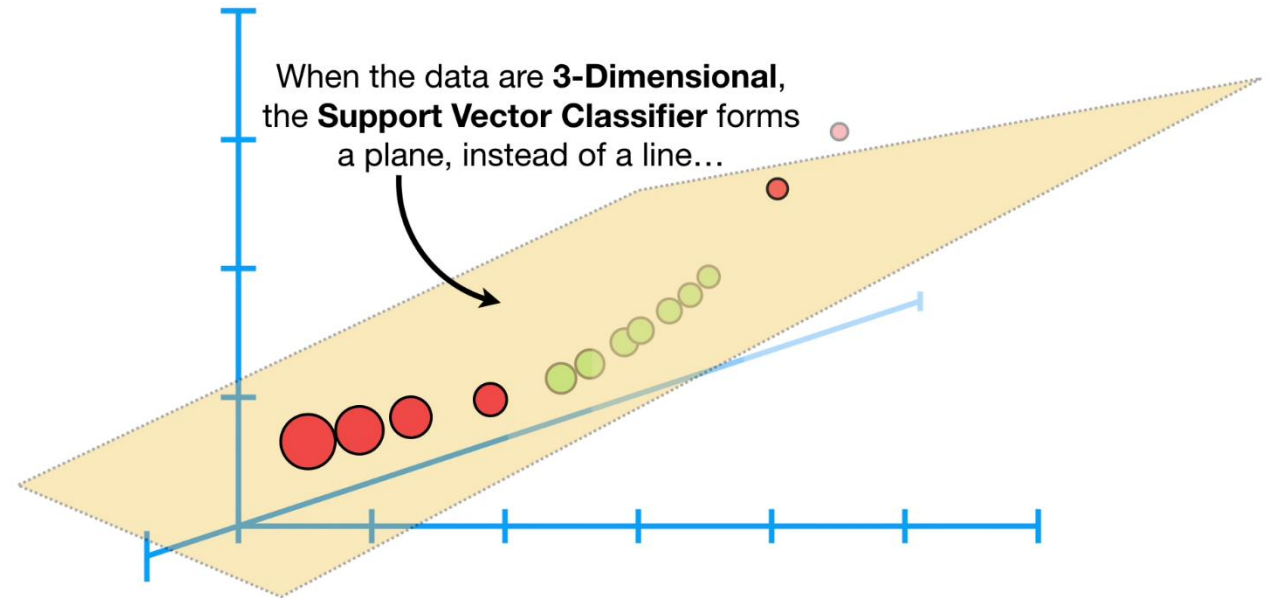
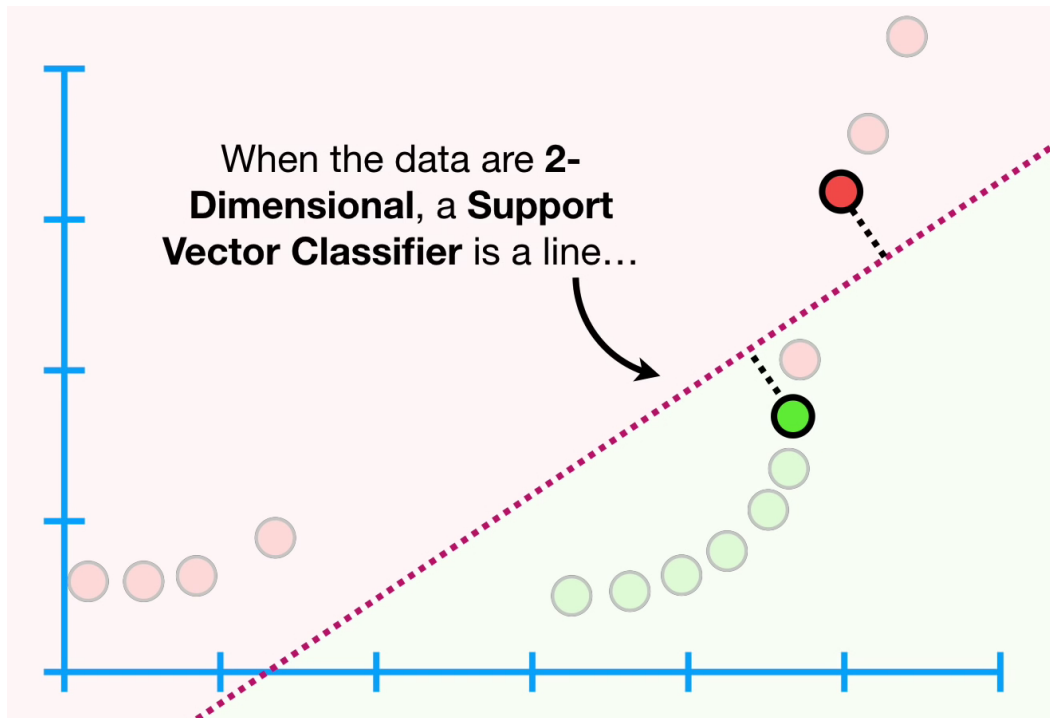
- Sometimes the threshold is hard to determine to ensure all the samples are classified properly, so misclassification is allowed, therefore soft margin appears.



The name **Support Vector Classifier** comes from the fact that the observations on the edge *and within* the **Soft Margin** are called **Support Vectors**.

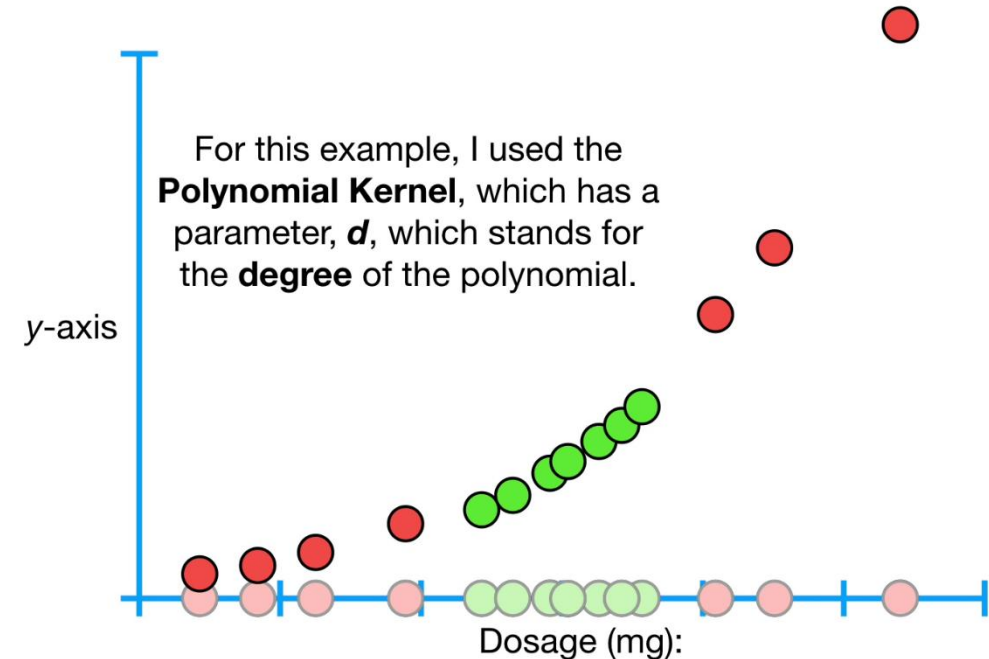
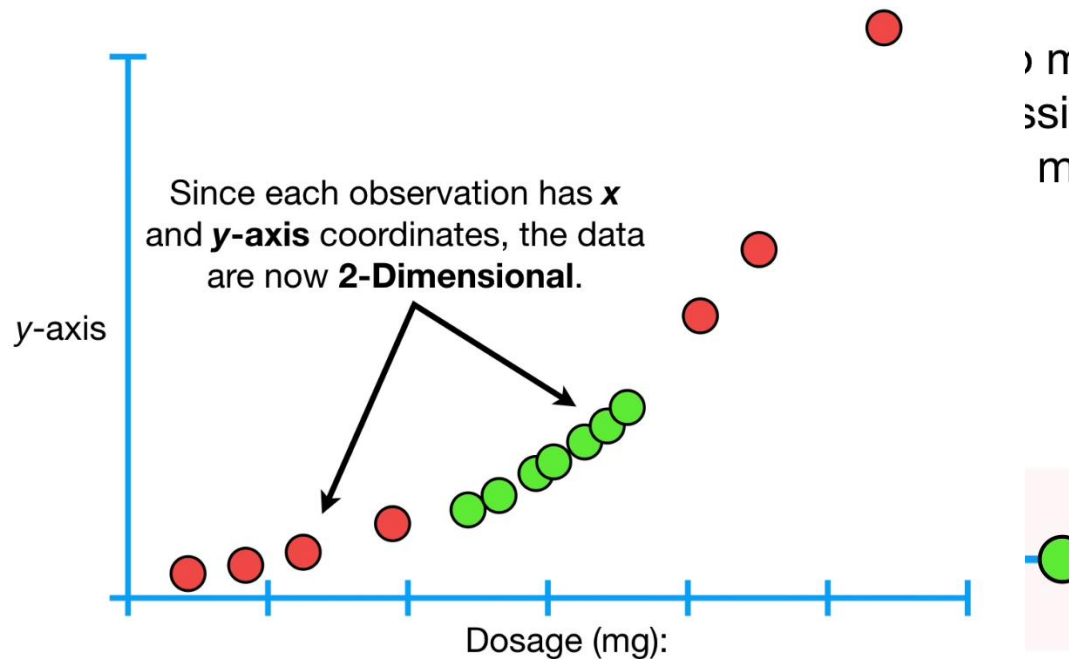
Support Vector Machine

- Support Vector Classifier is different when handling different dimension of dataset.
- When the data are in 4 or more dimensions, the SVC is a hyperplane, but it can't be drawn ☹️.



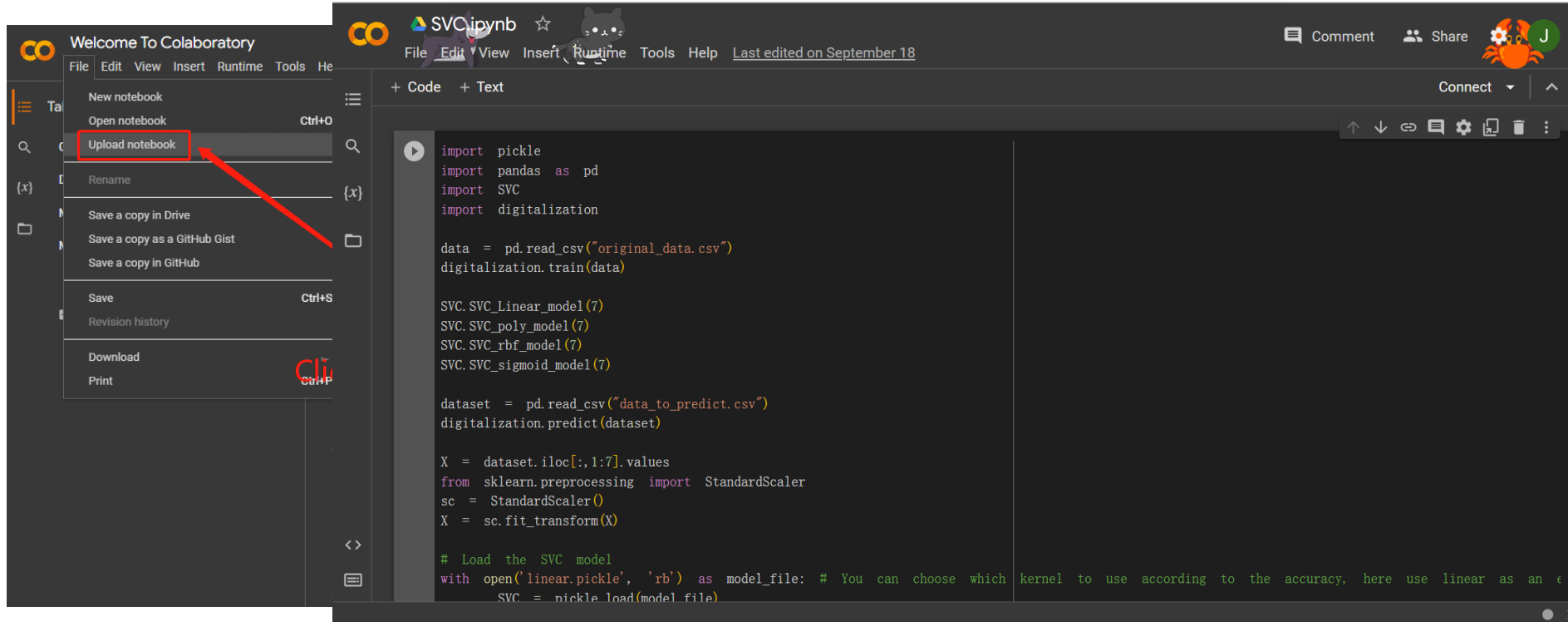
Support Vector Machine

- Support Vector Machine (SVM) can use different kernel functions to systematically find SVC in higher dimensions.
- Start with data in a relatively low dimension.
- Move the data into a higher dimension
- Find a support vector classifier that separates the higher dimensional data into two groups



Preparations

- Upload the “SVC.ipynb” from the file “Main_code”.



Upload the program

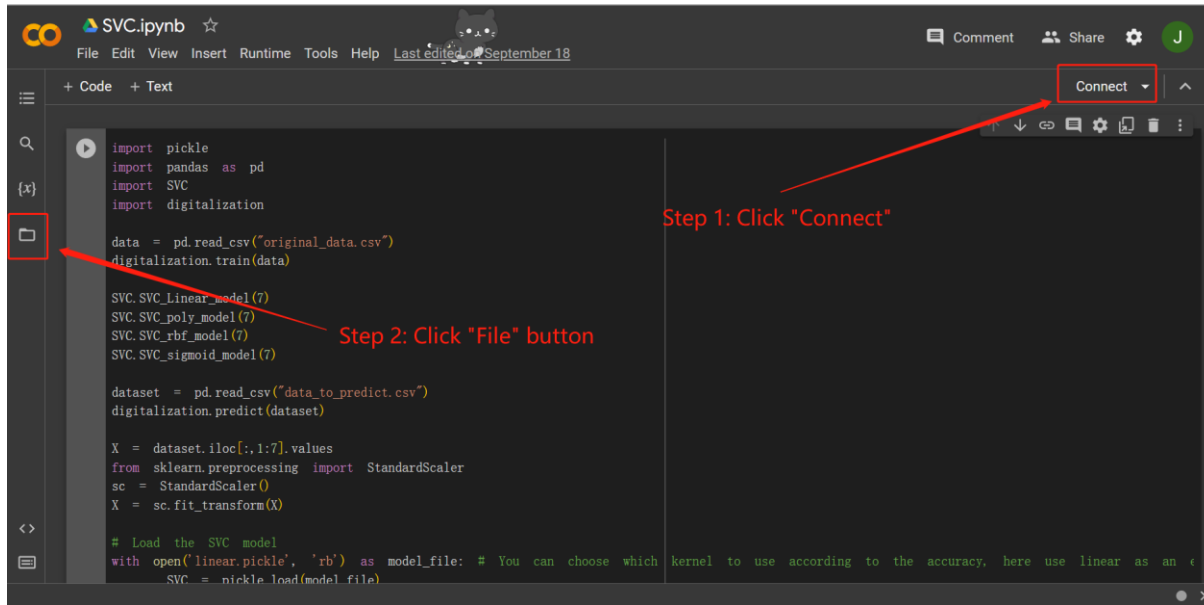
Upload complete

Select “SVC.ipynb”

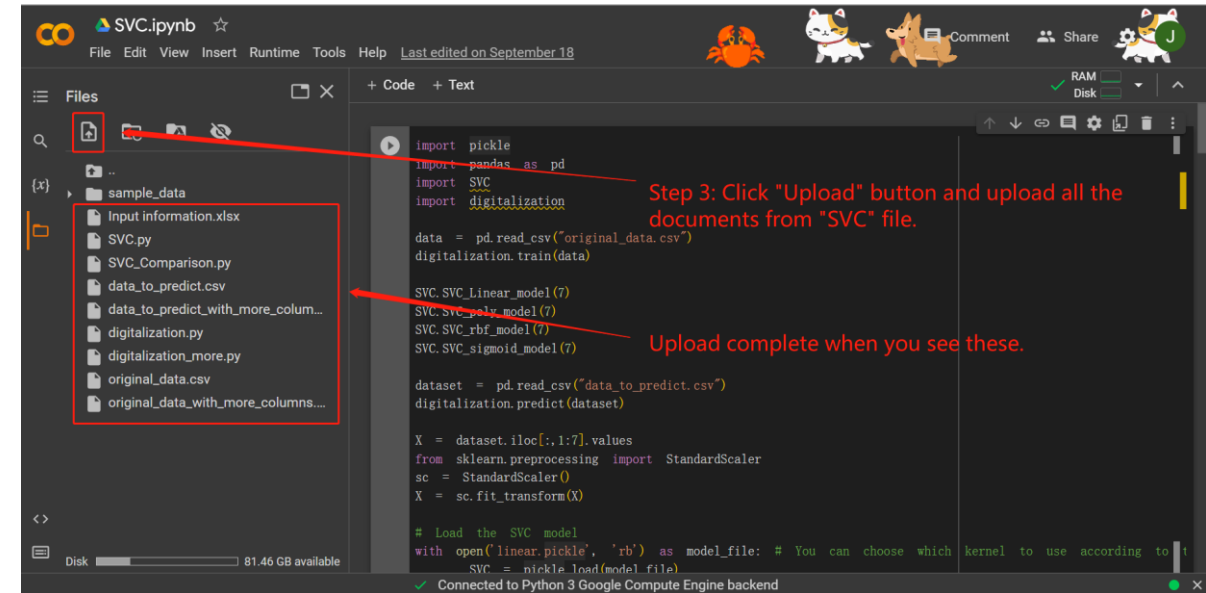
类型	大小
IPYNB 文件	15 KB
IPYNB 文件	137 KB
IPYNB 文件	84 KB
IPYNB 文件	82 KB

Preparations

- Connect the program to online processor.
- Upload all the documents from “SVC” file to the notebook.



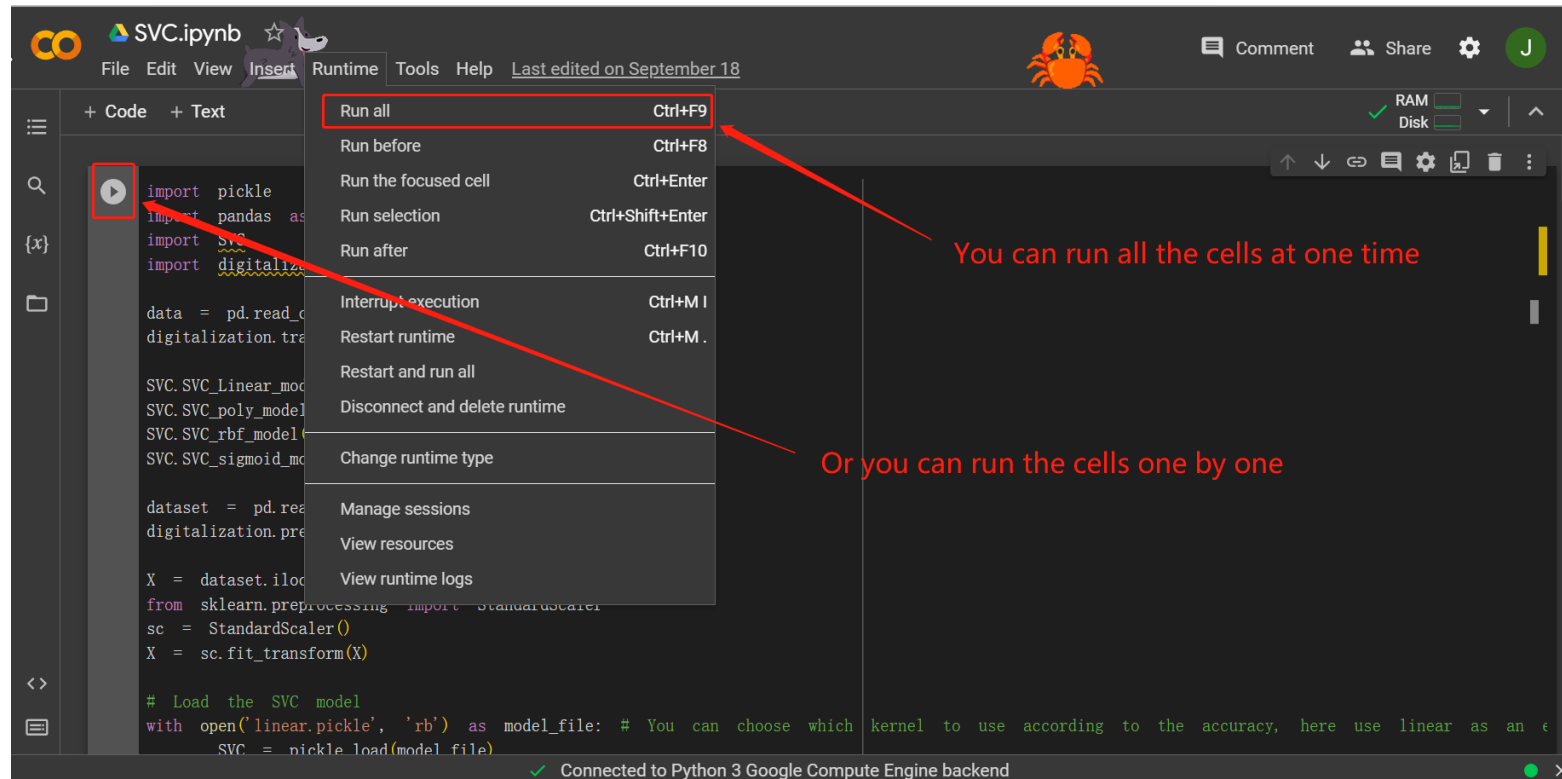
Connect the notebook



Upload our files

Run the program

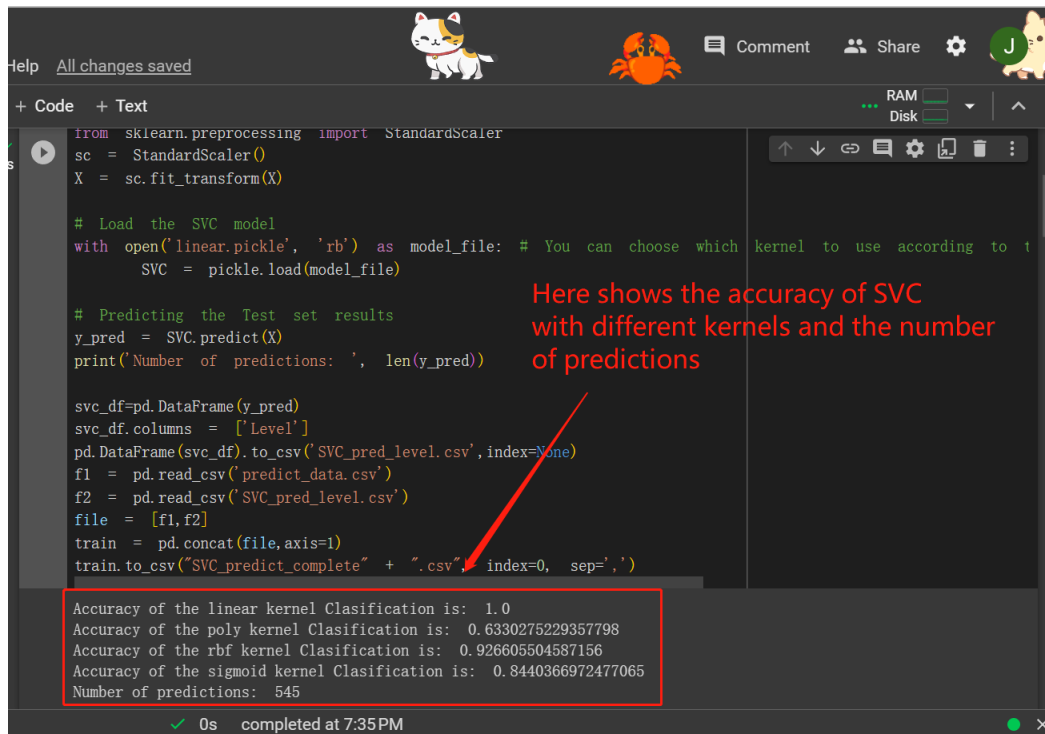
- Click “Run” button and run the cells **one by one** or you can run them all in one click. (One by one is recommended)



Run the SVC program

Check the result

- Click “Run” button and run the cells **one by one** or you can run them all in one click. (One by one is recommended)



```
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
X = sc.fit_transform(X)

# Load the SVC model
with open('linear.pickle', 'rb') as model_file: # You can choose which kernel to use according to t
    SVC = pickle.load(model_file)

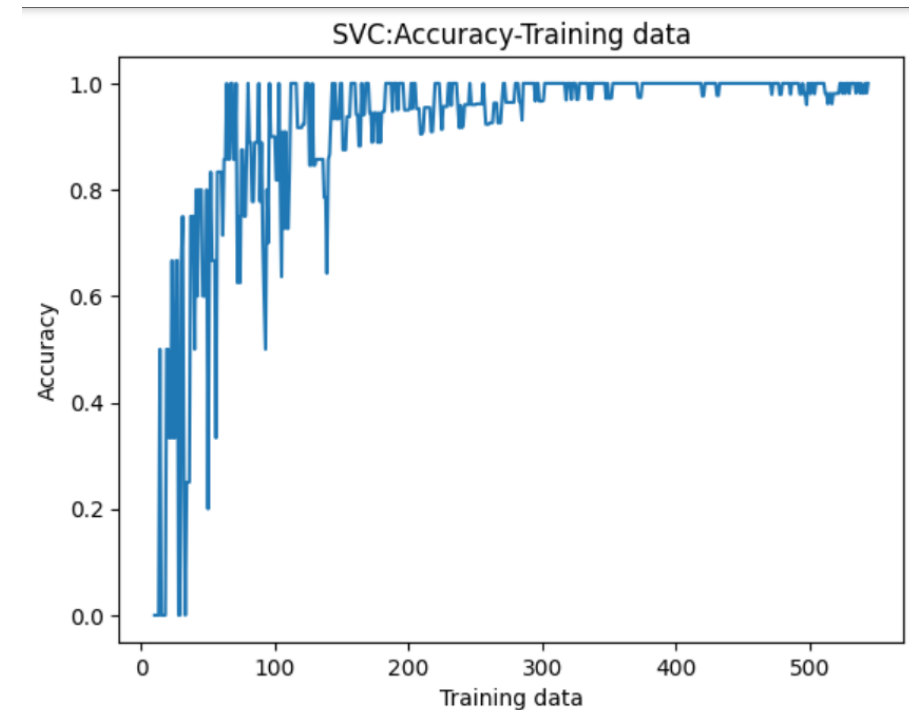
# Predicting the Test set results
y_pred = SVC.predict(X)
print('Number of predictions: ', len(y_pred))

svc_df=pd.DataFrame(y_pred)
svc_df.columns = ['Level']
pd.DataFrame(svc_df).to_csv('SVC_pred_level.csv',index=None)
f1 = pd.read_csv('predict_data.csv')
f2 = pd.read_csv('SVC_pred_level.csv')
file = [f1,f2]
train = pd.concat(file,axis=1)
train.to_csv("SVC_predict_complete" + ".csv", index=0, sep=',')
```

Accuracy of the linear kernel Clasification is: 1.0
Accuracy of the poly kernel Clasification is: 0.6330275229357798
Accuracy of the rbf kernel Clasification is: 0.926605504587156
Accuracy of the sigmoid kernel Clasification is: 0.8440366972477065
Number of predictions: 545

Here shows the accuracy of SVC with different kernels and the number of predictions

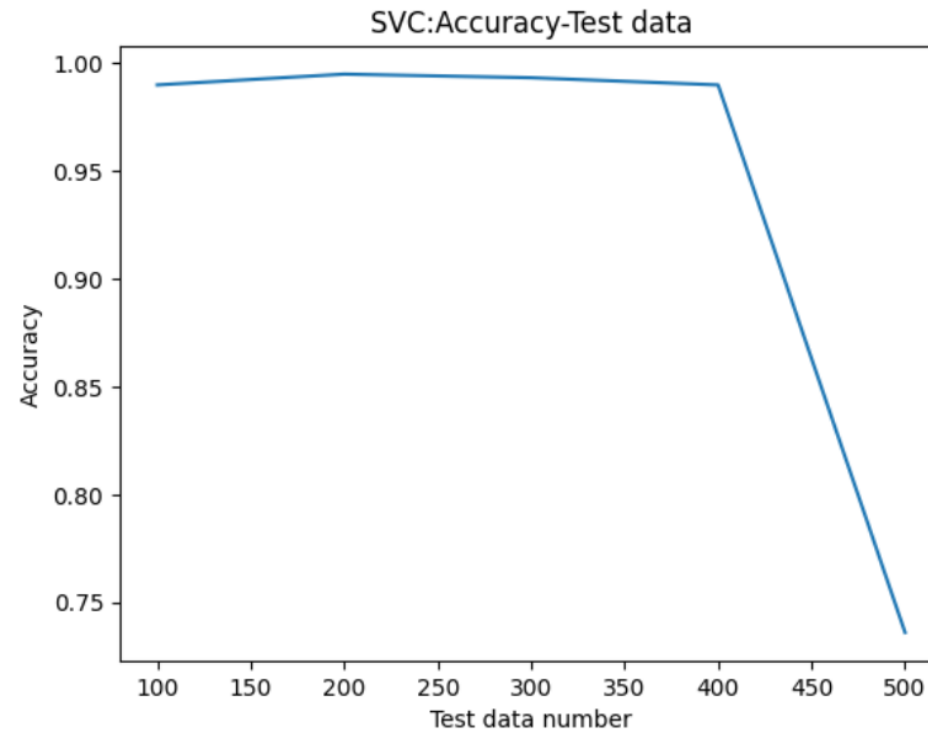
Output of first cell



Output of second cell

Check the result

- Different amount of test data versus accuracy.



Output of third cell

Check the output file

- The final output can be found on the left, named “**SVC_predict_complete.csv**”, after opening the file we can find that the predict level is on the rightest column.

The screenshot shows a Jupyter Notebook environment. On the left, a file explorer displays a list of files, with **SVC_predict_complete.csv** highlighted by a red box and a red arrow pointing to it. The main area contains a code cell with the following Python code:

```
import SVC_Comparison
import matplotlib.pyplot as plt

dataset = pd.read_csv('train_data.csv')
sample_number=len(dataset)
c = []
d = []
for r in range(100, sample_number, 100):
    c.append(r)
    d.append(SVC_Comparison.SVC_Linear_comparison(sample_number, r))
plt.plot(c, d)
plt.xlabel("Test data number")
plt.ylabel("Accuracy")
plt.title("SVC:Accuracy-Test data")
plt.show()
```

Below the code cell, a plot titled "SVC:Accuracy-Test data" is shown, displaying a line graph of accuracy versus test data number. The plot shows a high accuracy (around 1.00) for test data numbers up to approximately 400, followed by a sharp drop.

A red arrow points from the text "You can pick the final results in this file" to the file **SVC_predict_complete.csv** in the file explorer.

Outcome file

Select the best parameters (SVC)

- By **comparing** the predict data and the target, we can calculate the accuracy of the trained model.
- Also, for each algorithm, we can also find out in which set of **parameters** the model can reach the highest accuracy.

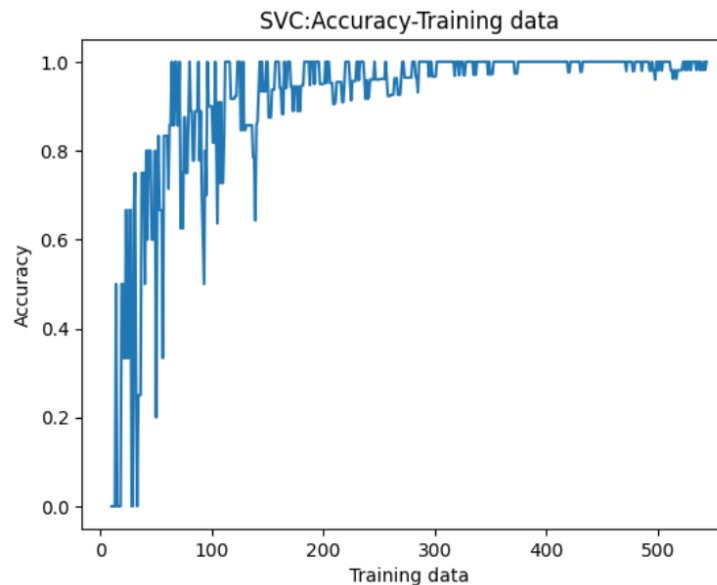
```
# Load the SVC model
with open('linear.pickle', 'rb') as model_file: # You can choose which kernel to use according to the data
    SVC = pickle.load(model_file)

# Predicting the Test set results
y_pred = SVC.predict(X)
print('Number of predictions: ', len(y_pred))

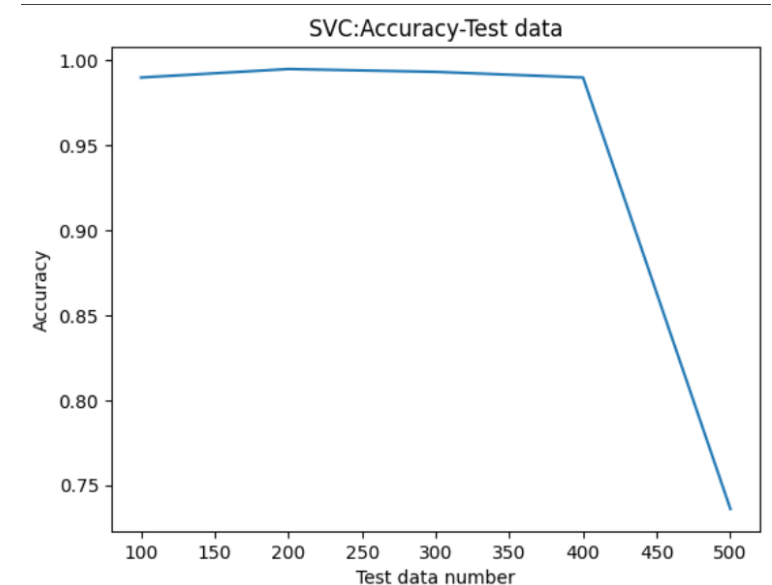
svc_df=pd.DataFrame(y_pred)
svc_df.columns = ['level']
pd.DataFrame(svc_df).to_csv('SVC_pred_level.csv', index=None)
f1 = pd.read_csv('predict_data.csv')
f2 = pd.read_csv('SVC_pred_level.csv')
file = [f1, f2]
train = pd.concat(file, axis=1)
train.to_csv('SVC_predict_complete' + '.csv', index=0, sep=',')
```

Accuracy of the linear kernel Classification is: 1.0
Accuracy of the poly kernel Classification is: 0.6330275229357798
Accuracy of the rbf kernel Classification is: 0.926605504587156
Accuracy of the sigmoid kernel Classification is: 0.8440366972477065
Number of predictions: 545

Linear kernel reaches
highest accuracy of 100%



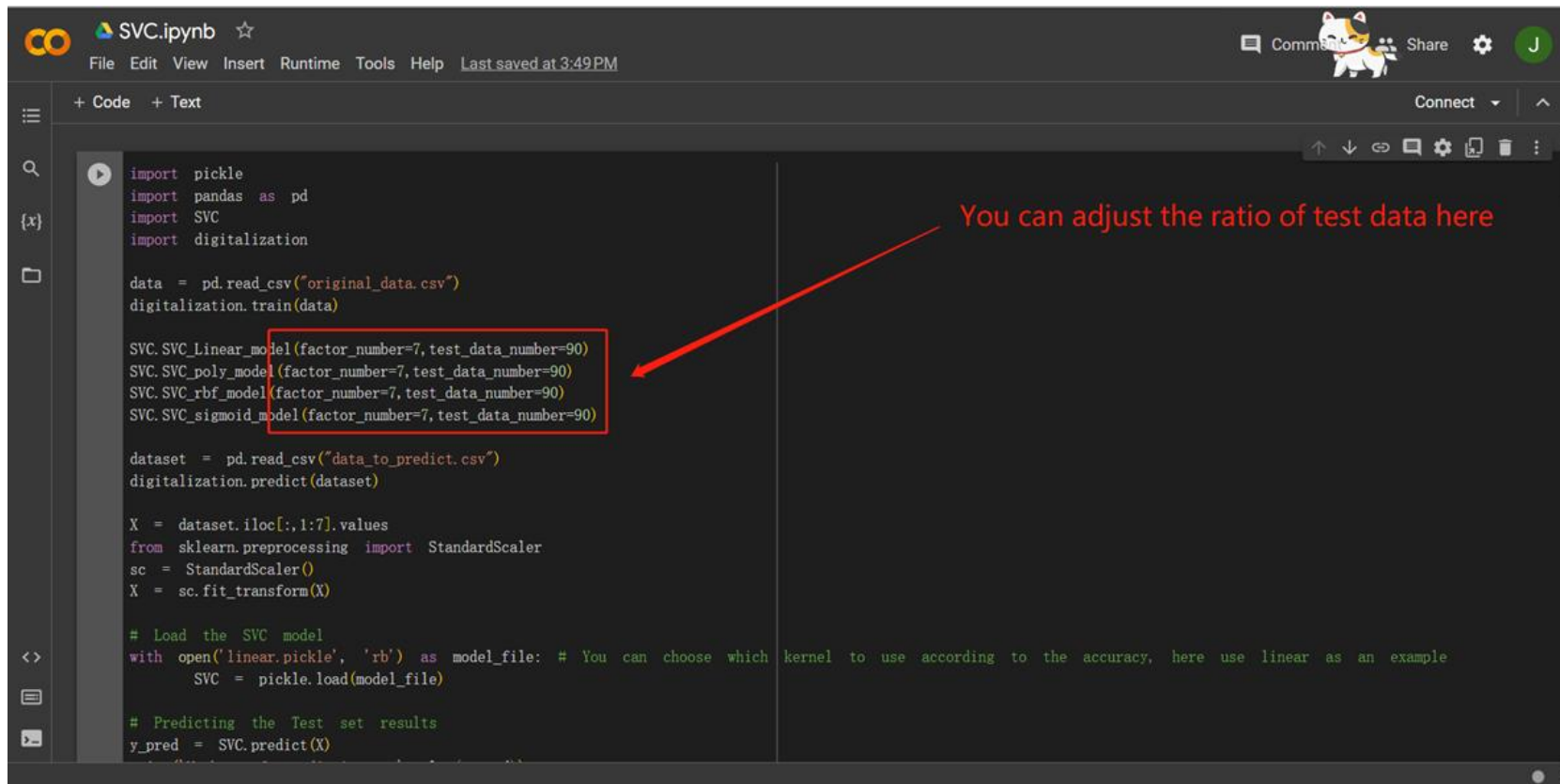
Generally, more training
data represents higher
accuracy



Generally, less test data
represents higher accuracy

Select the best parameters (SVC)

- After comparison, you can adjust the **ratio of test data** like in the picture below. (Since training data is generally the more the better so there's no need to change)



```
import pickle
import pandas as pd
import SVC
import digitalization

data = pd.read_csv("original_data.csv")
digitalization.train(data)

SVC.SVC_linear_model(factor_number=7, test_data_number=90)
SVC.SVC_poly_model(factor_number=7, test_data_number=90)
SVC.SVC_rbf_model(factor_number=7, test_data_number=90)
SVC.SVC_sigmoid_model(factor_number=7, test_data_number=90)

dataset = pd.read_csv("data_to_predict.csv")
digitalization.predict(dataset)

X = dataset.iloc[:, 1:7].values
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
X = sc.fit_transform(X)

# Load the SVC model
with open('linear.pickle', 'rb') as model_file: # You can choose which kernel to use according to the accuracy, here use linear as an example
    SVC = pickle.load(model_file)

# Predicting the Test set results
y_pred = SVC.predict(X)
```

You can adjust the parameters here based on the comparison

Want to try your own data?

■ Sure! Only need to follow my steps:

1. First, open the “[Input information.xlsx](#)” from the “SVC” file we have already downloaded, fill in your personal information. Notice that in “Exam” and “Math_Background” column, only Gaokao and HKDSE are accepted.

	A	B	C	D	E	F	G	H
1	Student_ID	Admission_Channel	Secondary_School	Exam	Math_Background	MATH1510	ENGG1130	ENGG2020
2	1	JUPAS	CCC HEEP WOH COLLEGE	Mainland Gaokao (JEE)	Mainland Gaokao (JEE):>140	A	A	
3	2	Non-JUPAS (Local)	SUEN WAN PUB HO CHUEN YIU MEMORIAL COL	HKDSE	HKDSE:5*	B+	C	
4								
5								
6			SHUN LEE CATHOLIC SECONDARY SCHOOL					
7			STFA LEE SHAU KEE COLLEGE					
8			STFA LEUNG KAU KUI COLLEGE					
9			STFA YUNG YAU COLLEGE					
10			STFA SEAWARD WOO COLLEGE					
11			STFA TAM PAK YU COLLEGE					
12			STFA CHENG YU TUNG SECONDARY SCHOOL					
13			ELEGANTIA COLLEGE (SPONSORED BY ED CONV)					
14			HEUNG TO MIDDLE SCHOOL					
15			HHCKLA BUDDHIST CHING KOK SEC SCH					
16			HONG KONG CHINESE WOMEN'S CLUB COLLEGE					
17			CHRISTIAN ALLIANCE CHENG WING GEE COLL					
18								
19								
20								
21								
22								
23								
24								
25								
26								
27								
28								
29								

Input your information

Want to try your own data?

2. Upload the “Input information.xlsx” to program like before and run the fourth cell.
3. Then pick up the “Personal_SVC_predict_complete.csv” file in the left to check the prediction of your grade!

The screenshot shows a Jupyter Notebook interface with the following components:

- Files Panel (Left):** A list of files including 'sample_data', 'Input information.xlsx', 'More_SVC_predict_complete.csv', 'Personal_Information.csv', 'Personal_SVC_predict_complet...', 'SVC.py', 'SVC_Comparison.py', 'SVC_pred_level.csv', 'SVC_predict_complete.csv', 'data_to_predict.csv', 'data_to_predict_with_more_colu...', 'digitalization.py', 'digitalization_more.py', 'linear.pickle', 'original_data.csv', 'original_data_with_more_colum...', 'poly.pickle', 'predict_data.csv', and 'rbf.pickle'. The file 'Personal_SVC_predict_complet...' is highlighted with a red box.
- Code Panel (Center):** A code cell with the following Python code:

```
student_number = int(input('Please input the number of students: '))
X = dataset.iloc[:student_number,1:7].values
# print(X)

from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
X = sc.fit_transform(X)
# Load the SVC model
with open('linear.pickle', 'rb') as model_file:
    SVC = pickle.load(model_file)

# Predicting the Test set results
y_pred = SVC.predict(X)
print('Number of predictions: ', len(y_pred))
print('Predict Level: ', y_pred)
svc_df=pd.DataFrame(y_pred)
svc_df.columns = ['Level']
pd.DataFrame(svc_df).to_csv('SVC_pred_level.csv', index=None)
f1 = pd.read_csv('predict_data.csv')
f2 = pd.read_csv('SVC_pred_level.csv')
file = [f1,f2]
train = pd.concat(file,axis=1)
train.to_csv("Personal_SVC_predict_complete" + ".csv", index=0, sep=',')
```

Below the code, the output is displayed:

```
Please input the number of students: 2
Number of predictions: 2
Predict Level: ['A' 'C']
```
- Annotations:** Red arrows point to specific parts of the interface:
 - One arrow points to the 'Personal_SVC_predict_complet...' file in the Files panel, with the text "You can check the result in this file."
 - Another arrow points to the model loading code block, with the text "You can choose to load the kernel with the highest accuracy."
 - A third arrow points to the output text, with the text "Or you can just see the predictions below".

Run the fourth cell.

Two ways to see the predictions

Want to add some extra factors?

■ Sure! Only need to follow my steps:

1. We can take a look at “original_data_with_more_columns.csv” in “SVC” file.

A1

Original No.

We add two more courses as factors for example

	A	B	C	D	E	F	G	H	I	J
1	Original No.	Admission_Channel	Secondary_School	Exam	Math_Background	MATH1510	ENGG1130	MATH1010	ENGG1120	ENGG2020
2	17	JUPAS	TSUEN WAN PUB HO CHUEN YIU MEMORIAL COLL	HKDSE	HKDSE:5	B+	B+	A-	B	B+
3	20	JUPAS	CHRISTIAN ALLIANCE S C CHAN MEM COLL	HKDSE	HKDSE:5*	A-	B	B	B+	B
4	25	JUPAS	MUNSANG COLLEGE (HONG KONG ISLAND)	HKDSE	HKDSE:4	B	A-	A-	B+	A
5	29	JUPAS	PO LEUNG KUK LAWS FOUNDATION COLLEGE	HKDSE	HKDSE:4	B+	B-	B+	A-	B-
6	36	JUPAS	STFA LEUNG KAU KUI COLLEGE	HKDSE	HKDSE:5	B-	B+	B+	B	B+
7	37	JUPAS	METHODIST CHURCH HK WESLEY COLLEGE	HKDSE	HKDSE:5	D	D+	B+	B+	D
8	50	JUPAS	Others	HKDSE	HKDSE:5	B-	C+	D	D+	C
9	55	JUPAS	KWOK TAK SENG CATHOLIC SECONDARY SCHOOL	HKDSE	HKDSE:5	B	B	B	C+	B
10	62	JUPAS	YWCA HIOE TJO YOENG COLLEGE	HKDSE	HKDSE:4	B	B+	B-	B	B+
11	82	JUPAS	CCC CHUEN YUEN COLLEGE	HKDSE	HKDSE:5	B	F	B	B+	F
12	88	JUPAS	HKMA K S LO COLLEGE	HKDSE	HKDSE:4	C	C+	C-	F	C
13	92	JUPAS	LIONS COLLEGE	HKDSE	HKDSE:4	B	C-	C+	C+	C
14	94	JUPAS	YCH LIM POR YEN SECONDARY SCHOOL	HKDSE	HKDSE:5**	B+	A-	C-	C-	A
15	98	JUPAS	HK SKH BISHOP HALL SECONDARY SCHOOL	HKDSE	HKDSE:5**	B+	B+	A-	A-	B+
16	101	JUPAS	CHRISTIAN ALLIANCE S W CHAN MEM COLL	HKDSE	HKDSE:5	B-	B-	B+	B+	B-
17	109	JUPAS	VALTORTA COLLEGE	HKDSE	HKDSE:5**	A-	A	B-	B+	A
18	119	JUPAS	AOG HEBRON SECONDARY SCHOOL	HKDSE	HKDSE:5	B-	A-	A	A	A
19	129	JUPAS	PUI CHING MIDDLE SCHOOL	HKDSE	HKDSE:5**	A-	A	A-	A-	A
20	130	JUPAS	TWGH LI KA SHING COLLEGE	HKDSE	HKDSE:5	C+	C-	A	A	C
21	142	JUPAS	TANG KING PO SCHOOL	HKDSE	HKDSE:4	B+	B-	C-	C+	B-
22	144	JUPAS	PLK MRS MA KAM MING CHEUNG FOOK SIEN COL	HKDSE	HKDSE:4	B+	B-	B-	B-	B-
23	147	JUPAS Appeal-in	STFA YUNG YAU COLLEGE	HKDSE	HKDSE:5	B-	C	B-	B-	C
24	168	JUPAS	FUKIEN SECONDARY SCHOOL	HKDSE	HKDSE:5*	A-	B+	C	C	B+
25	172	JUPAS	SKH LAM WOO MEMORIAL SECONDARY SCHOOL	HKDSE	HKDSE:4	B	C+	B+	B+	C

You can replace the content as you like

Want to add some extra factors?

2. After **training** the model with the dataset in previous slide.
3. You can make a csv document to **contain the students' information** you want to predict their level.

A1 Student_ID

After running the cell, the level of these students will be added

	A	B	C	D	E	F	G	H	I	J
1	Student_ID	Admission_Channel	Secondary_School	Exam	Math_Background	MATH1510	ENGG1130	MATH1010	ENGG1120	ENGG2020
2	627	JUPAS	CHAN SUI KI (LA SALLE) COLLEGE	HKDSE	HKDSE:5*	A-	A-	B-	B-	
3	882	JUPAS	CHRISTIAN ALLIANCE CHENG WING GEE COLL	HKDSE	HKDSE:4	D	F	B	B+	
4	801	JUPAS	MA ON SHAN TSUNG TSIN SECONDARY SCHOOL	HKDSE	HKDSE:5*	B-	B	C	C+	
5	1087	Mainland Fee-paying	Others	Mainland	(Mainland Gaokao (JEE):<110	B-	B	B-	C+	
6	224	JUPAS	ELEGANTIA COLLEGE (SPONSORED BY ED CONV)	HKDSE	HKDSE:5*	B-	F	B-	B-	
7	240	JUPAS	SKH BISHOP MOK SAU TSENG SEC SCH	HKDSE	HKDSE:5*	A-	A-	B-	B+	
8	911	JUPAS	Others	HKDSE	HKDSE:5**	A	A	B-	B+	
9	917	JUPAS	DIOCESAN BOYS SCHOOL	HKDSE	HKDSE:4	D	B-	B	B+	
10	624	Mainland	Others	Mainland	(Mainland Gaokao (JEE):120-129	A-	A	B	A	
11	724	JUPAS	SING YIN SECONDARY SCHOOL	HKDSE	HKDSE:4	B	B	B-	B+	
12	546	JUPAS	KWUN TONG MARYKNOLL COLLEGE	HKDSE	HKDSE:5*	A	A	B	B+	
13	410	Mainland	Others	Mainland	(Mainland Gaokao (JEE):130-139	B+	B	A-	B	
14	492	JUPAS	Others	HKDSE	HKDSE:5	B-	A-	B	B-	
15	986	JUPAS	Others	HKDSE	HKDSE:4	C+	B+	A-	B+	
16	1046	JUPAS	HELEN LIANG MEMORIAL SEC SCH (SHATIN)	HKDSE	HKDSE:5	A-	B+	B	B	
17	272	JUPAS	HEUNG TO MIDDLE SCHOOL	HKDSE	HKDSE:4	B-	C-	A-	A-	
18	803	JUPAS	Others	HKDSE	HKDSE:4	B-	B+	B	B-	
19	383	JUPAS	LAI KING CATHOLIC SECONDARY SCHOOL	HKDSE	HKDSE:4	C-	B	B-	B	
20	881	JUPAS	YAN OI TONG TIN KA PING SECONDARY SCHOOL	HKDSE	HKDSE:4	B-	C+	B+	B	
21	932	JUPAS	SKH TANG SHIU KIN SECONDARY SCHOOL	HKDSE	HKDSE:4	B	B+	B-	A-	
22	120	Mainland	Others	Mainland	(Mainland Gaokao (JEE):120-129	A	A	A-	A	
23	732	JUPAS	HKSYC&IA WONG TAI SHAN MEM COLL	HKDSE	HKDSE:4	C	B	B+	B+	
24	474	JUPAS	SING YIN SECONDARY SCHOOL	HKDSE	HKDSE:4	B	A	B-	C-	
25	776	JUPAS	STFA LEUNG KAU KUI COLLEGE	HKDSE	HKDSE:5*	A	A	B-	B-	

The level of students will be added automatically

Want to add some extra factors?

4. Then give the file to our model.

Files

- sample_data
- Input information.xlsx
- More_SVC_predict_complete.csv
- Personal_Information.csv
- Personal_SVC_predict_complet...
- SVC.py
- SVC_Comparison.py
- SVC_pred_level.csv
- SVC_predict_complete.csv
- data_to_predict.csv
- data_to_predict_with_more_colu...
- digitalization.py
- digitalization_more.py
- linear.pickle
- original_data.csv
- original_data_with_more_colum...
- poly.pickle
- predict_data.csv
- rbf.pickle

Code

```
[5] SVC.SVC_Linear_model(8)

dataset = pd.read_csv("data_to_predict_with_more_columns.csv")
digitalization_more.predict(dataset)

X = dataset.iloc[:,1:8].values
# print(X)
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
X = sc.fit_transform(X)
# Load the SVC model
with open('linear.pickle', 'rb') as model_file:
    SVC = pickle.load(model_file)

# Predicting the Test set results
y_pred = SVC.predict(X)
print('Number of predictions: ', len(y_pred))
svc_df=pd.DataFrame(y_pred)
svc_df.columns = ['Level']
pd.DataFrame(svc_df).to_csv('SVC_pred_level.csv', index=False)
f1 = pd.read_csv('predict_data.csv')
f2 = pd.read_csv('SVC_pred_level.csv')
file = [f1,f2]
train = pd.concat(file,axis=1)
train.to_csv("More_SVC_predict_complete" + ".csv", index=0, sep=',')
```

Accuracy of the linear kernel Clasification is: 0.9908256880733946
Number of predictions: 545

Step 1: Upload two csv files to the program.

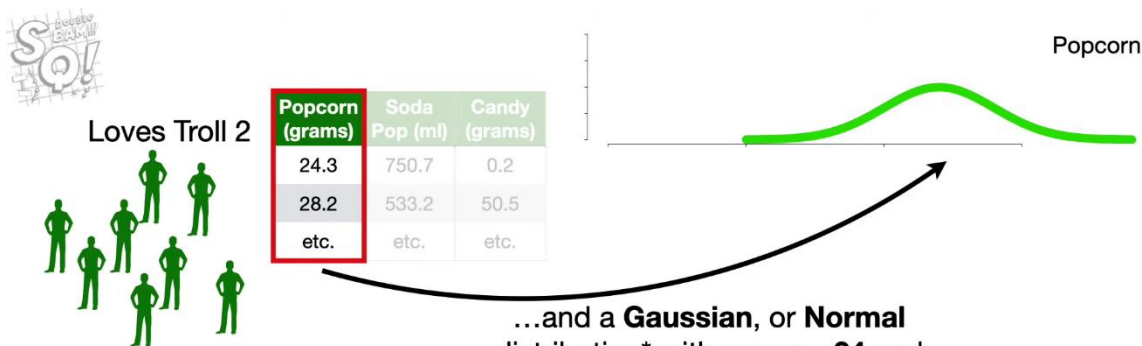
Step 2: You can check the accuracy of the predictions.

Step 3: You can pick the final results in this csv file after running the cell.

Run the fifth cell and get your results

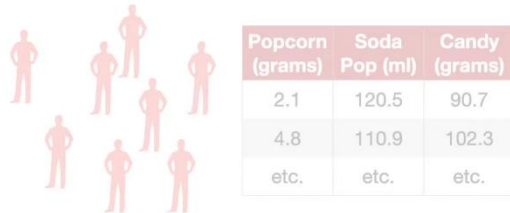
Gaussian Naïve Bayes Classification

- To simplify the explanation, we assume that there are two groups of people, one of them loves “Troll 2”, another group does not, and we collect some of their habits in eat and drink.

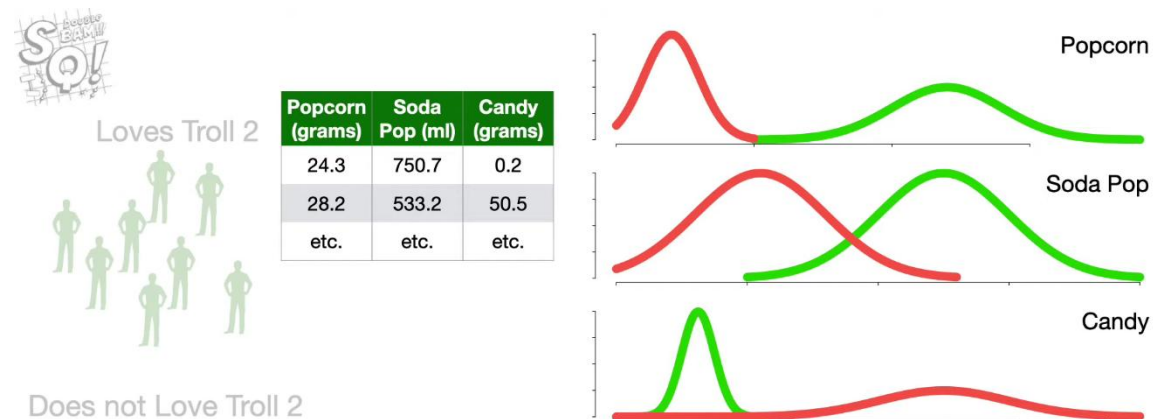


...and a **Gaussian**, or **Normal** distribution* with **mean = 24** and **standard deviation = 4** looks like this.

Does not Love Troll 2



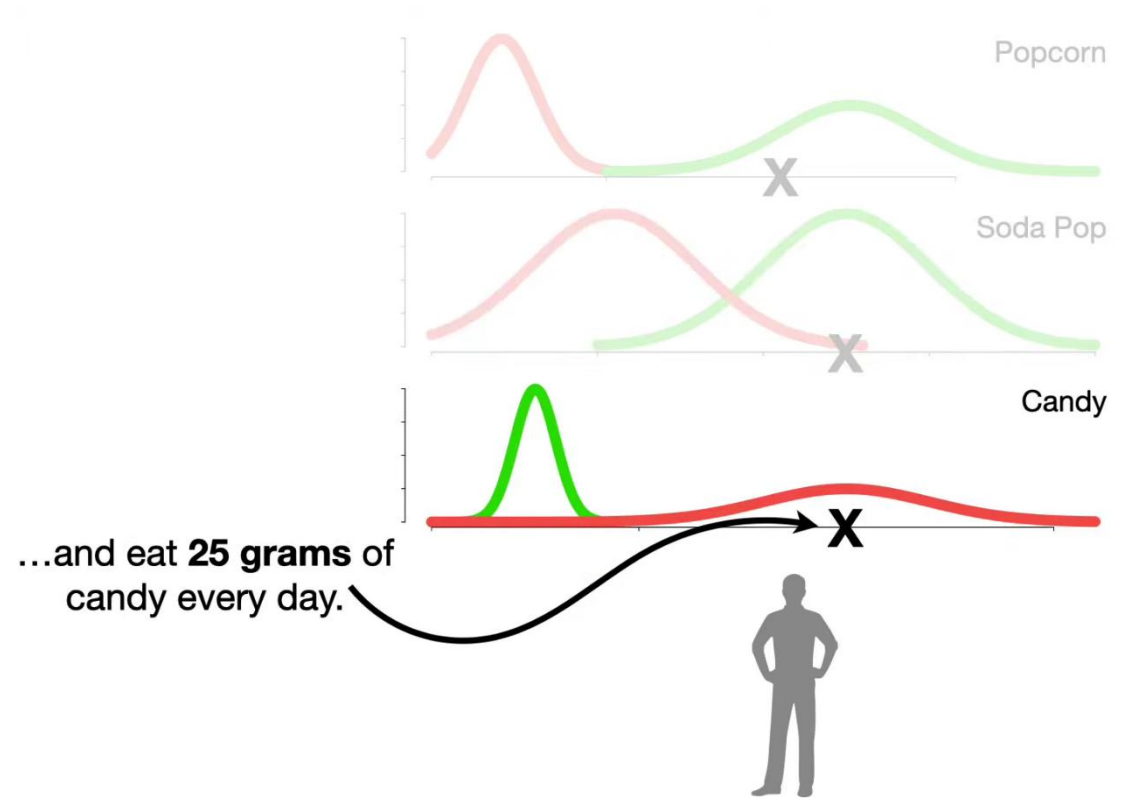
*Gaussian Distributions are the same as Normal Distributions.



Gaussian Naive Bayes is named after the **Gaussian** distributions that represent the data in the **Training Dataset**.

Gaussian Naïve Bayes Classification

- Now comes a new sample who eat 20 grams of popcorn, 25 grams of candy and drink 500 ml of soda pop everyday.



Gaussian Naïve Bayes Classification

- We find the values in each cluster and get the corresponding likelihood. Multiply X axis value (amount of popcorn, pop soda or candy) by Y axis value (Likelihood) and add the three results together we can get the total score.
- The sample will be classified to the group with higher score.

Now do it by yourself!



$$p(\text{Loves Troll 2}) = 0.5$$

Loves Troll 2 Score =

...times the
Likelihood that
eat **25 grams**
candy, given tha
Love Troll 2

$$\log(\text{Loves Troll 2 Score}) = -124$$



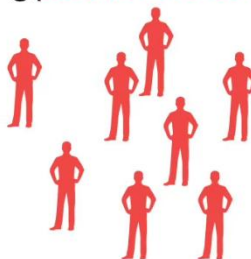
$p(\text{Loves})$

$$\times L(\text{popcorn} = 20 \mid \text{Loves})$$

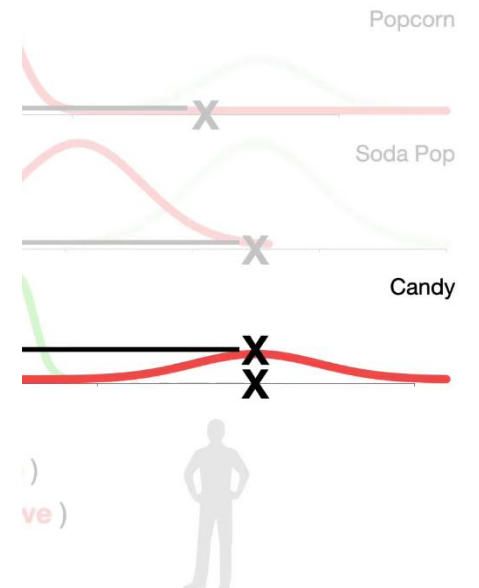
$$\times L(\text{soda pop} = 500 \mid \text{Love})$$

$$\times L(\text{candy} = 25 \mid \text{Loves})$$

$$\log(\text{Does not Love Troll 2 Score}) = -47.66$$

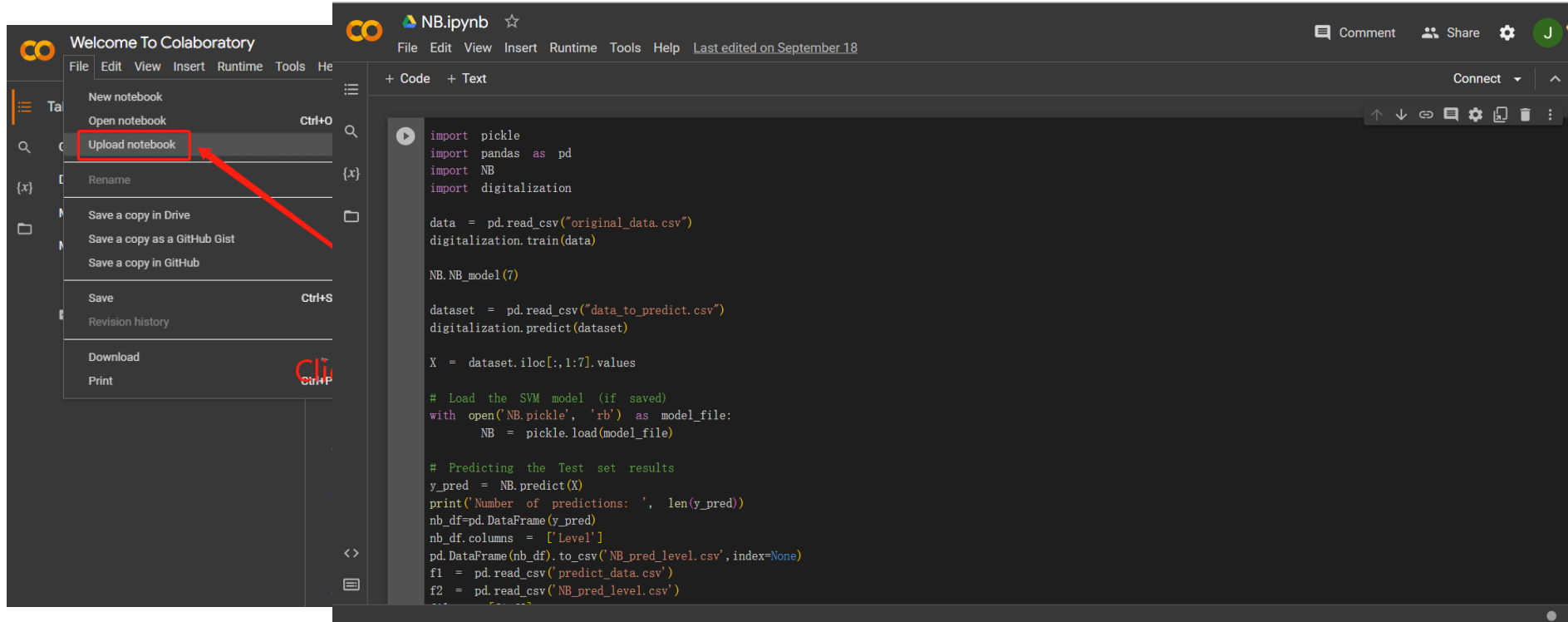


We will classify this person as someone who
Does Not Love Troll 2.



Preparations

- Upload the “NB.ipynb” from the file “Main_code”.



类型	大小
IPYNB 文件	15 KB
IPYNB 文件	137 KB
IPYNB 文件	84 KB
IPYNB 文件	82 KB

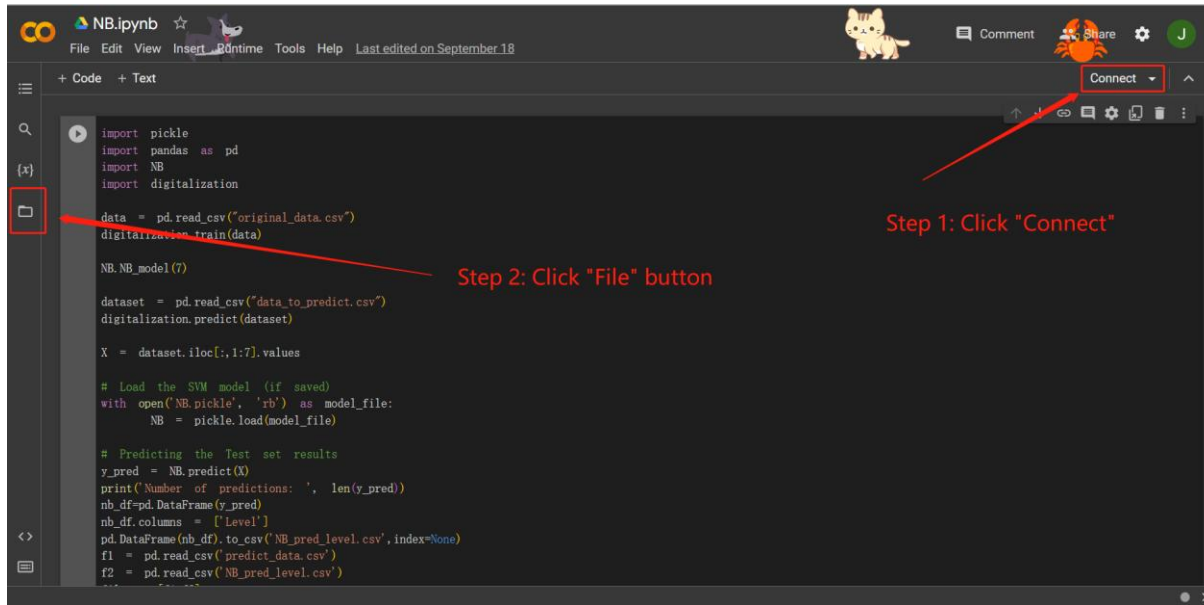
Upload the program

Upload complete

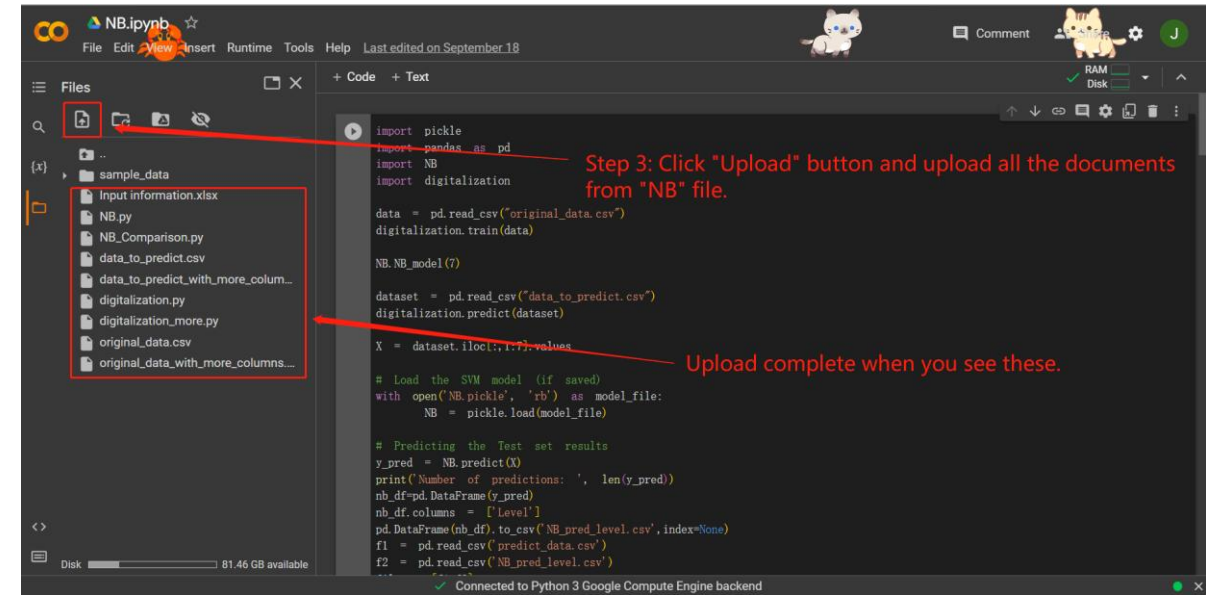
Select “NB.ipynb”

Preparations

- Connect the program to online processor.
- Upload all the documents from “NB” file to the notebook.



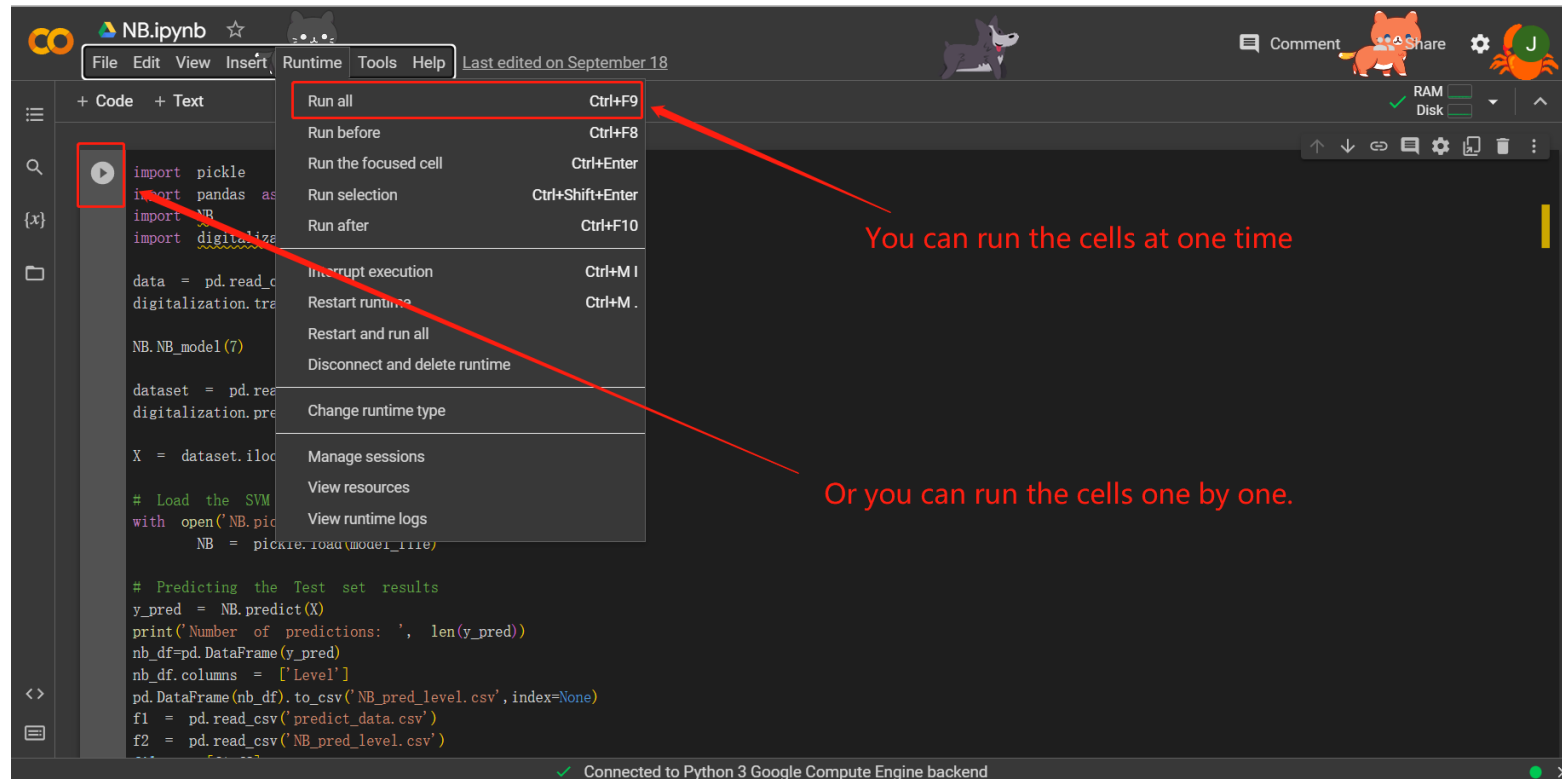
Connect the notebook



Upload our files

Run the program

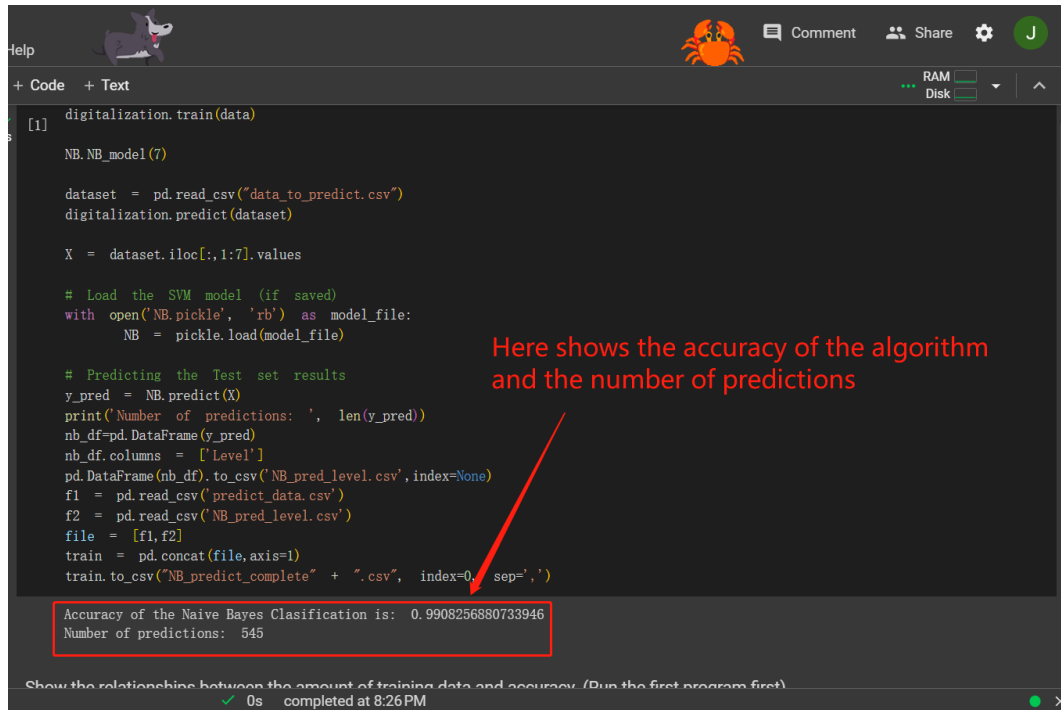
- Click “Run” button and run the cells **one by one** or you can run them all in one click. (One by one is recommended)



Run the NB program

Check the result

- Click “Run” button and run the cells **one by one** or you can run them all in one click. (One by one is recommended)



The screenshot shows a Jupyter Notebook cell with the following code:

```
[1] digitalization.train(data)
    NB.NB_model(7)

    dataset = pd.read_csv("data_to_predict.csv")
    digitalization.predict(dataset)

    X = dataset.iloc[:, 1:7].values

    # Load the SVM model (if saved)
    with open('NB.pickle', 'rb') as model_file:
        NB = pickle.load(model_file)

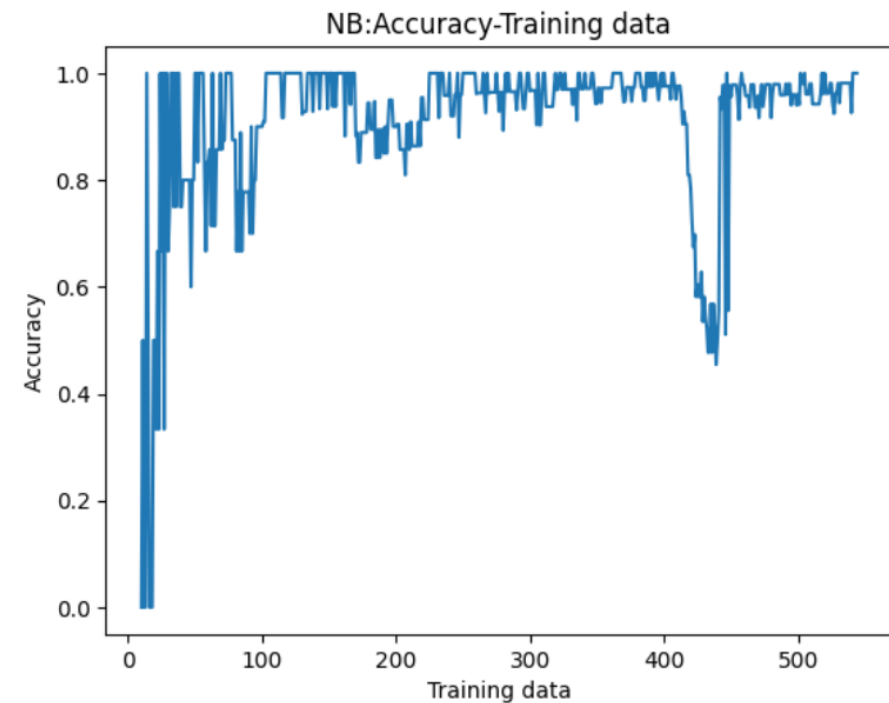
    # Predicting the Test set results
    y_pred = NB.predict(X)
    print('Number of predictions: ', len(y_pred))
    nb_df=pd.DataFrame(y_pred)
    nb_df.columns = ['Level']
    pd.DataFrame(nb_df).to_csv('NB_pred_level.csv', index=None)
    f1 = pd.read_csv('predict_data.csv')
    f2 = pd.read_csv('NB_pred_level.csv')
    file = [f1, f2]
    train = pd.concat(file,axis=1)
    train.to_csv("NB_predict_complete" + ".csv", index=0, sep=',')
```

The output of the cell is:

```
Accuracy of the Naive Bayes Clasification is: 0.9908256880733946
Number of predictions: 545
```

A red arrow points from the text "Here shows the accuracy of the algorithm and the number of predictions" to the output text.

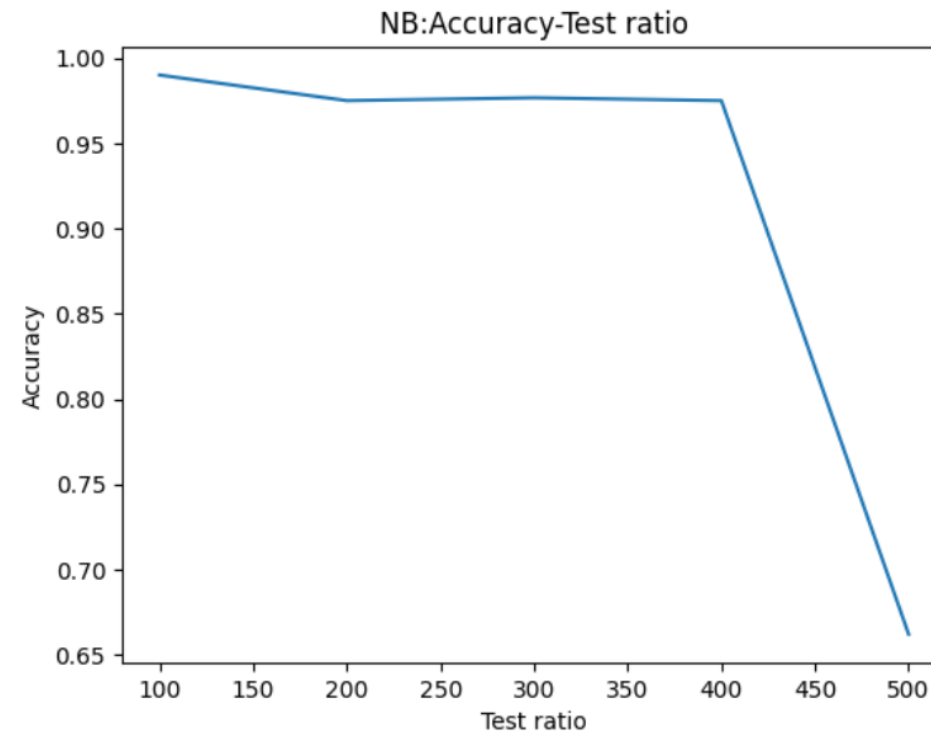
Output of first cell



Output of second cell

Check the result

- Different amount of test data versus accuracy.



Output of third cell

Check the output file

- The final output can be found on the left, named “NB_predict_complete.csv”, after opening the file we can find that the predict level is on the rightest column.

The screenshot shows a Jupyter Notebook environment. On the left, the 'Files' panel lists various files, with 'NB_predict_complete.csv' highlighted by a red box. A red arrow points from the text 'You can pick the final results in this file' to this file. The central code cell contains Python code for loading data, training a model, and plotting accuracy against the test ratio. The top plot shows 'Training data' with a single bar at 0.0. The bottom plot is titled 'NB:Accuracy-Test ratio' and shows a horizontal bar at 1.00. The status bar at the bottom indicates the notebook is completed at 8:26 PM.

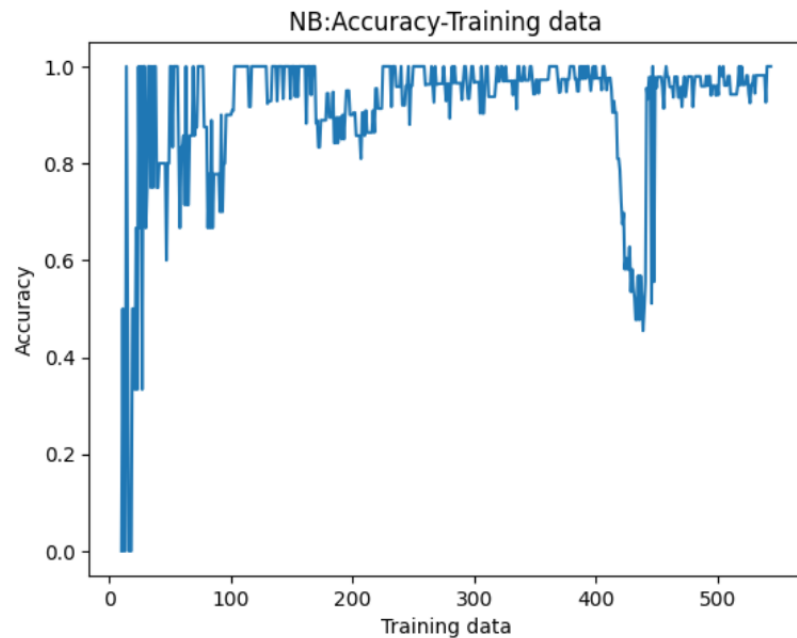
```
[3] import NB_Comparison
import matplotlib.pyplot as plt

dataset = pd.read_csv('train_data.csv')
sample_number=len(dataset)
c = []
d = []
for r in range(100, sample_number, 100):
    c.append(r)
    d.append(NB_Comparison.NB_comparison(sample_number, r))
plt.plot(c, d)
plt.xlabel("Test ratio")
plt.ylabel("Accuracy")
plt.title("NB:Accuracy-Test ratio")
plt.show()
```

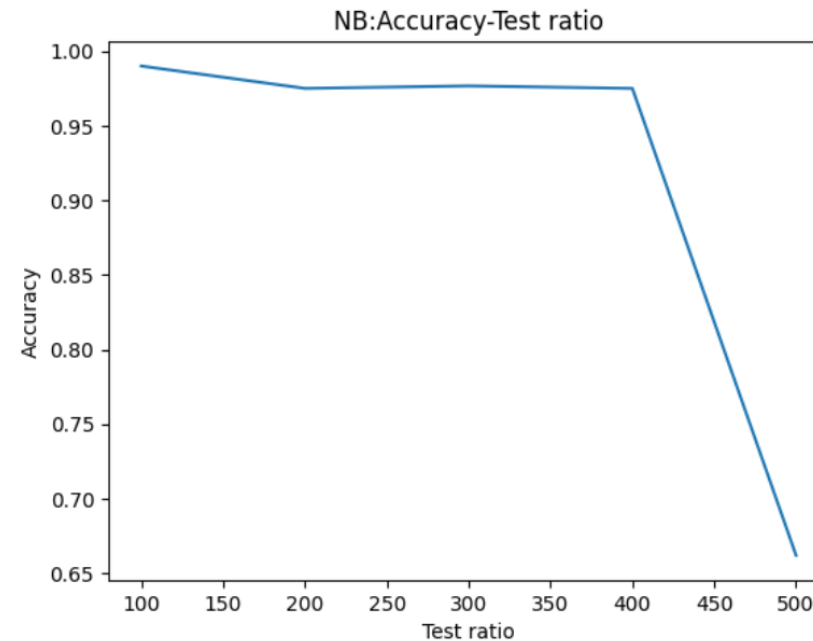
Outcome file

Select the best parameters (NB)

- By **comparing** the predict data and the target, we can calculate the accuracy of the trained model.
- Also, for each algorithm, we can also find out in which set of parameters the model can reach the highest accuracy.



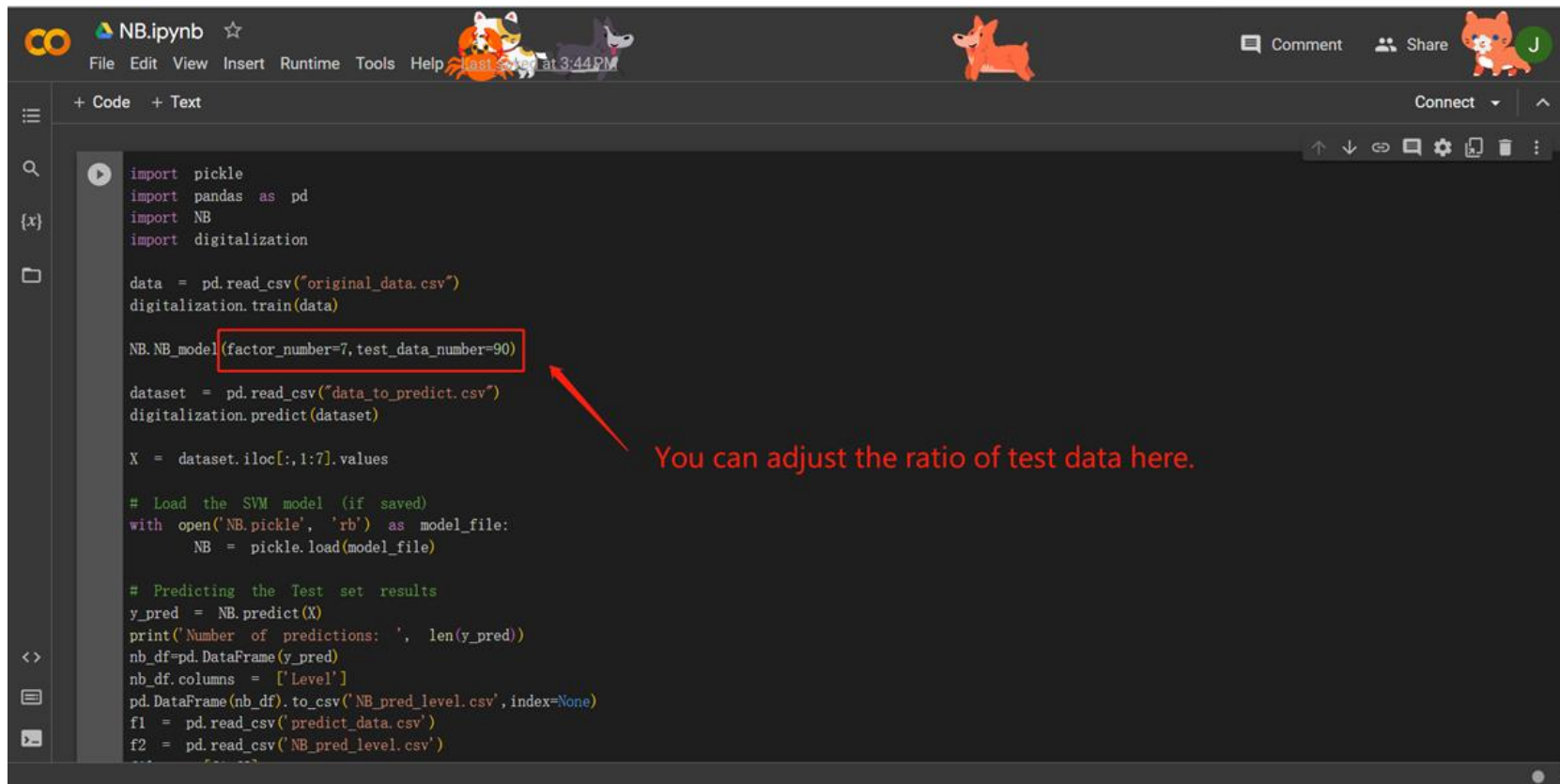
Generally, more training data represents higher accuracy



Generally, less test data represents higher accuracy

Select the best parameters (NB)

- After comparison, you can adjust the **ratio of test data** like in the picture below. (Since training data is generally the more the better so there's no need to change)



```
import pickle
import pandas as pd
import NB
import digitalization

data = pd.read_csv("original_data.csv")
digitalization.train(data)

NB.NB_model(factor_number=7, test_data_number=90)

dataset = pd.read_csv("data_to_predict.csv")
digitalization.predict(dataset)

X = dataset.iloc[:, 1:7].values

# Load the SVM model (if saved)
with open('NB.pickle', 'rb') as model_file:
    NB = pickle.load(model_file)

# Predicting the Test set results
y_pred = NB.predict(X)
print('Number of predictions: ', len(y_pred))
nb_df=pd.DataFrame(y_pred)
nb_df.columns = ['Level']
pd.DataFrame(nb_df).to_csv('NB_pred_level.csv', index=None)
f1 = pd.read_csv('predict_data.csv')
f2 = pd.read_csv('NB_pred_level.csv')
```

You can adjust the parameters here based on the comparison

Want to try your own data?

■ Sure! Only need to follow my steps:

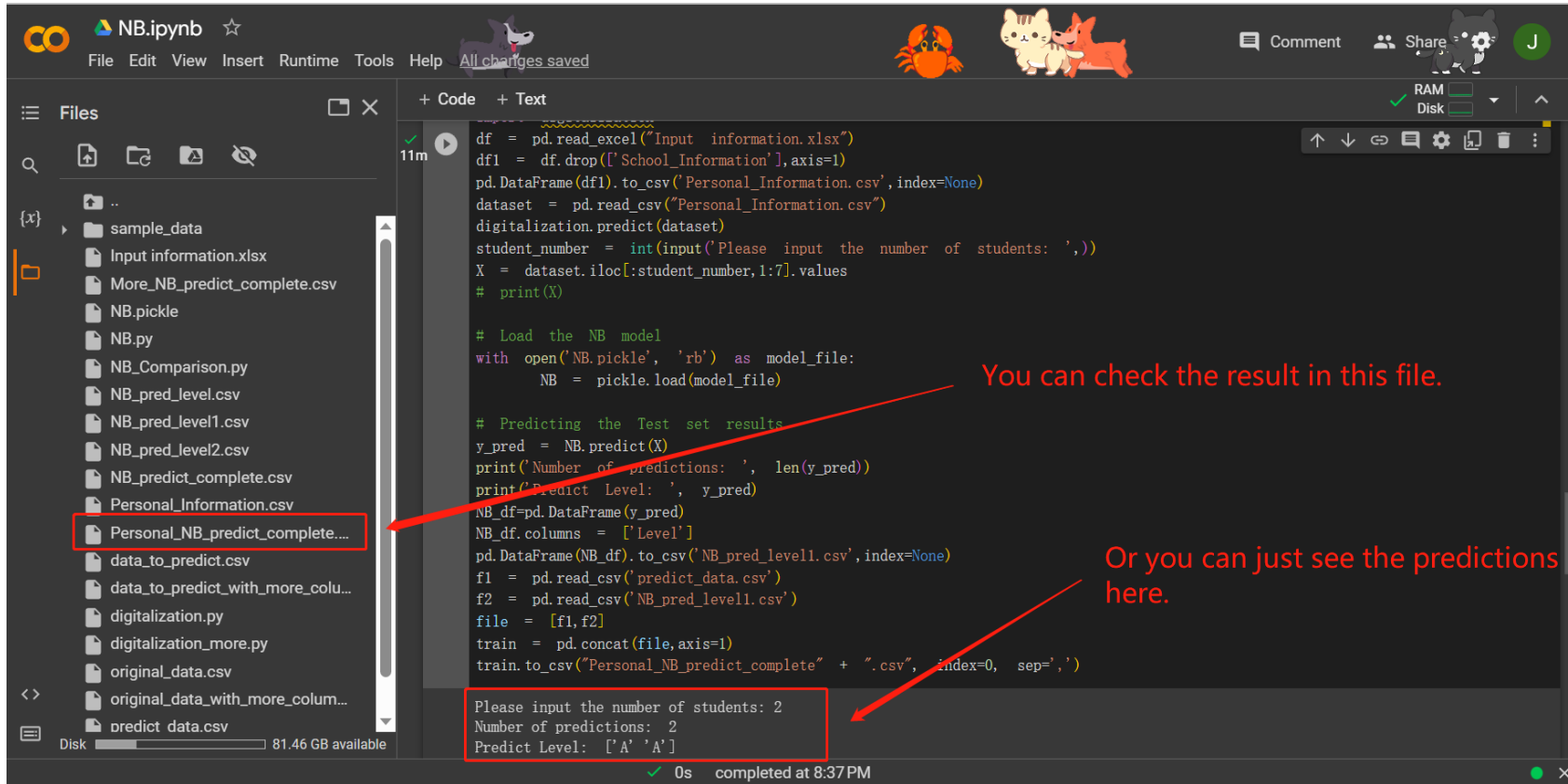
1. First, open the “[Input information.xlsx](#)” from the “NB” file we have already downloaded, fill in your personal information. Notice that in “Exam” and “Math_Background” column, only Gaokao and HKDSE are accepted.

	A	B	C	D	E	F	G	H
1	Student_ID	Admission_Channel	Secondary_School	Exam	Math_Background	MATH1510	ENGG1130	ENGG2020
2	1	JUPAS	CCC HEEP WOH COLLEGE	Mainland Gaokao (JEE)	Mainland Gaokao (JEE):>140	A	A	
3	2	Non-JUPAS (Local)	SUEN WAN PUB HO CHUEN YIU MEMORIAL COL	HKDSE	HKDSE:5*	B+	C	
4								
5								
6			SHUN LEE CATHOLIC SECONDARY SCHOOL					
7			STFA LEE SHAU KEE COLLEGE					
8			STFA LEUNG KAU KUI COLLEGE					
9			STFA YUNG YAU COLLEGE					
10			STFA SEAWARD WOO COLLEGE					
11			STFA TAM PAK YU COLLEGE					
12			STFA CHENG YU TUNG SECONDARY SCHOOL					
13			ELEGANTIA COLLEGE (SPONSORED BY ED CONV)					
14			HEUNG TO MIDDLE SCHOOL					
15			HHCKLA BUDDHIST CHING KOK SEC SCH					
16			HONG KONG CHINESE WOMEN'S CLUB COLLEGE					
17			CHRISTIAN ALLIANCE CHENG WING GEE COLL					
18								
19								
20								
21								
22								
23								
24								
25								
26								
27								
28								
29								

Input your information

Want to try your own data?

2. Upload the “Input information.xlsx” to program like before and run the fourth cell.
3. Then pick up the “Personal_NB_predict_complete.csv” file in the left to check the prediction of your grade!



The screenshot displays a Jupyter Notebook environment. On the left, the file explorer shows a list of files, with 'Personal_NB_predict_complete.csv' highlighted. The main area contains a code cell with the following Python code:

```
df = pd.read_excel("Input information.xlsx")
df1 = df.drop(['School_Information'], axis=1)
pd.DataFrame(df1).to_csv('Personal_Information.csv', index=None)
dataset = pd.read_csv("Personal_Information.csv")
digitalization.predict(dataset)
student_number = int(input('Please input the number of students: '))
X = dataset.iloc[:student_number, 1:7].values
# print(X)

# Load the NB model
with open('NB.pickle', 'rb') as model_file:
    NB = pickle.load(model_file)

# Predicting the Test set results
y_pred = NB.predict(X)
print('Number of predictions: ', len(y_pred))
print('Predict Level: ', y_pred)
NB_df = pd.DataFrame(y_pred)
NB_df.columns = ['Level']
pd.DataFrame(NB_df).to_csv('NB_pred_level1.csv', index=None)
f1 = pd.read_csv('predict_data.csv')
f2 = pd.read_csv('NB_pred_level1.csv')
file = [f1, f2]
train = pd.concat(file, axis=1)
train.to_csv("Personal_NB_predict_complete" + ".csv", index=0, sep=',')
```

Below the code, the terminal output shows the execution results:

```
Please input the number of students: 2
Number of predictions: 2
Predict Level: ['A' 'A']
```

Red arrows point from the text annotations to the file explorer and the terminal output.

Run the fourth cell.

Two ways to see the predictions

Want to add some extra factors?

■ Sure! Only need to follow my steps:

1. We can take a look at “original_data_with_more_columns.csv” in “NB” file.

A1

Original No.

We add two more courses as factors for example

	A	B	C	D	E	F	G	H	I	J
1	Original No.	Admission_Channel	Secondary_School	Exam	Math_Background	MATH1510	ENGG1130	MATH1010	ENGG1120	ENGG2020
2	17	JUPAS	TSUEN WAN PUB HO CHUEN YIU MEMORIAL COLL	HKDSE	HKDSE:5	B+	B+	A-	B	B+
3	20	JUPAS	CHRISTIAN ALLIANCE S C CHAN MEM COLL	HKDSE	HKDSE:5*	A-	B	B	B+	B
4	25	JUPAS	MUNSANG COLLEGE (HONG KONG ISLAND)	HKDSE	HKDSE:4	B	A-	A-	B+	A
5	29	JUPAS	PO LEUNG KUK LAWS FOUNDATION COLLEGE	HKDSE	HKDSE:4	B+	B-	B+	A-	B-
6	36	JUPAS	STFA LEUNG KAU KUI COLLEGE	HKDSE	HKDSE:5	B-	B+	B+	B	B+
7	37	JUPAS	METHODIST CHURCH HK WESLEY COLLEGE	HKDSE	HKDSE:5	D	D+	B+	B+	D
8	50	JUPAS	Others	HKDSE	HKDSE:5	B-	C+	D	D+	C
9	55	JUPAS	KWOK TAK SENG CATHOLIC SECONDARY SCHOOL	HKDSE	HKDSE:5	B	B	B	C+	B
10	62	JUPAS	YWCA HIOE TJO YOENG COLLEGE	HKDSE	HKDSE:4	B	B+	B-	B	B+
11	82	JUPAS	CCC CHUEN YUEN COLLEGE	HKDSE	HKDSE:5	B	F	B	B+	F
12	88	JUPAS	HKMA K S LO COLLEGE	HKDSE	HKDSE:4	C	C+	C-	F	C
13	92	JUPAS	LIONS COLLEGE	HKDSE	HKDSE:4	B	C-	C+	C+	C
14	94	JUPAS	YCH LIM POR YEN SECONDARY SCHOOL	HKDSE	HKDSE:5**	B+	A-	C-	C-	A
15	98	JUPAS	HK SKH BISHOP HALL SECONDARY SCHOOL	HKDSE	HKDSE:5**	B+	B+	A-	A-	B+
16	101	JUPAS	CHRISTIAN ALLIANCE S W CHAN MEM COLL	HKDSE	HKDSE:5	B-	B-	B+	B+	B-
17	109	JUPAS	VALTORTA COLLEGE	HKDSE	HKDSE:5**	A-	A	B-	B+	A
18	119	JUPAS	AOG HEBRON SECONDARY SCHOOL	HKDSE	HKDSE:5	B-	A-	A	A	A
19	129	JUPAS	PUI CHING MIDDLE SCHOOL	HKDSE	HKDSE:5**	A-	A	A-	A-	A
20	130	JUPAS	TWGH LI KA SHING COLLEGE	HKDSE	HKDSE:5	C+	C-	A	A	C
21	142	JUPAS	TANG KING PO SCHOOL	HKDSE	HKDSE:4	B+	B-	C-	C+	B-
22	144	JUPAS	PLK MRS MA KAM MING CHEUNG FOOK SIEN COL	HKDSE	HKDSE:4	B+	B-	B-	B-	B-
23	147	JUPAS Appeal-in	STFA YUNG YAU COLLEGE	HKDSE	HKDSE:5	B-	C	B-	B-	C
24	168	JUPAS	FUKIEN SECONDARY SCHOOL	HKDSE	HKDSE:5*	A-	B+	C	C	B+
25	172	JUPAS	SKH LAM WOO MEMORIAL SECONDARY SCHOOL	HKDSE	HKDSE:4	B	C+	B+	B+	C

You can replace the content as you like

Want to add some extra factors?

2. After **training** the model with the dataset in previous slide.
3. You can make a csv document to **contain the students' information** you want to predict their level.

Student_ID									
After running the cell, the level of these students will be added									
A	B	C	D	E	F	G	H	I	J
Student_ID	Admission_Channel	Secondary_School	Exam	Math_Background	MATH1510	ENGG1130	MATH1010	ENGG1120	ENGG2020
627	JUPAS	CHAN SUI KI (LA SALLE) COLLEGE	HKDSE	HKDSE:5*	A-	A-	B-	B-	
882	JUPAS	CHRISTIAN ALLIANCE CHENG WING GEE COLL	HKDSE	HKDSE:4	D	F	B	B+	
801	JUPAS	MA ON SHAN TSUNG TSIN SECONDARY SCHOOL	HKDSE	HKDSE:5*	B-	B	C	C+	
1087	Mainland Fee-paying	Others	Mainland	(Mainland Gaokao (JEE):<110	B-	B	B-	C+	
224	JUPAS	ELEGANTIA COLLEGE (SPONSORED BY ED CONV)	HKDSE	HKDSE:5*	B-	F	B-	B-	
240	JUPAS	SKH BISHOP MOK SAU TSENG SEC SCH	HKDSE	HKDSE:5*	A-	A-	B-	B+	
911	JUPAS	Others	HKDSE	HKDSE:5**	A	A	B-	B+	
917	JUPAS	DIOCESAN BOYS SCHOOL	HKDSE	HKDSE:4	D	B-	B	B+	
624	Mainland	Others	Mainland	(Mainland Gaokao (JEE):120-129	A-	A	B	A	
724	JUPAS	SING YIN SECONDARY SCHOOL	HKDSE	HKDSE:4	B	B	B-	B+	
546	JUPAS	KWUN TONG MARYKNOLL COLLEGE	HKDSE	HKDSE:5*	A	A	B	B+	
410	Mainland	Others	Mainland	(Mainland Gaokao (JEE):130-139	B+	B	A-	B	
492	JUPAS	Others	HKDSE	HKDSE:5	B-	A-	B	B-	
986	JUPAS	Others	HKDSE	HKDSE:4	C+	B+	A-	B+	
1046	JUPAS	HELEN LIANG MEMORIAL SEC SCH (SHATIN)	HKDSE	HKDSE:5	A-	B+	B	B	
272	JUPAS	HEUNG TO MIDDLE SCHOOL	HKDSE	HKDSE:4	B-	C-	A-	A-	
803	JUPAS	Others	HKDSE	HKDSE:4	B-	B+	B	B-	
383	JUPAS	LAI KING CATHOLIC SECONDARY SCHOOL	HKDSE	HKDSE:4	C-	B	B-	B	
881	JUPAS	YAN OI TONG TIN KA PING SECONDARY SCHOOL	HKDSE	HKDSE:4	B-	C+	B+	B	
932	JUPAS	SKH TANG SHIU KIN SECONDARY SCHOOL	HKDSE	HKDSE:4	B	B+	B-	A-	
120	Mainland	Others	Mainland	(Mainland Gaokao (JEE):120-129	A	A	A-	A	
732	JUPAS	HKSYC&IA WONG TAI SHAN MEM COLL	HKDSE	HKDSE:4	C	B	B+	B+	
474	JUPAS	SING YIN SECONDARY SCHOOL	HKDSE	HKDSE:4	B	A	B-	C-	
776	JUPAS	STFA LEUNG KAU KUI COLLEGE	HKDSE	HKDSE:5*	A	A	B-	B-	

The level of students will be added automatically

Want to add some extra factors?

4. Then give the file to our model.

The screenshot shows a Jupyter Notebook environment with the following components:

- File Explorer (Left):** Lists files including `sample_data`, `Input information.xlsx`, `More_NB_predict_complete.csv` (highlighted with a yellow box), `NB.pickle`, `NB.py`, `NB_Comparison.py`, `NB_pred_level.csv`, `NB_pred_level1.csv`, `NB_pred_level2.csv`, `NB_predict_complete.csv`, `Personal_Information.csv`, `Personal_NB_predict_complete...`, `data_to_predict.csv`, `data_to_predict_with_more_colu...` (highlighted with a red box), `digitalization.py`, `digitalization_more.py`, `original_data.csv`, `original_data_with_more_colum...` (highlighted with a red box), and `predict_data.csv`.
- Code Cell (Center):** Contains Python code for a Naive Bayes model:

```
[5]: digitalization_more.train(data)

NB.NB_model(factor_number=9, test_data_number=90)

dataset = pd.read_csv("data_to_predict_with_more_columns.csv")
digitalization_more.predict(dataset)

X = dataset.iloc[:,1:9].values

# Load the NB model
with open('NB.pickle', 'rb') as model_file:
    NB = pickle.load(model_file)

# Predicting the Test set results
y_pred = NB.predict(X)
print('Number of predictions: ', len(y_pred))
NB_df=pd.DataFrame(y_pred)
NB_df.columns = ['Level1']
pd.DataFrame(NB_df).to_csv('NB_pred_level2.csv', index=None)
f1 = pd.read_csv('predict_data.csv')
f2 = pd.read_csv('NB_pred_level2.csv')
file = [f1, f2]
train = pd.concat(file, axis=1)
train.to_csv("More_NB_predict_complete" + ".csv", index=0, sep=',')
```
- Output (Bottom):** A box showing the results:

```
Accuracy of the Naive Bayes Clasification is: 0.9888888888888889
Number of predictions: 545
```
- Annotations:**
 - Step 1:** "Upload two csv file to the program." (Red arrow pointing to the file explorer).
 - Step 2:** "You can check the accuracy of the predictions" (Blue arrow pointing to the output box).
 - Step 3:** "You can pick the final results in this csv file after running the cell." (Yellow arrow pointing to `More_NB_predict_complete.csv`).

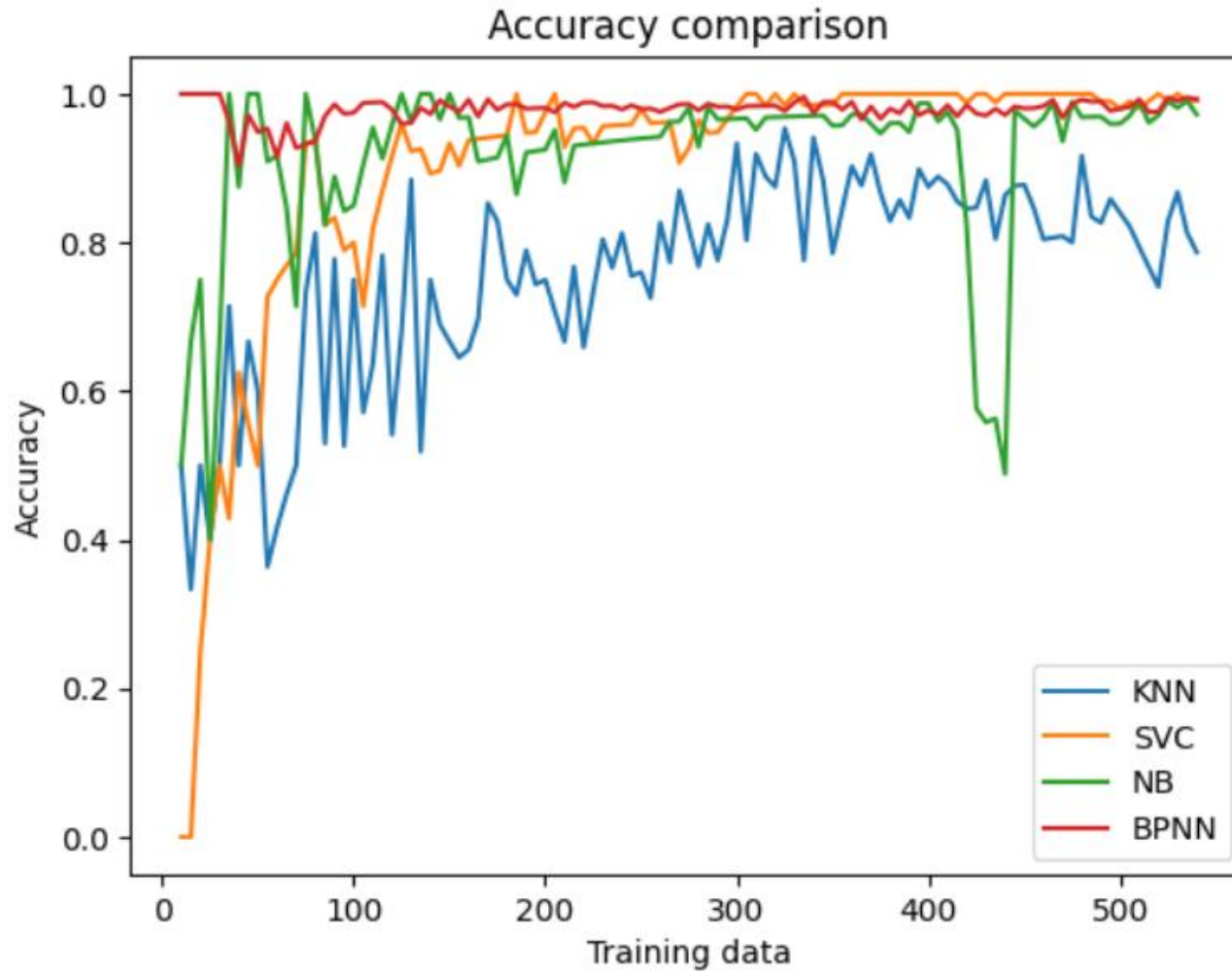
Run the fifth cell and get your results



Methods comparison

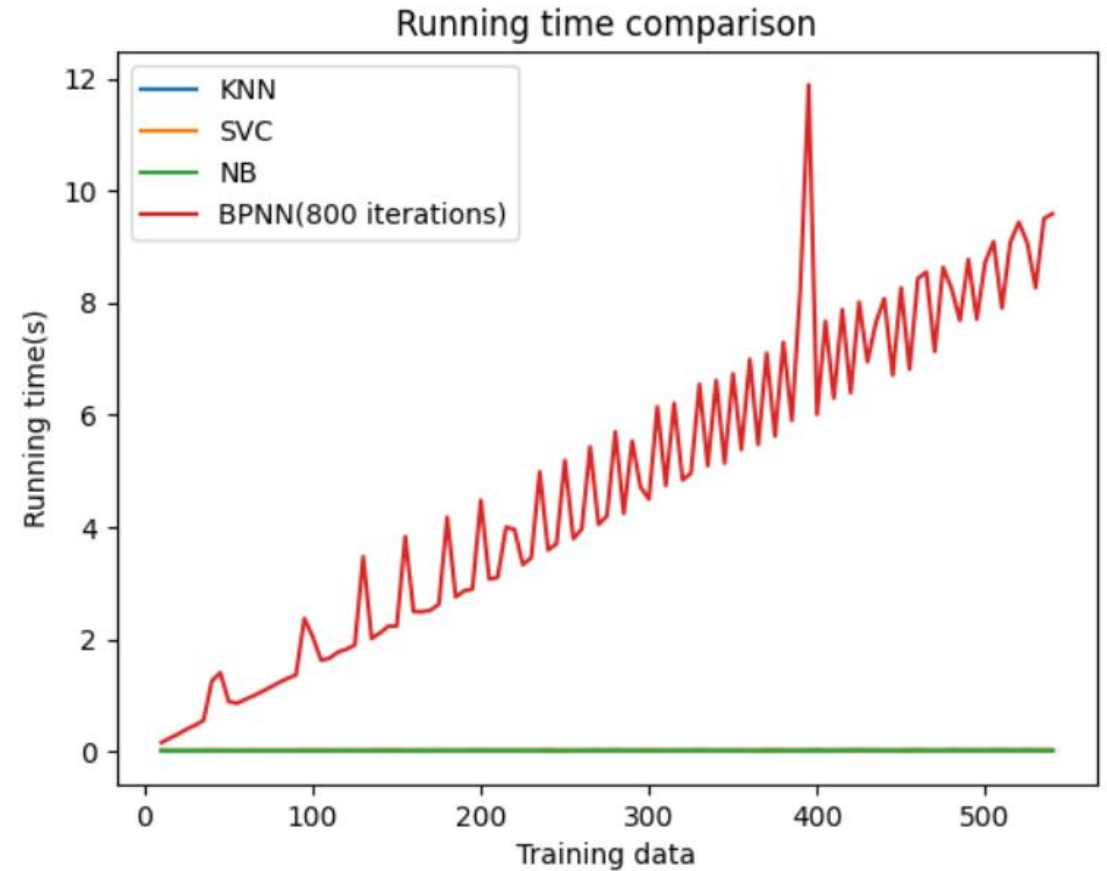
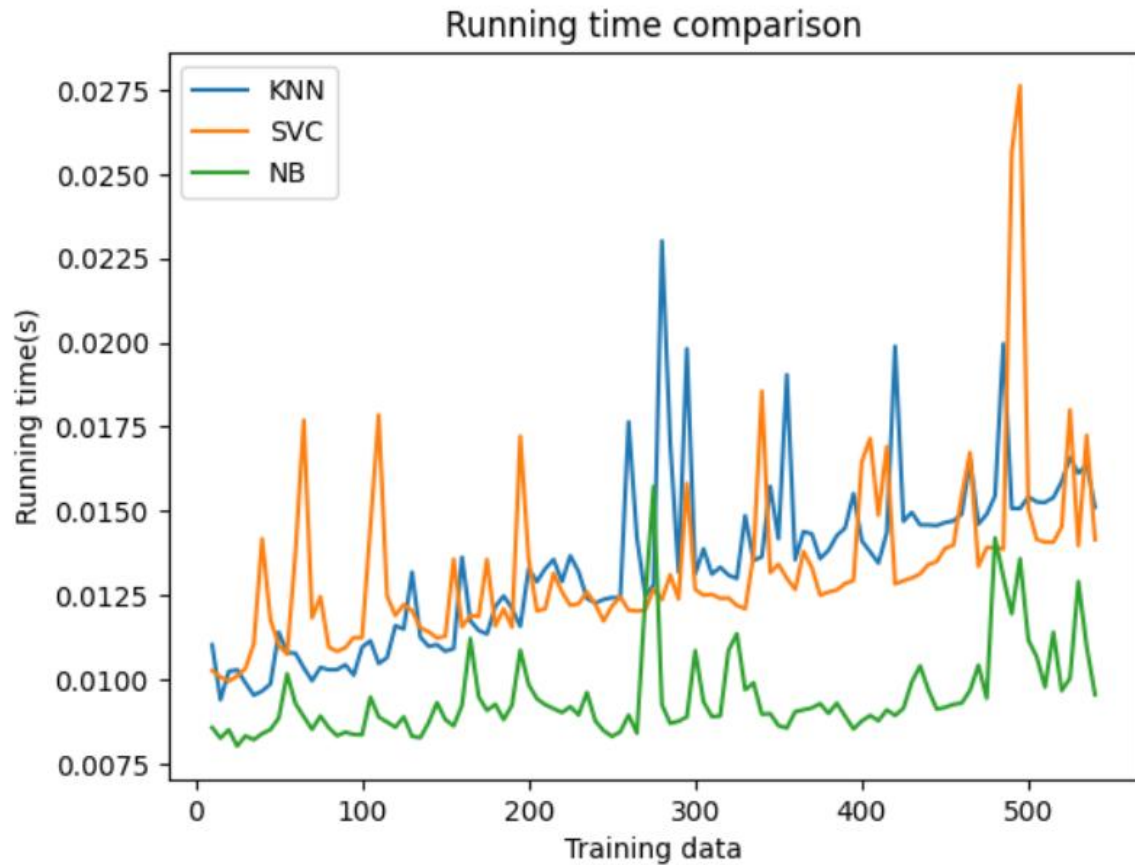
Select the best methods for
learner level prediction

Comparison in accuracy



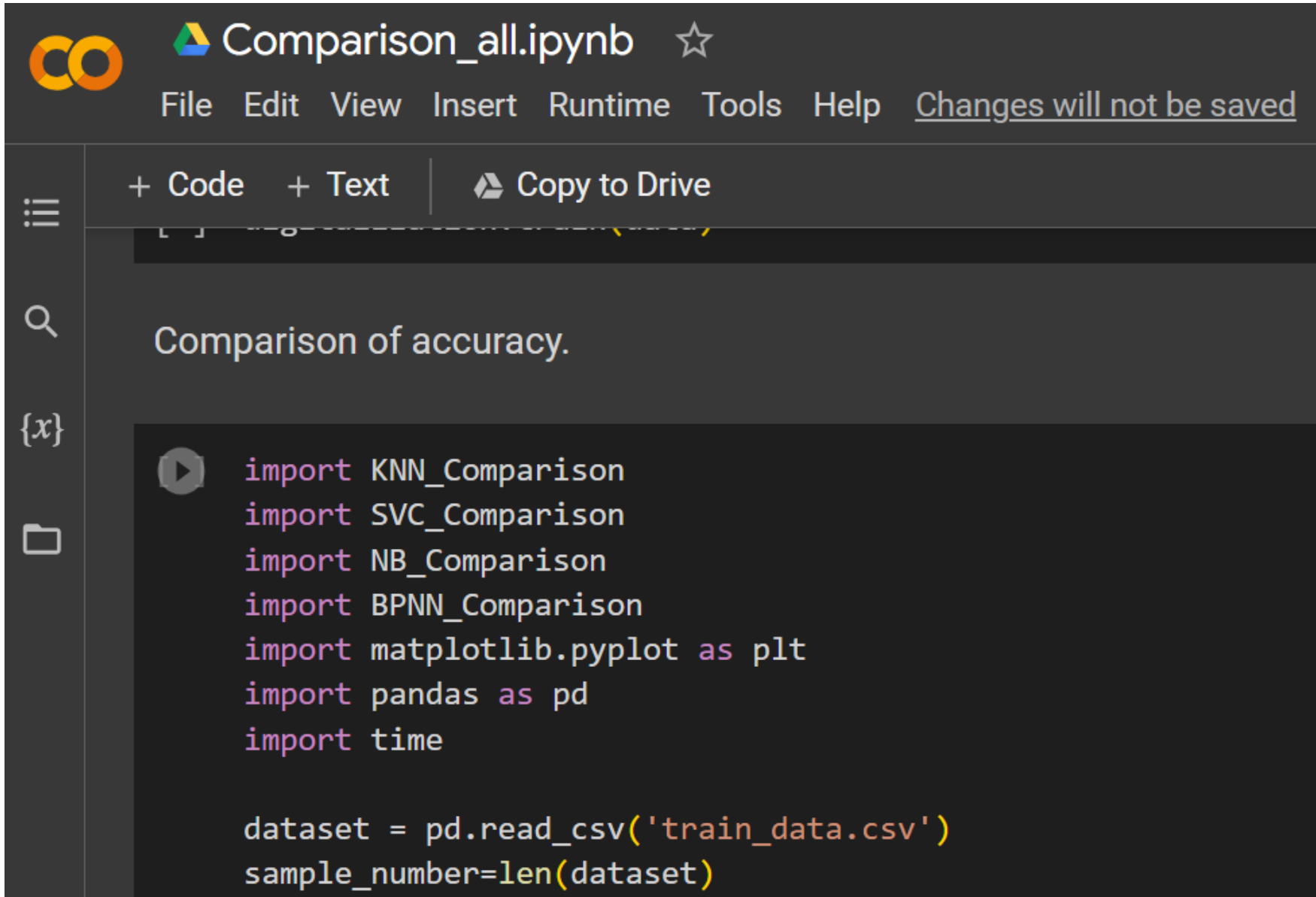
Comparison of four algorithms in accuracy

Comparison in accuracy



Comparison of four algorithms in computational time

Code for comparison

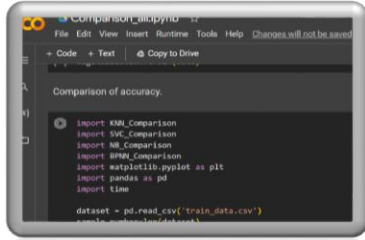


The image shows a Jupyter Notebook interface with a dark theme. The title bar at the top reads "Comparison_all.ipynb" with a star icon to its right. Below the title bar is a menu bar with options: File, Edit, View, Insert, Runtime, Tools, Help, and a status message "Changes will not be saved". Under the menu bar, there are buttons for "+ Code", "+ Text", and "Copy to Drive". On the left side, there is a sidebar with icons for a list, search, a variable "{x}", and a folder. The main area of the notebook contains the text "Comparison of accuracy." followed by a code cell. The code cell has a play button icon and contains the following Python code:

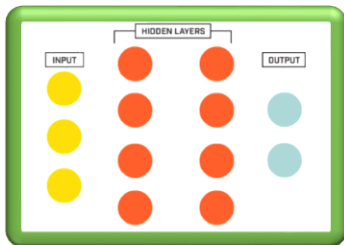
```
import KNN_Comparison
import SVC_Comparison
import NB_Comparison
import BPNN_Comparison
import matplotlib.pyplot as plt
import pandas as pd
import time

dataset = pd.read_csv('train_data.csv')
sample_number=len(dataset)
```

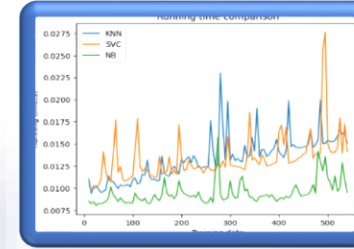
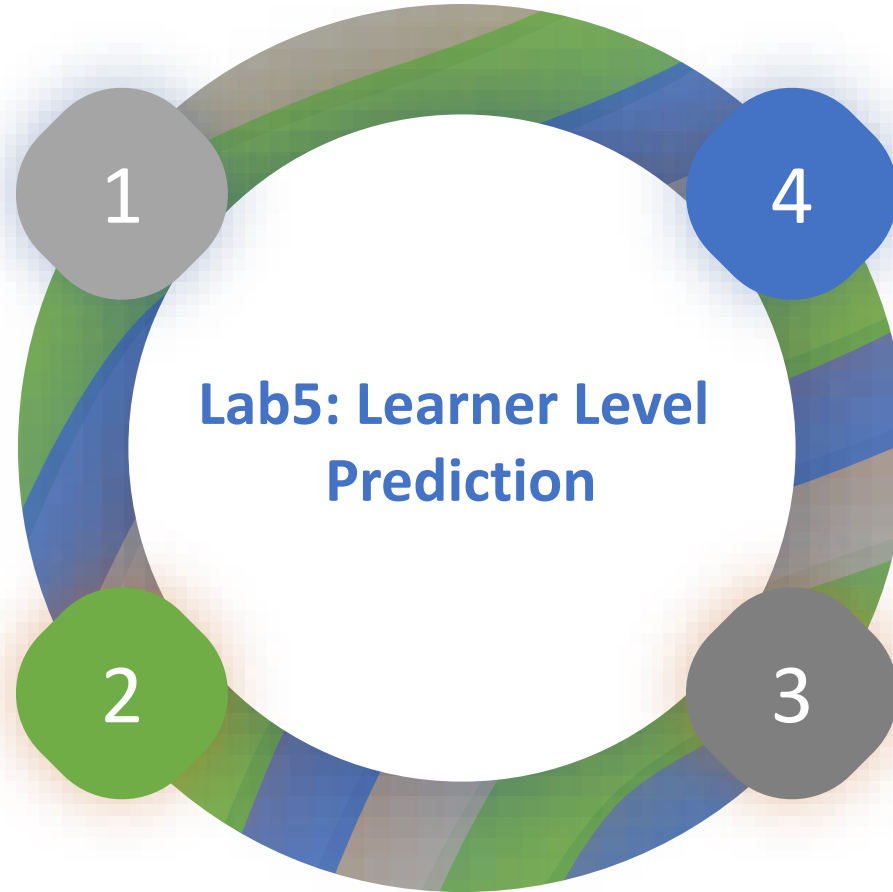
Takeaways



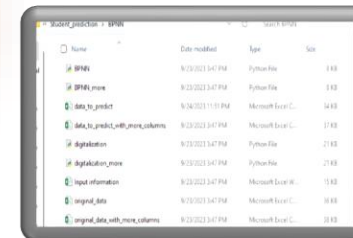
Operation
on Colab



4 methods
introduction

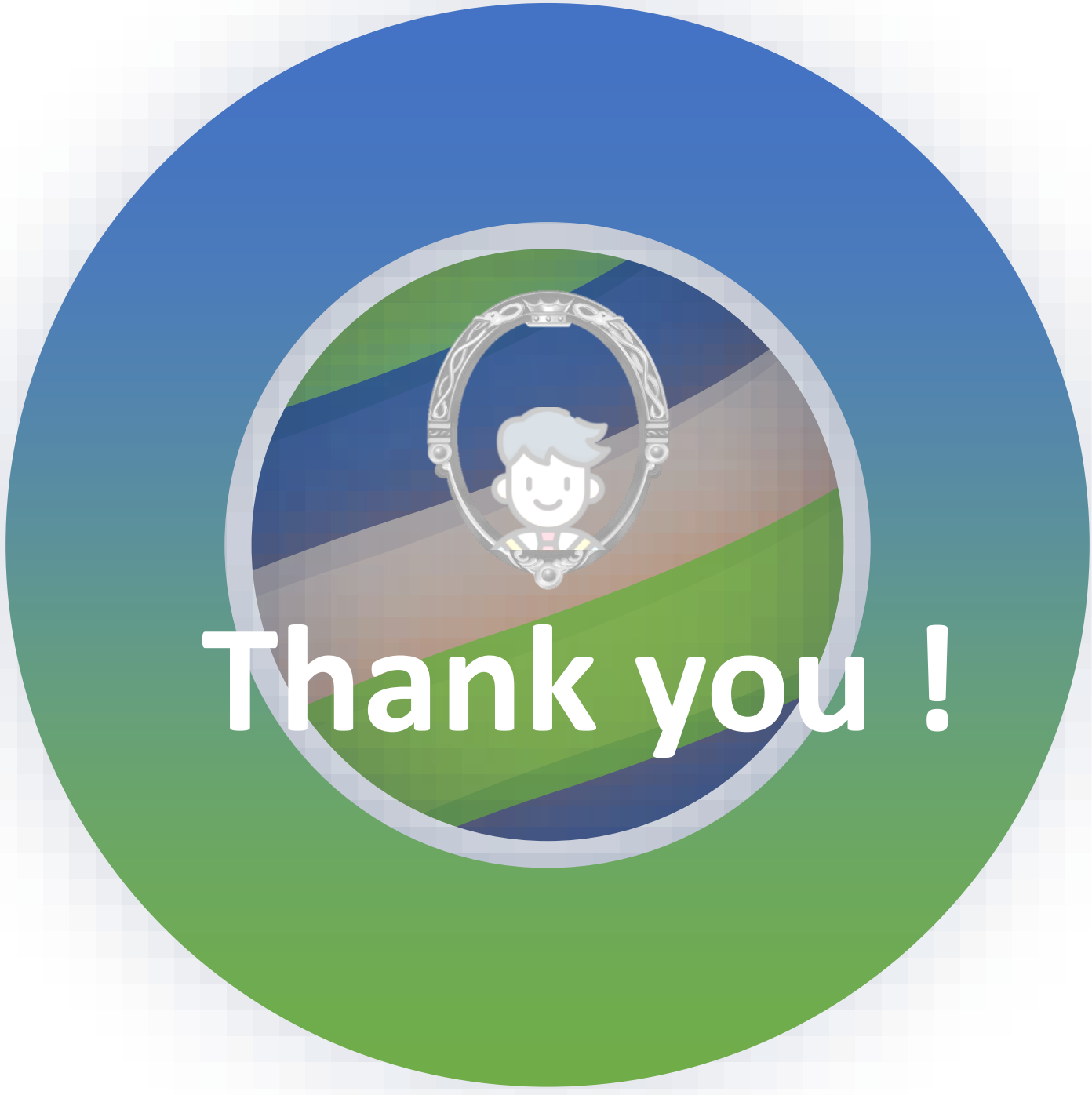


Best method
selection strategy



Name	Date modified	Type	Size
RFM	9/11/2023 14:7 PM	Python File	1 KB
RFM more	9/11/2023 14:7 PM	Python File	1 KB
data_to_predict	9/11/2023 11:11 PM	Microsoft Excel C...	14 KB
data_to_predict_with_more_columns	9/11/2023 14:7 PM	Microsoft Excel C...	17 KB
digitization	9/11/2023 14:7 PM	Python File	21 KB
digitization more	9/11/2023 14:7 PM	Python File	21 KB
input information	9/11/2023 14:7 PM	Microsoft Excel W...	15 KB
input_data	9/11/2023 14:7 PM	Microsoft Excel C...	10 KB
input_data_with_more_columns	9/11/2023 14:7 PM	Microsoft Excel C...	10 KB

Sample data and
sample codes



Thank you !